



Setting Up OpenShift

In Your Lab or Homelab

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Welcome! I'm Glad You're Here!

Agenda

Introduction

Credit

Network

DHCP

DNS

HAProxy

NFS

Installation

Post-installation

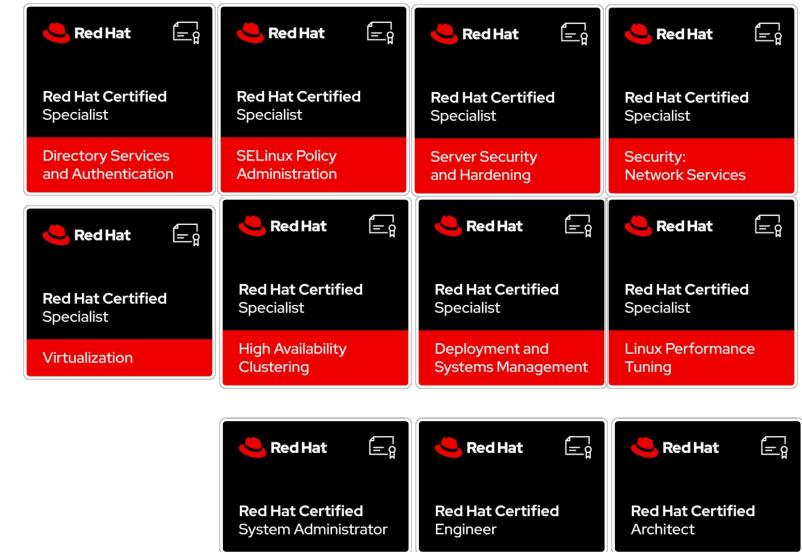
Introduction

Introduction

Who am I?

I'm Thomas Cameron. Senior Principal Solution Architect.

- I've been doing this since 1993 🧐
- Started as a Novell CNE
- Went to work for Microsoft, became an MCSE/MCT
- Started using Linux in 1995 (while working at MSFT)
- Worked for Red Hat for about 14 years - RHCA level 5
- Went to AWS for 4 years back in 2019 - Solution Architect Professional, Solution Architect Associate, Security Specialty, SysOps Associate, AAI
- Saw the error of my ways, and I'm **back at Red Hat!**



Credit

Bryan Hay on Youtube

I used a lot of his information

HUGE credit goes to Ryan Hay (<https://github.com/ryanhay>)

I shamelessly used his video:

<https://www.youtube.com/watch?v=d03xg2PKOPg>

And his instructions:

<https://github.com/ryanhay/ocp4-metal-install>

...to prepare this demo. I had to update a fair amount for OpenShift 4.19, but the bones are from his work.

Network

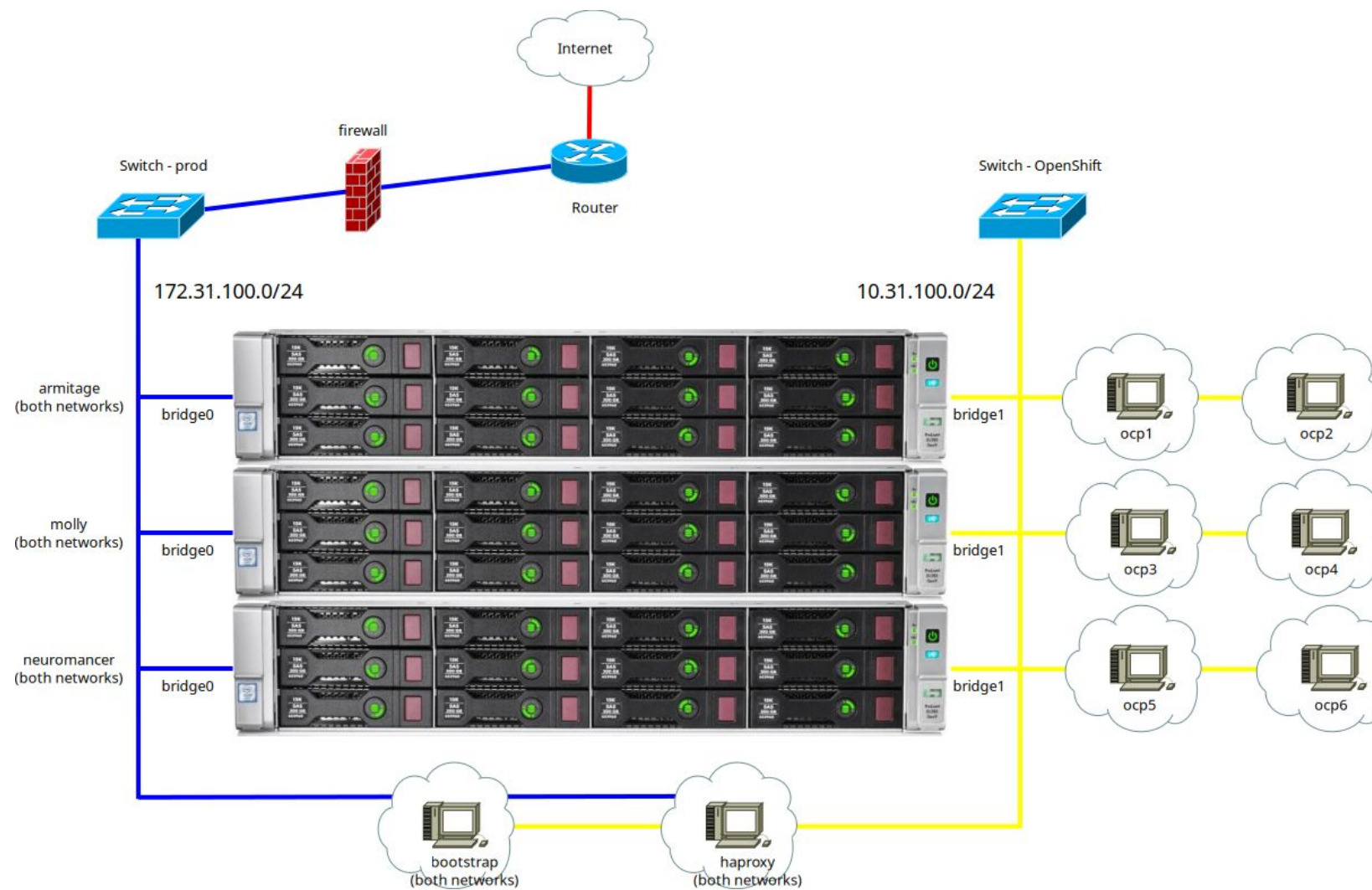


Red Hat

Network

I tried to make it as close to “production” as possible

- Three Proliant DL380 Servers running RHEL 10, connected to the production network (172.31.100.0/24) and the private network (10.31.100.0/24). Each physical interface is bridged.
- Load balancer KVM virtual machine (HAProxy) connected to both networks
 - Also runs DNS and DHCP for the private network
 - The load balancer is used for intercluster communication between nodes and for access from the production network
- Bootstrap VM connected to both networks for installation
- 6 VMs attached to the bridge on the private network only for OpenShift



DHCP

The HAProxy Machine

Connect the VM to both networks

I kickstarted the VM on the production network

Then added a second interface on the OpenShift network

On the big physical machines, they're connected to each network

Each interface is bridged so we can assign IP addresses to each of the VMs

```
(sysadmin) armitage — Konsole
New Tab Split View Copy Paste Find...
tcameron@case:~$ ssh sysadmin@armitage
Web console: https://armitage.aus.redhat.com:9090/ or https://172.31.100.16:9090/

Last login: Fri May 8 12:18:24 2026 from ::ffff:172.31.100.4
sysadmin@armitage:~$ ip -br a sh
lo                UNKNOWN      127.0.0.1/8 ::1/128
eno1              UP
eno2              DOWN
eno3              DOWN
eno4              DOWN
eno49             UP
eno50             DOWN
bridge0           UP            172.31.100.16/24 fe80::6b53:7681:6f9:978/64
vnet8             UNKNOWN      fe80::fc54:ff:fe03:c6b9/64
vnet17            UNKNOWN      fe80::fc54:ff:fe45:506e/64
vnet27            UNKNOWN      fe80::fc54:ff:fe6a:6733/64
bridge1           UP            10.31.100.16/24 fe80::a2a3:77f2:4a3e:d9f4/64
vnet30            UNKNOWN      fe80::fc54:ff:fe1d:97f1/64
sysadmin@armitage:~$
```


Networking - sysadmin@neuromancer.aus.redhat.com - Google Chrome

Networking - sysadmin@neuromancer.aus.redhat.com | Networking - sysadmin@neuromancer.aus.redhat.com | Networking - sysadmin@neuromancer.aus.redhat.com | +

Not secure https://neuromancer:9090/network

Personal

sysadmin@neuromancer.aus.redhat.com

Administrative access ? Help ⚙ Session

Search

Apps

Edit

Image Builder

System

Overview

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Podman containers

Virtual machines

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Tools

Applications

Disconnections

Kbps Transmitting

17:24 17:25 17:26 17:27 17:28

Mbps Receiving

17:24 17:25 17:26 17:27 17:28

Firewall ☒ Enabled

Edit rules and zones

2 active zones

Interfaces

Add VPN

Add bond

Add bridge

Add VLAN

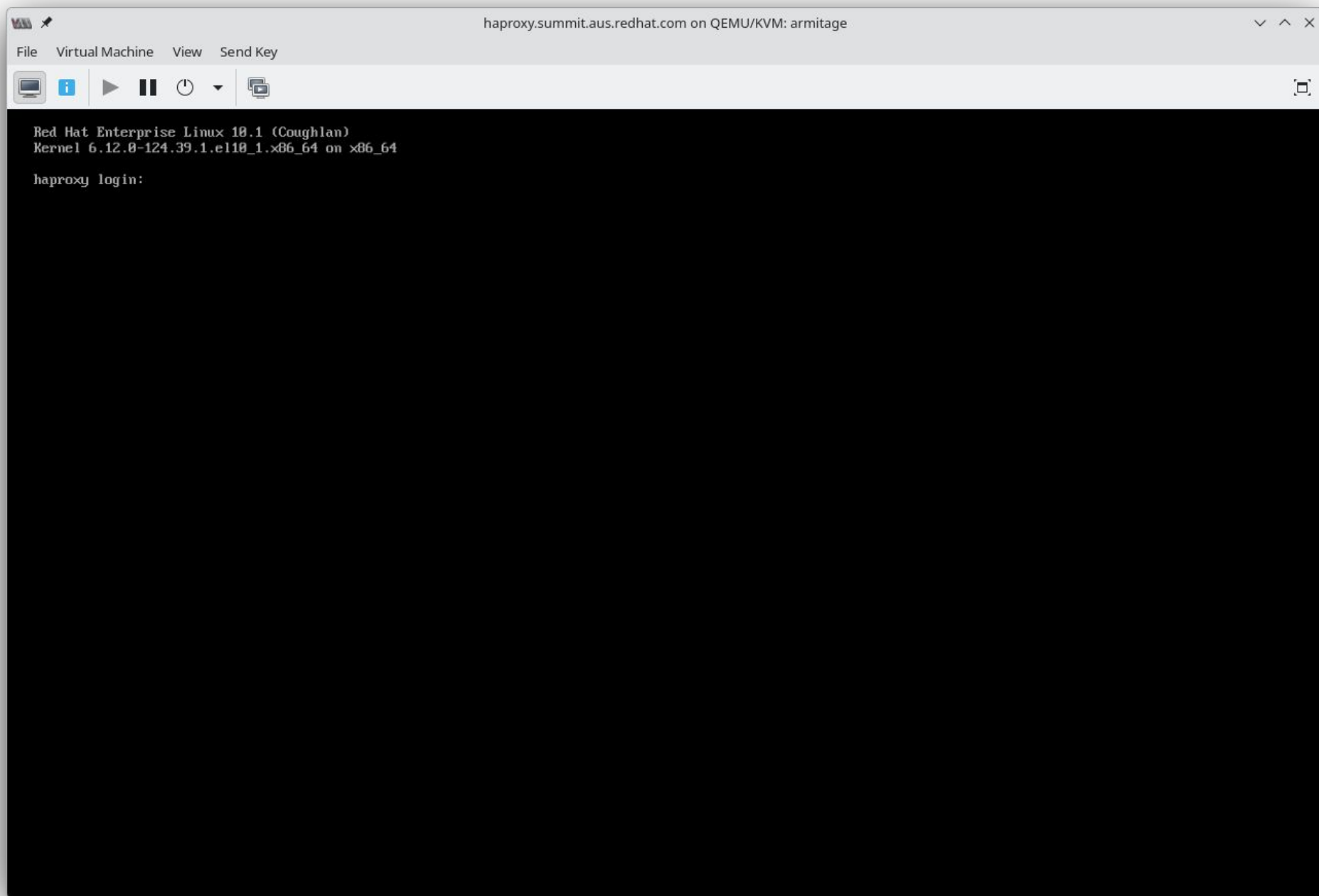
Name	IP address	Sending	Receiving
bridge0	172.31.100.3/24	36.0 Kbps	753 Kbps
bridge1	10.31.100.3/24	0 bps	367 bps
eno2		Not available	
eno3		Not available	

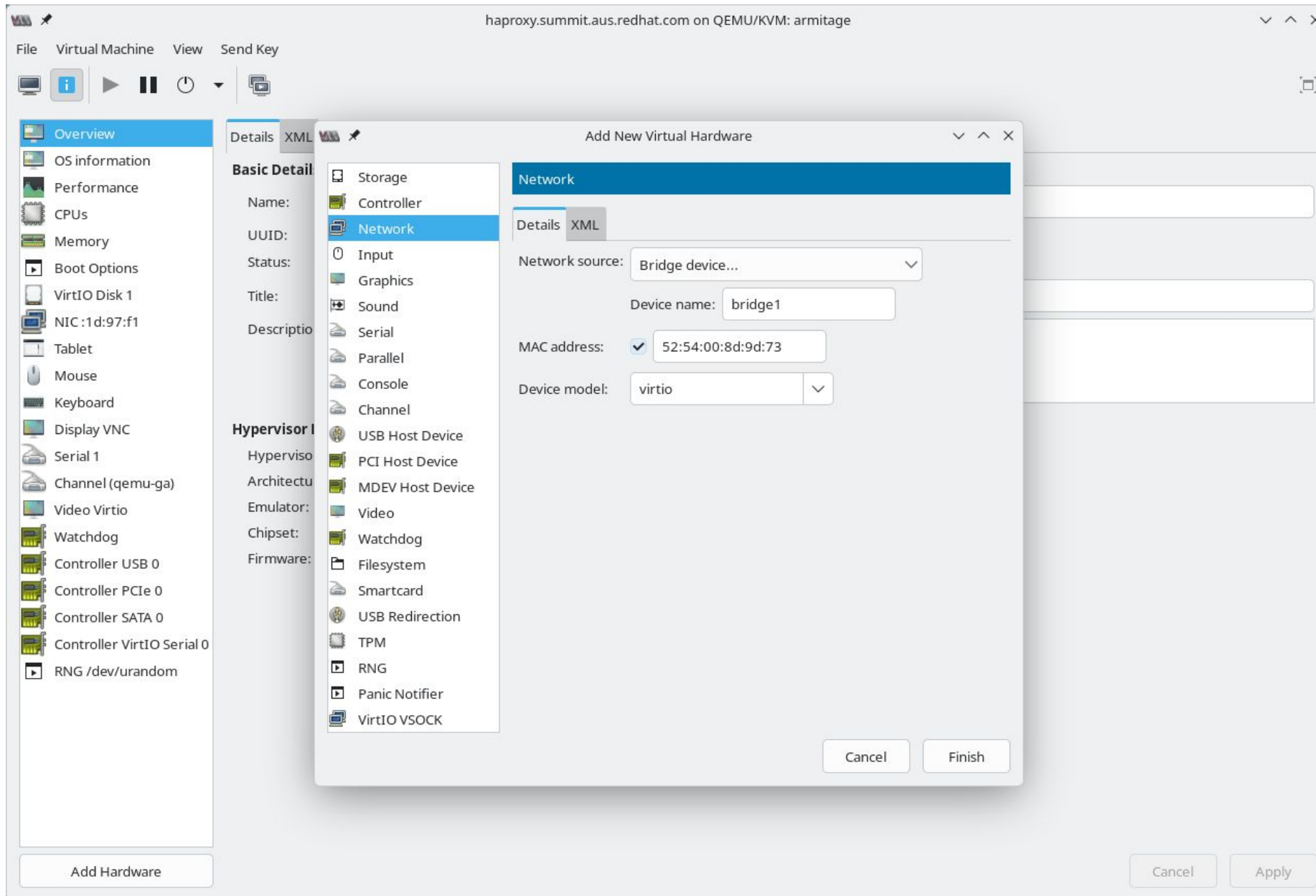
The HAProxy Machine

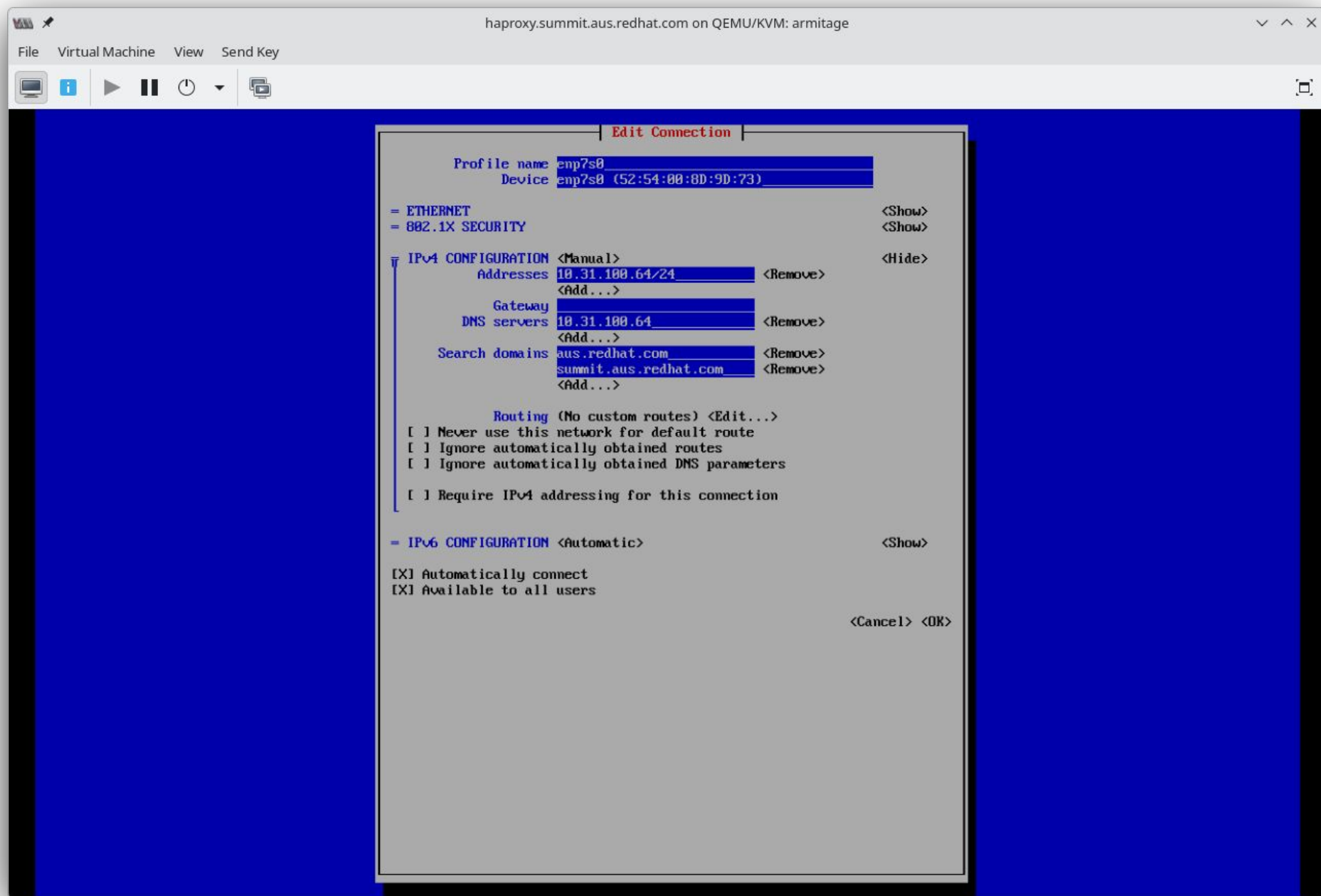
Connect the VM to both networks

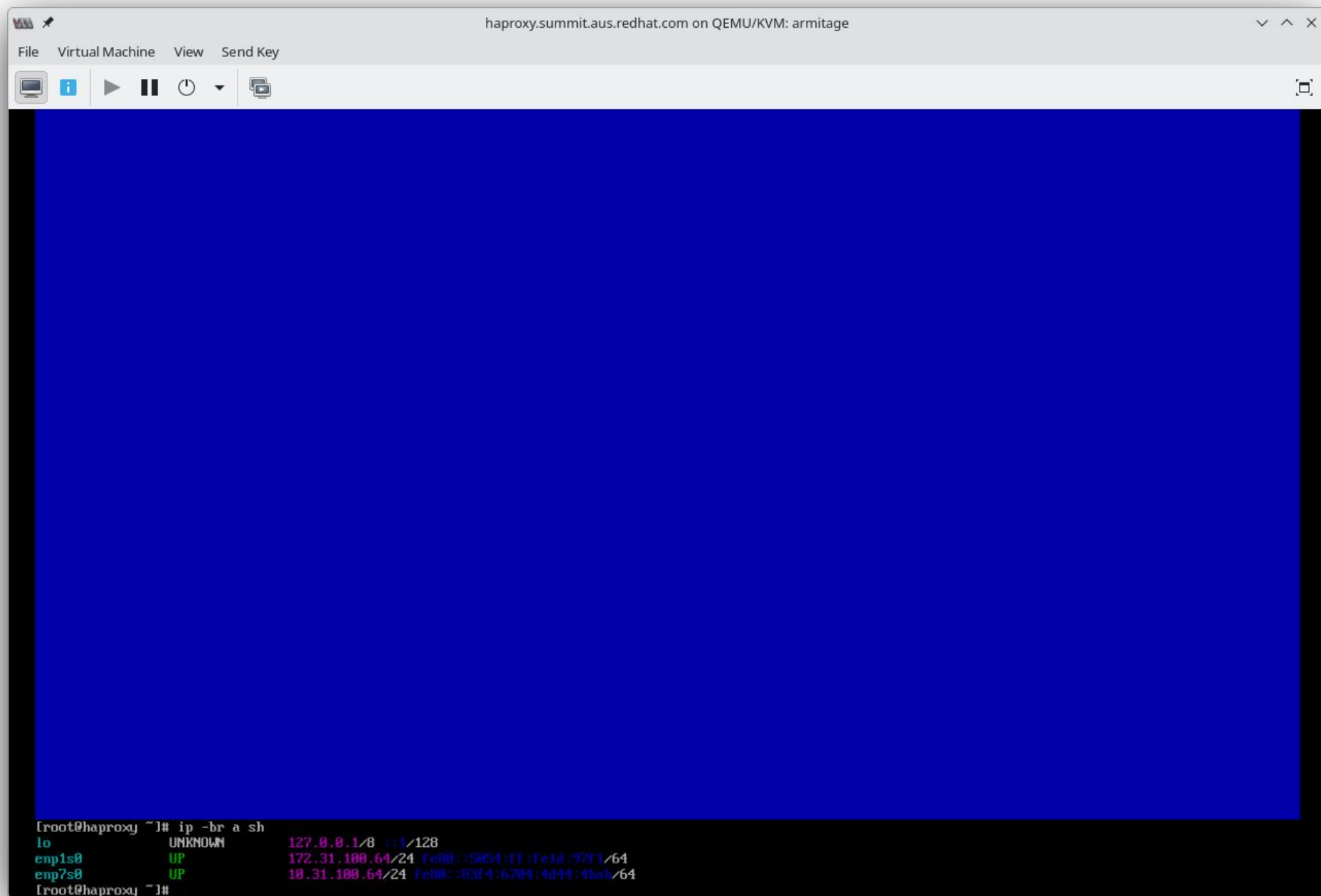
I am adding another interface to the VM, and attaching it to the bridged network on the private (10.31.100.0/24) network

This proxy machine is going to run DNS, DHCP, and a web server, so I will set the IP address statically.





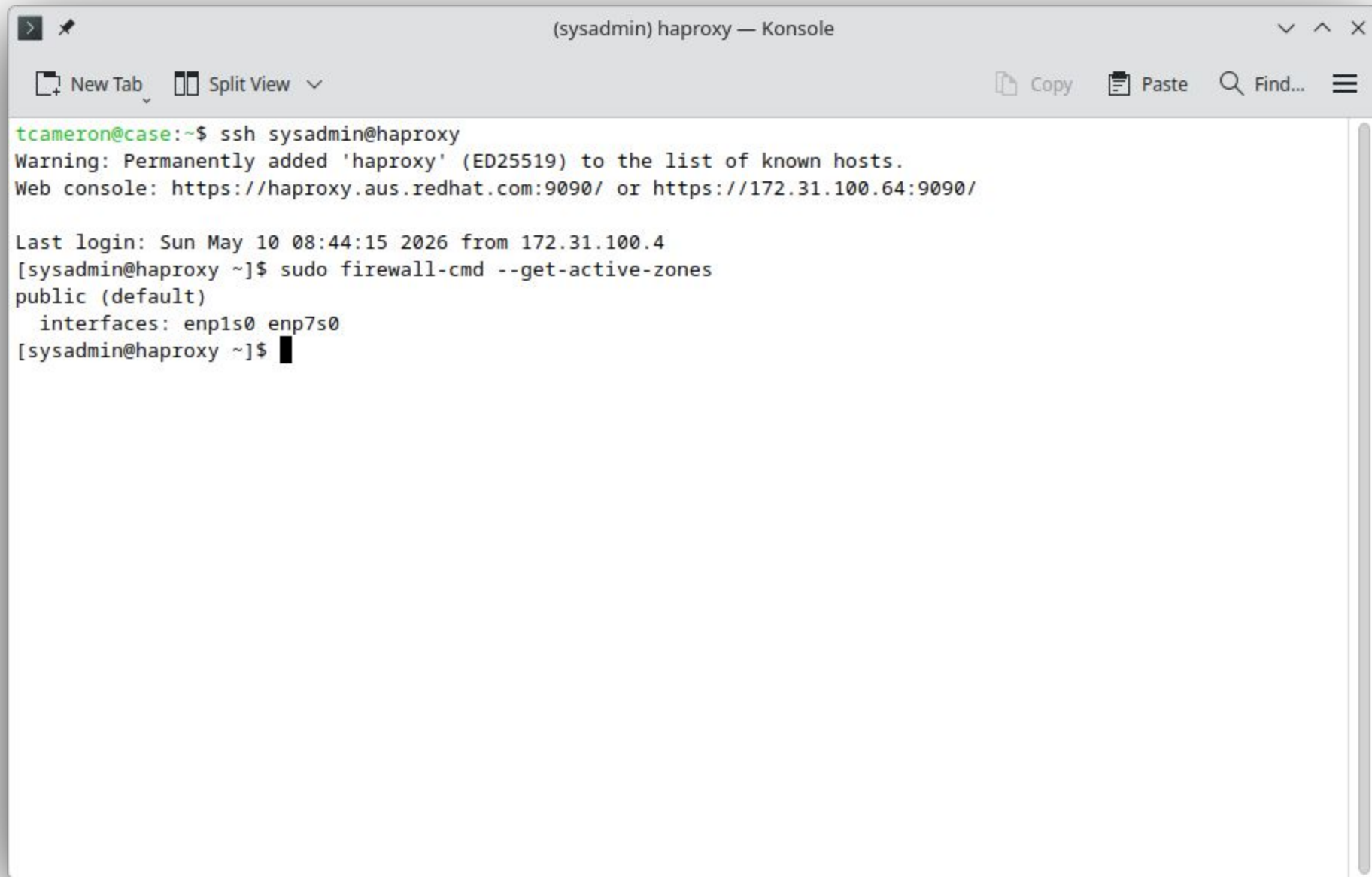




The HAProxy Machine

Connect the VM to both networks

Now we need to configure the firewall. We need to associate each interface with a firewall zone.



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal text shows a successful SSH login from "tcameron@case" to "sysadmin@haproxy". It displays a warning about adding the host to known hosts, a web console URL, the last login time, and the output of the "sudo firewall-cmd --get-active-zones" command, which shows the "public" zone with interfaces "enp1s0" and "enp7s0".

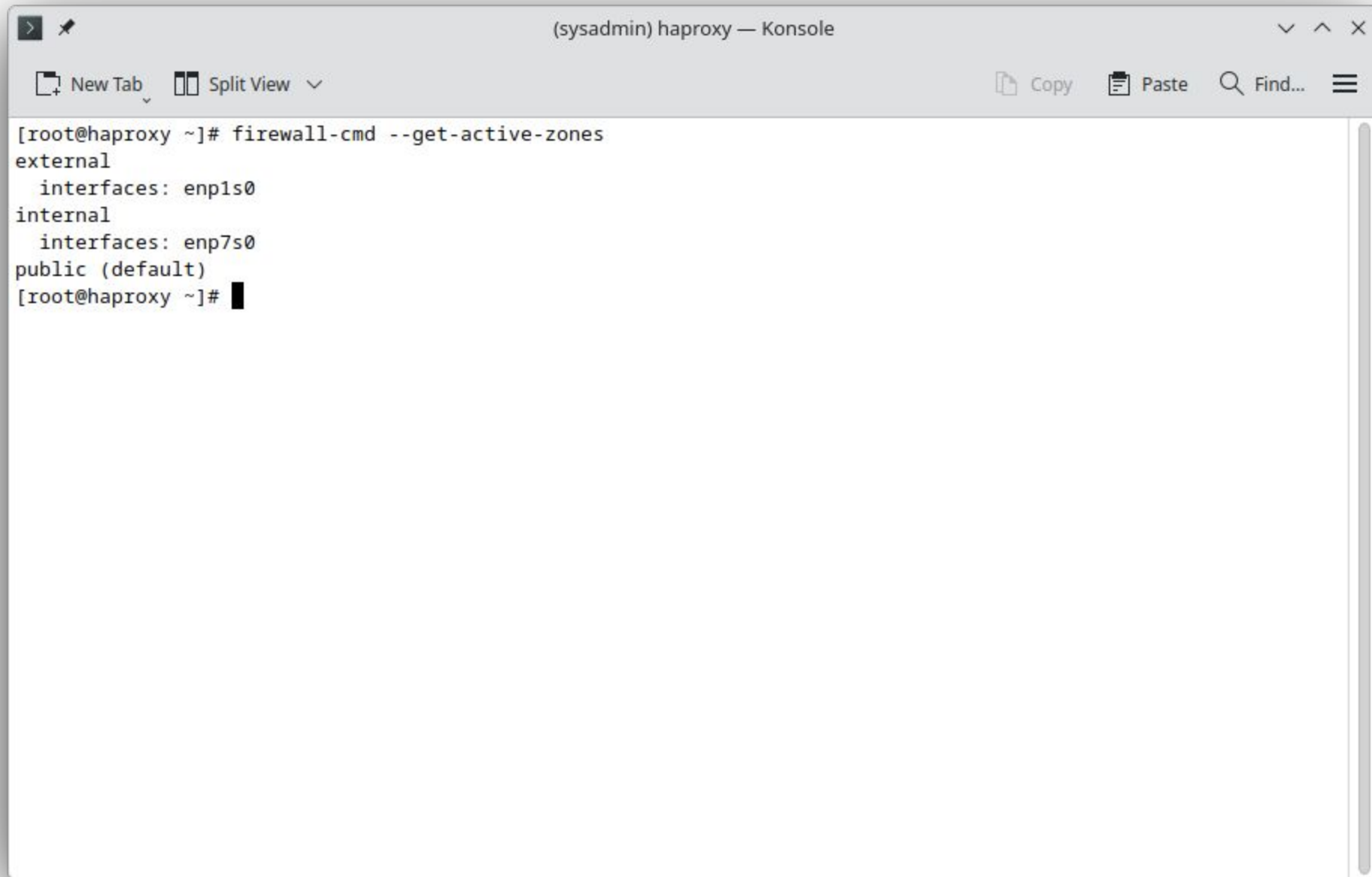
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...

tcameron@case:~$ ssh sysadmin@haproxy
Warning: Permanently added 'haproxy' (ED25519) to the list of known hosts.
Web console: https://haproxy.aus.redhat.com:9090/ or https://172.31.100.64:9090/

Last login: Sun May 10 08:44:15 2026 from 172.31.100.4
[sysadmin@haproxy ~]$ sudo firewall-cmd --get-active-zones
public (default)
  interfaces: enp1s0 enp7s0
[sysadmin@haproxy ~]$
```



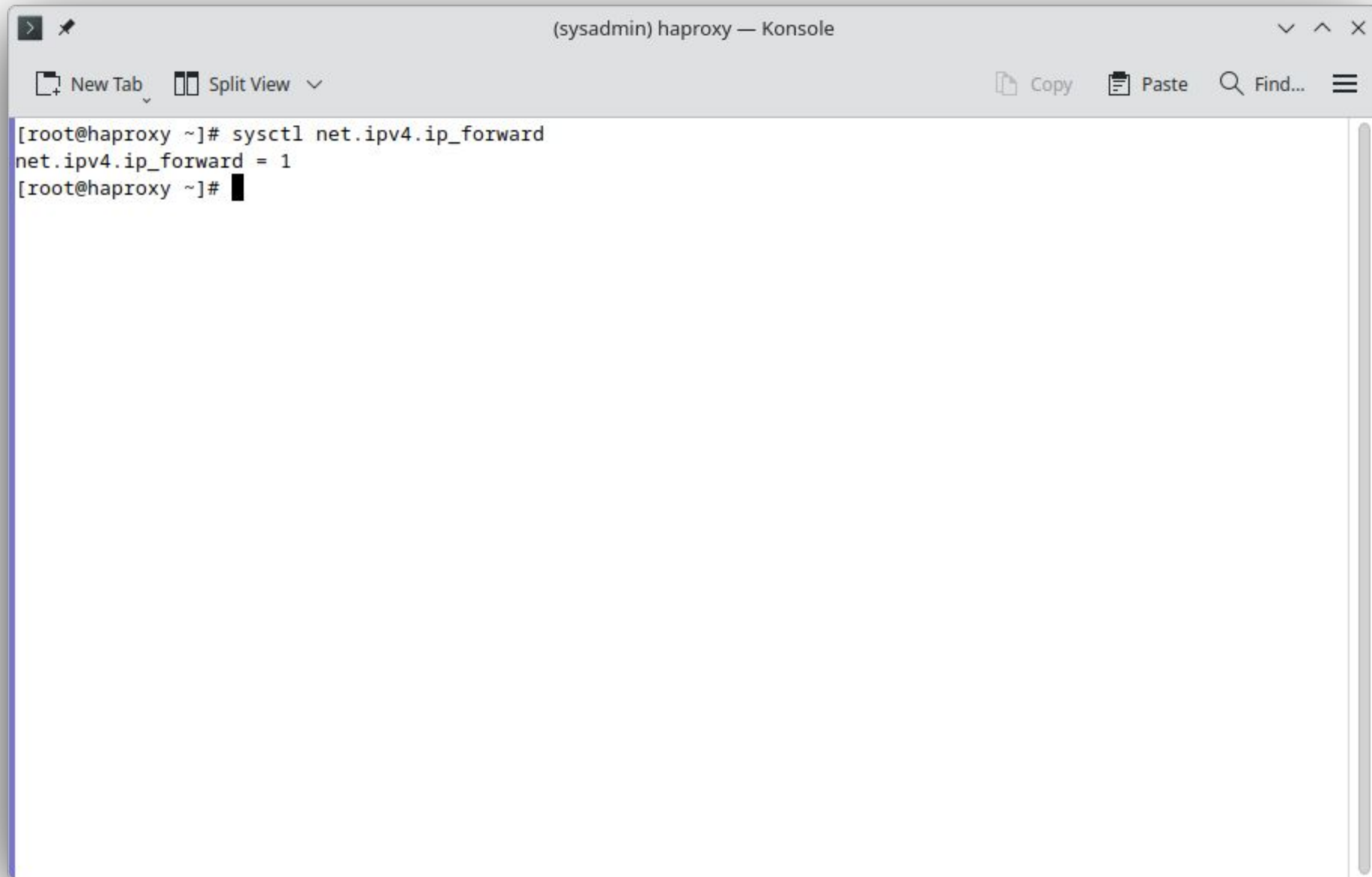
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# ip -br a sh
lo UNKNOWN 127.0.0.1/8 ::1/128
enp1s0 UP 172.31.100.64/24 fe80::5054:ff:fe1d:97f1/64
enp7s0 UP 10.31.100.64/24 fe80::83f4:6704:4d44:4bab/64
[root@haproxy ~]# nmcli connection modify enp1s0 connection.zone external
[root@haproxy ~]# nmcli connection modify enp7s0 connection.zone internal
[root@haproxy ~]#
```

A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows the output of the command "firewall-cmd --get-active-zones". The output lists three zones: "external" with interface "enp1s0", "internal" with interface "enp7s0", and "public (default)". The prompt returns to the root user at the haproxy host.

```
[root@haproxy ~]# firewall-cmd --get-active-zones
external
  interfaces: enp1s0
internal
  interfaces: enp7s0
public (default)
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --zone=internal --list-all
internal (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp7s0
  sources:
  services: cockpit dhcpv6-client mdns samba-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --zone=external --list-all
external (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp1s0
  sources:
  services: ssh
  ports:
  protocols:
  forward: yes
  masquerade: yes
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal interface includes a top bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The command history shows the execution of "sysctl net.ipv4.ip_forward" resulting in "net.ipv4.ip_forward = 1".

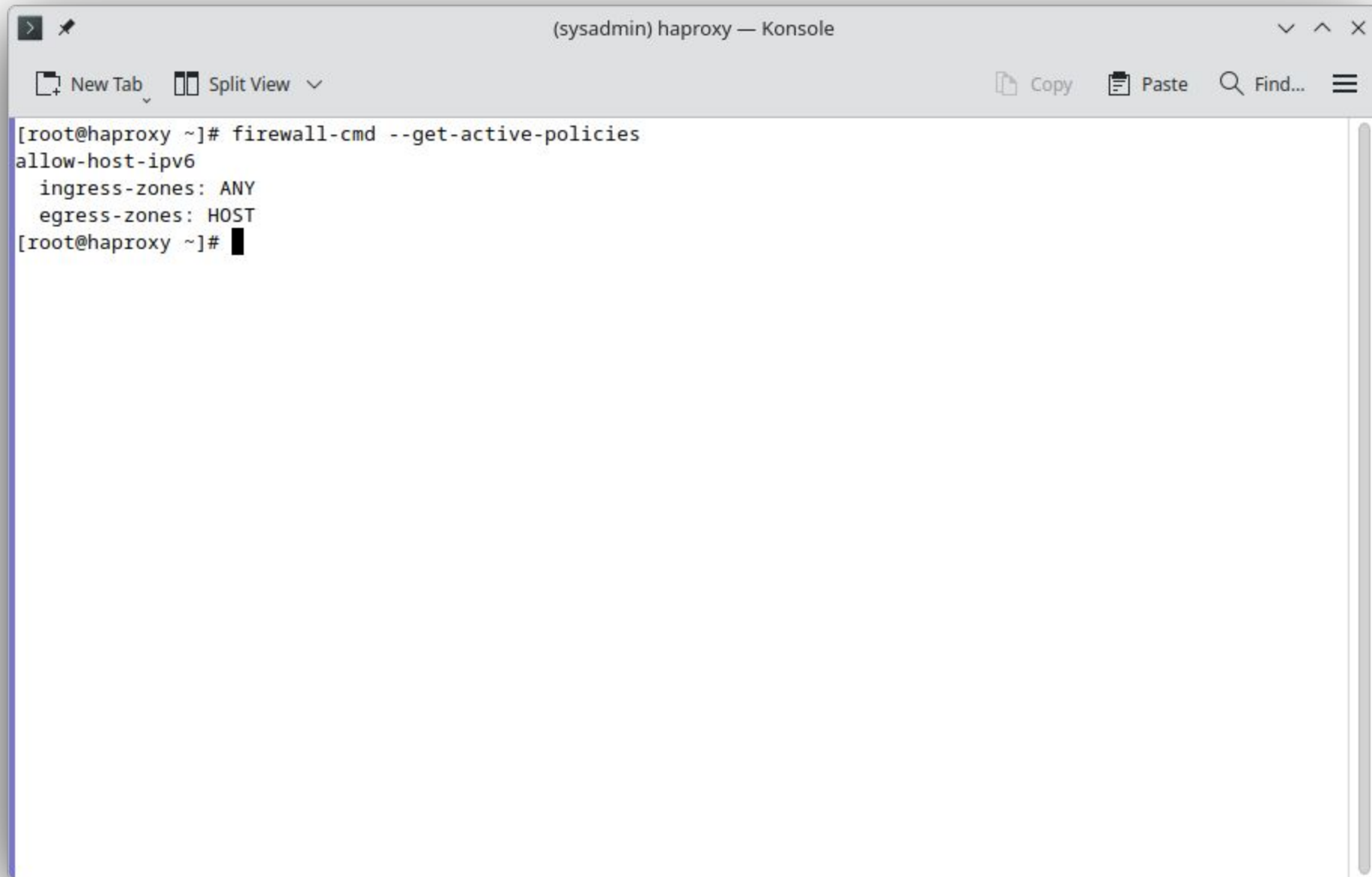
```
[root@haproxy ~]# sysctl net.ipv4.ip_forward
net.ipv4.ip_forward = 1
[root@haproxy ~]#
```

The HAProxy Machine

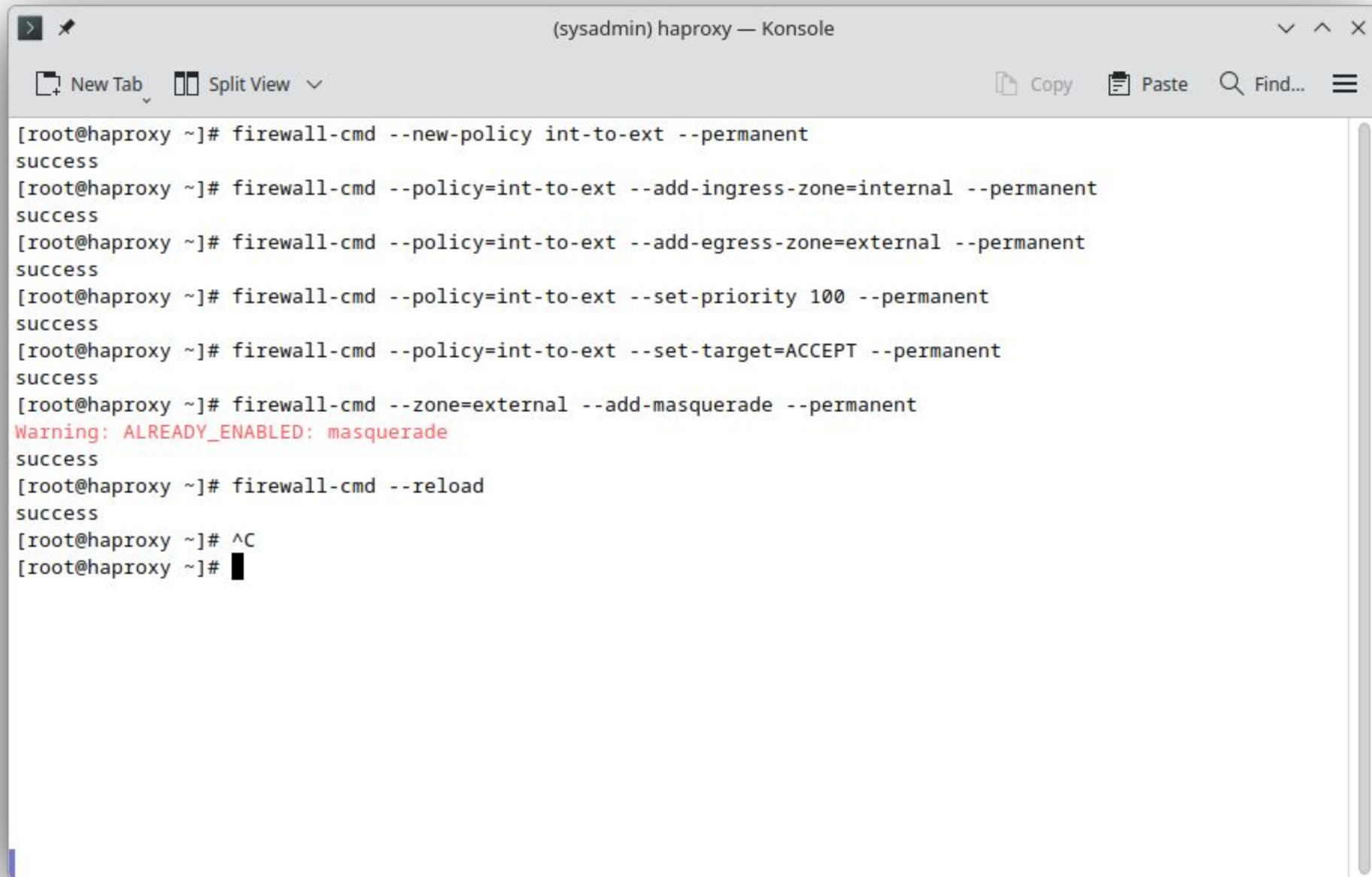
Connect the VM to both networks

We're going to set up a policy to allow all the traffic from the internal network get out via a masquerade rule:

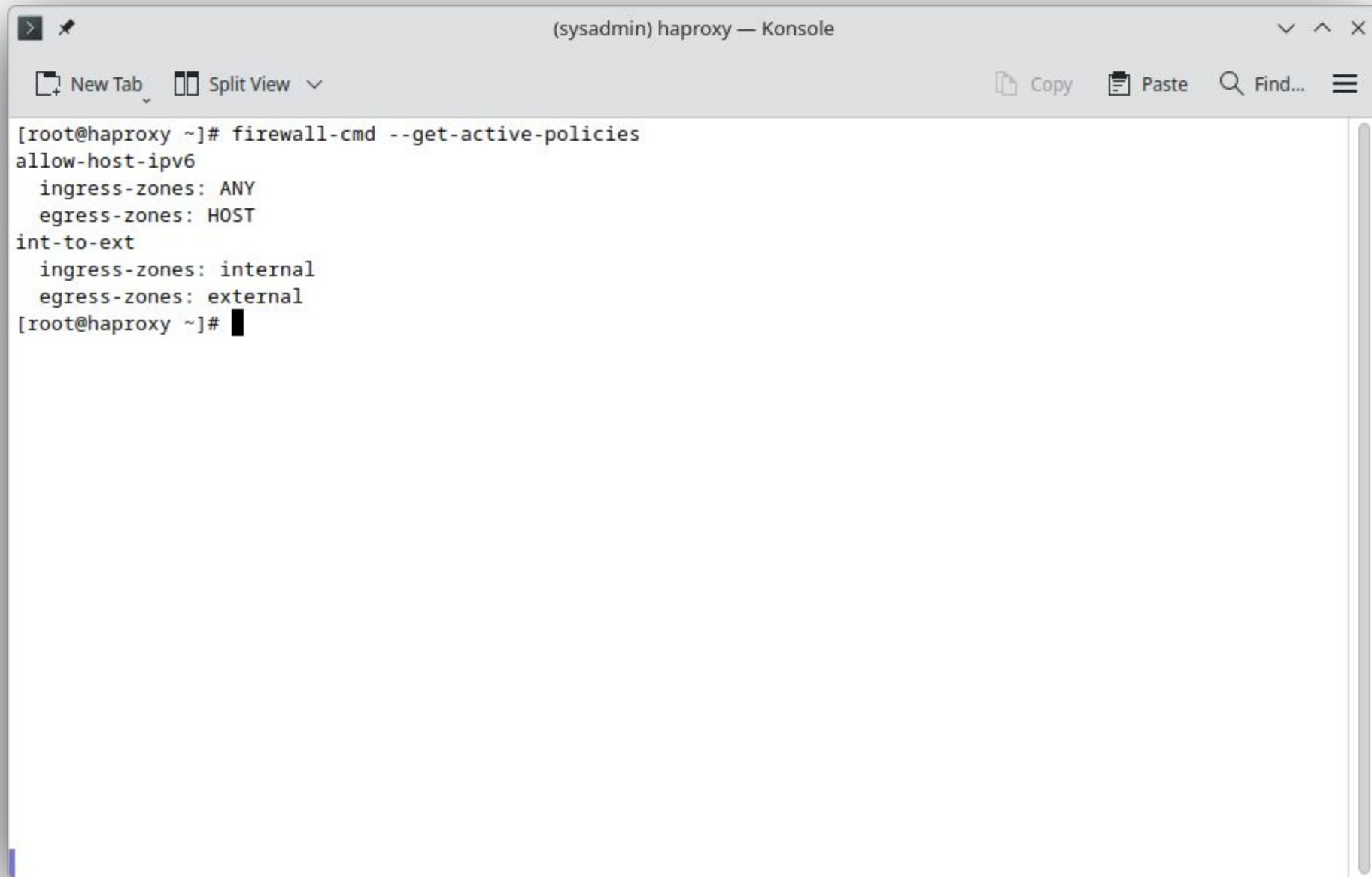
```
firewall-cmd --new-policy internal-to-external --permanent  
  
firewall-cmd --policy=int-to-ext --add-ingress-zone=internal --permanent  
  
firewall-cmd --policy=int-to-ext --add-egress-zone=external --permanent  
  
firewall-cmd --policy=int-to-ext --add-egress-zone=external --permanent  
  
firewall-cmd --policy=int-to-ext --set-target=ACCEPT --permanent  
  
firewall-cmd --zone=external --add-masquerade --permanent  
  
firewall-cmd --reload
```

A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows the command "firewall-cmd --get-active-policies" being executed, which returns "allow-host-ipv6" followed by "ingress-zones: ANY" and "egress-zones: HOST". The prompt returns to "[root@haproxy ~]#". The terminal interface includes a top bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --get-active-policies
allow-host-ipv6
  ingress-zones: ANY
  egress-zones: HOST
[root@haproxy ~]#
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --new-policy int-to-ext --permanent
success
[root@haproxy ~]# firewall-cmd --policy=int-to-ext --add-ingress-zone=internal --permanent
success
[root@haproxy ~]# firewall-cmd --policy=int-to-ext --add-egress-zone=external --permanent
success
[root@haproxy ~]# firewall-cmd --policy=int-to-ext --set-priority 100 --permanent
success
[root@haproxy ~]# firewall-cmd --policy=int-to-ext --set-target=ACCEPT --permanent
success
[root@haproxy ~]# firewall-cmd --zone=external --add-masquerade --permanent
Warning: ALREADY_ENABLED: masquerade
success
[root@haproxy ~]# firewall-cmd --reload
success
[root@haproxy ~]# ^C
[root@haproxy ~]#
```

A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows the output of the command "firewall-cmd --get-active-policies". The output lists two active policies: "allow-host-ipv6" and "int-to-ext", each with its ingress and egress zones. The terminal interface includes a top bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. A vertical scrollbar is on the right side of the terminal area.

```
[root@haproxy ~]# firewall-cmd --get-active-policies
allow-host-ipv6
  ingress-zones: ANY
  egress-zones: HOST
int-to-ext
  ingress-zones: internal
  egress-zones: external
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --get-active-policies
allow-host-ipv6
  ingress-zones: ANY
  egress-zones: HOST
int-to-ext
  ingress-zones: internal
  egress-zones: external
[root@haproxy ~]# firewall-cmd --info-policy=int-to-ext
int-to-ext (active)
  disable: no
  priority: 100
  target: ACCEPT
  ingress-zones: internal
  egress-zones: external
  services:
  ports:
  protocols:
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```

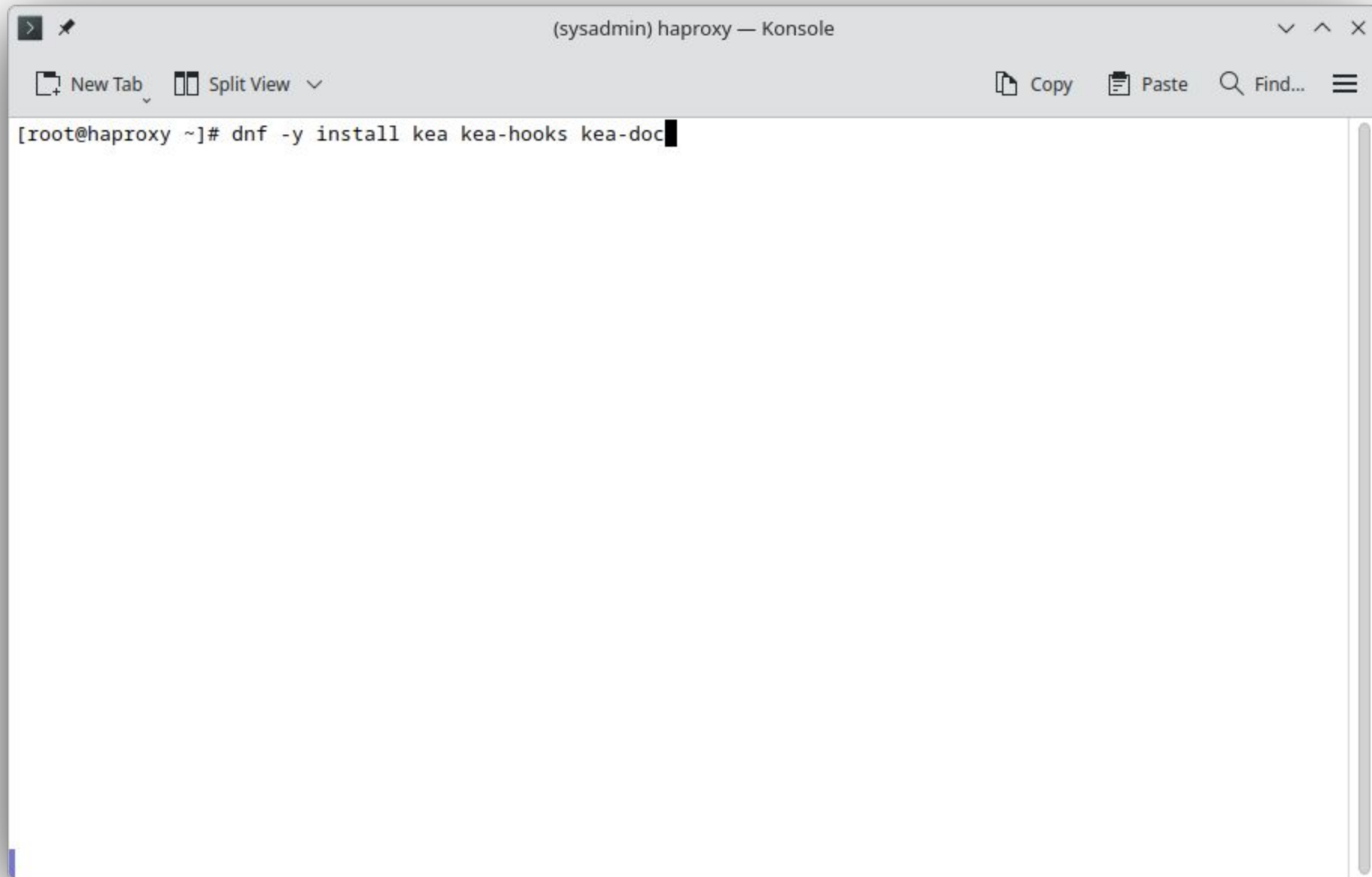
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --zone=external --list-all
external (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp1s0
  sources:
  services: ssh
  ports:
  protocols:
  forward: yes
  masquerade: yes
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --zone=internal --list-all
internal (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp7s0
  sources:
  services: cockpit dhcpv6-client mdns samba-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```

The HAProxy Machine

Set up DHCP for the private network

I install kea, kea-hooks, and kea-doc



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal content shows a root user at the haproxy machine typing a dnf command to install kea, kea-hooks, and kea-doc.

```
[root@haproxy ~]# dnf -y install kea kea-hooks kea-doc
```

The HAProxy Machine

Set up DHCP for the private network

A few updates to the `/etc/kea/kea-dhcp4.conf` file

Set the DNS server to be the private interface of the proxy machine, since it's serving DHCP and DNS to the internal network

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
{
  "Dhcp4": {
    "interfaces-config": {
      "interfaces": [
        "enp7s0"
      ]
    },
    "control-socket": {
      "socket-type": "unix",
      "socket-name": "kea4-ctrl-socket"
    },
    "lease-database": {
      "type": "memfile",
      "lfc-interval": 3600
    },
    "expired-leases-processing": {
      "reclaim-timer-wait-time": 10,
      "flush-reclaimed-timer-wait-time": 25,
      "hold-reclaimed-time": 3600,
      "max-reclaim-leases": 100,
      "max-reclaim-time": 250,
      "unwarned-reclaim-cycles": 5
    },
    -- INSERT --
  }
}
```

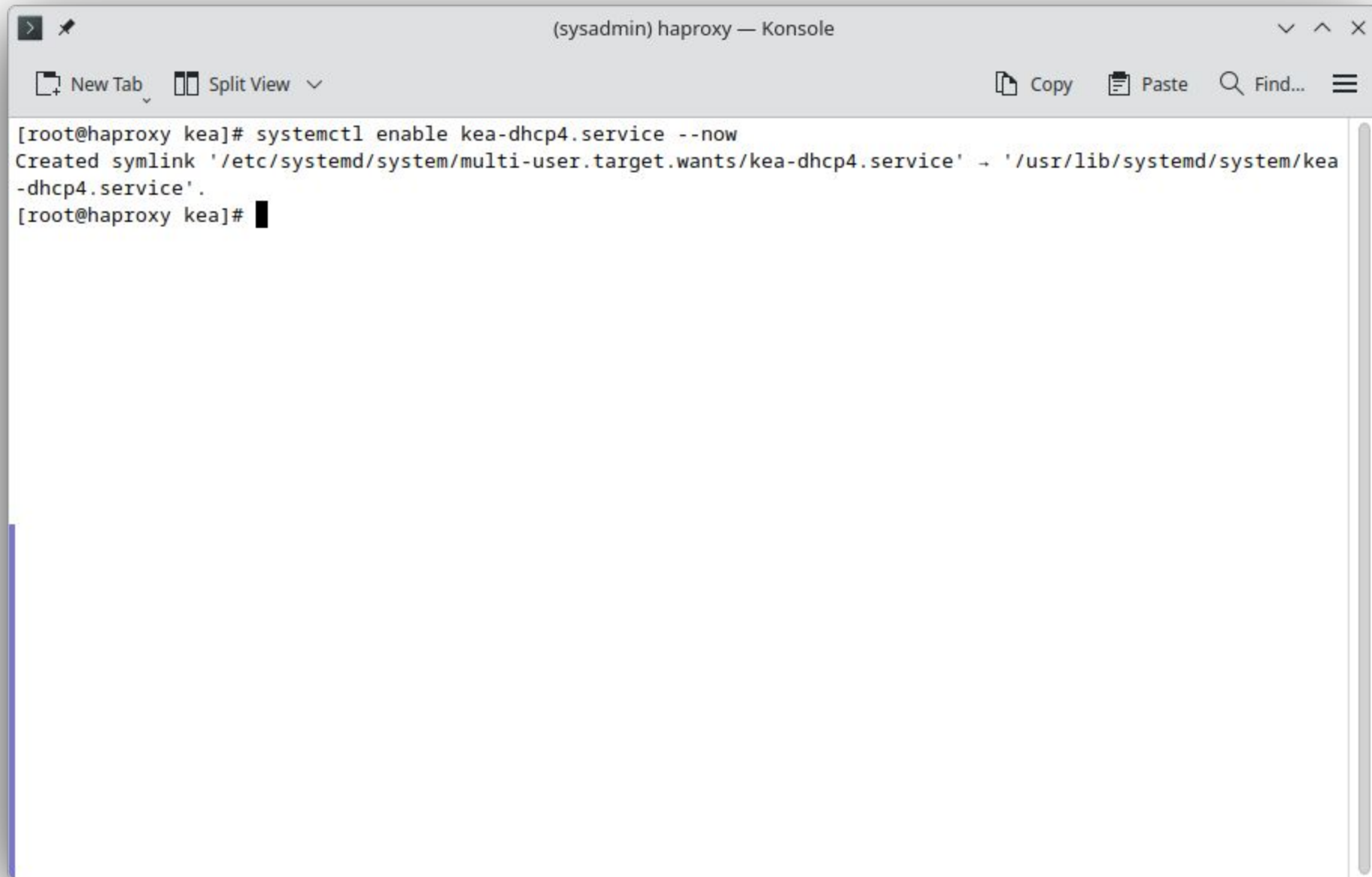


```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
},
"renew-timer": 900,
"rebind-timer": 1800,
"valid-lifetime": 3600,
"option-data": [
  {
    "name": "domain-name-servers",
    "data": "10.31.100.65"
  },
  {
    "code": 15,
    "data": "summit.aus.redhat.com"
  },
  {
    "name": "domain-search",
    "data": "aus.redhat.com, summit.aus.redhat.com"
  },
  {
    "name": "boot-file-name",
    "data": "EST5EDT4\\,M3.2.0/02:00\\,M11.1.0/02:00"
  },
],
43,1 19%
```

```
(sysadmin) haproxy — Konsole
New Tab Split View
Copy Paste Find...

    "server-hostname": "hal9000",
    "boot-file-name": "/dev/null"
  },
  "subnet4": [
    {
      "id": 1,
      "subnet": "10.31.100.0/24",
      "pools": [ { "pool": "10.31.100.128 - 10.31.100.254" } ],
      "option-data": [
        {
          "name": "routers",
          "data": "10.31.100.64"
        }
      ],
      "reservations": [
        {
          "hw-address": "1a:1b:1c:1d:1e:1f",
          "ip-address": "192.0.2.201"
        }
      ]
    }
  ]
}
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy kea]# kea-dhcp4 -t /etc/kea/kea-dhcp4.conf
2026-05-10 09:40:15.781 WARN [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_MT_DISABLED_QUEUE_CONTROL disabling dhcp queue control when multi-threading is enabled.
2026-05-10 09:40:15.781 WARN [kea-dhcp4.dhcp4/2665.140167936829568] DHCP4_RESERVATIONS_LOOKUP_FIRST_ENABLED Multi-threading is enabled and host reservations lookup is always performed first.
2026-05-10 09:40:15.782 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_CFGMGR_NEW_SUBNET4 a new subnet has been added to configuration: 10.31.100.0/24 with params: t1=900, t2=1800, valid-lifetime=3600
2026-05-10 09:40:15.782 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_CFGMGR_SOCKET_TYPE_SELECT using socket type raw
2026-05-10 09:40:15.782 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_CFGMGR_ADD_IFACE listening on interface enp7s0
2026-05-10 09:40:15.783 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_CFGMGR_SOCKET_TYPE_DEFAULT "dhcp-socket-type" not specified, using default socket type raw
2026-05-10 09:40:15.783 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_LEASE_MGR_BACKENDS_REGISTERED the following lease backend types are available: memfile
2026-05-10 09:40:15.783 INFO [kea-dhcp4.hosts/2665.140167936829568] HOSTS_BACKENDS_REGISTERED the following host backend types are available:
2026-05-10 09:40:15.783 INFO [kea-dhcp4.dhcpsrv/2665.140167936829568] DHCP4_SRV_FORENSIC_BACKENDS_REGISTERED the following forensic backend types are available:
2026-05-10 09:40:15.783 INFO [kea-dhcp4.database/2665.140167936829568] CONFIG_BACKENDS_REGISTERED the following config backend types are available:
[root@haproxy kea]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows a root prompt at "haproxy" where the command "systemctl enable kea-dhcp4.service --now" has been executed. The output indicates that a symlink was created to enable the service. The prompt is now ready for the next command.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy kea]# systemctl enable kea-dhcp4.service --now
Created symlink '/etc/systemd/system/multi-user.target.wants/kea-dhcp4.service' -> '/usr/lib/systemd/system/kea-dhcp4.service'.
[root@haproxy kea]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy kea]# systemctl status kea-dhcp4.service
• kea-dhcp4.service - Kea DHCPv4 Server
  Loaded: loaded (/usr/lib/systemd/system/kea-dhcp4.service; enabled; preset: disabled)
  Active: active (running) since Sun 2026-05-10 09:40:48 UTC; 36s ago
  Invocation: 3d176ee228ed4275bf1edc13d610cab4
  Docs: man:kea-dhcp4(8)
  Main PID: 2732 (kea-dhcp4)
  Status: "Dispatching packets..."
  Tasks: 9 (limit: 23120)
  Memory: 2.5M (peak: 6.2M)
  CPU: 67ms
  CGroup: /system.slice/kea-dhcp4.service
          └─2732 /usr/sbin/kea-dhcp4 -c /etc/kea/kea-dhcp4.conf

May 10 09:40:48 haproxy.aus.redhat.com systemd[1]: Starting kea-dhcp4.service - Kea DHCPv4 Server...
May 10 09:40:48 haproxy.aus.redhat.com kea-dhcp4[2732]: 2026-05-10 09:40:48.387 INFO [kea-dhcp4.dhcp4/2732.1>
May 10 09:40:48 haproxy.aus.redhat.com kea-dhcp4[2732]: 2026-05-10 09:40:48.388 INFO [kea-dhcp4.commands/273>
May 10 09:40:48 haproxy.aus.redhat.com systemd[1]: Started kea-dhcp4.service - Kea DHCPv4 Server.
lines 1-17/17 (END)
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy kea]# firewall-cmd --zone=internal --add-service=dhcp --permanent
success
[root@haproxy kea]# firewall-cmd --reload
success
[root@haproxy kea]# firewall-cmd --zone=internal --list-all
internal (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp7s0
  sources:
  services: cockpit dhcp dhcpv6-client mdns samba-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy kea]#
```

DNS

DNS

Configure DNS for the private network

External DNS is already set up, but the machines on the private network need to be able to resolve themselves and each other.

Remember that OpenShift really needs wildcard entries. If not, you have to create an `/etc/hosts` file with a ridiculous amount of names in it. Don't do it.

DNS

Configure DNS for the private network

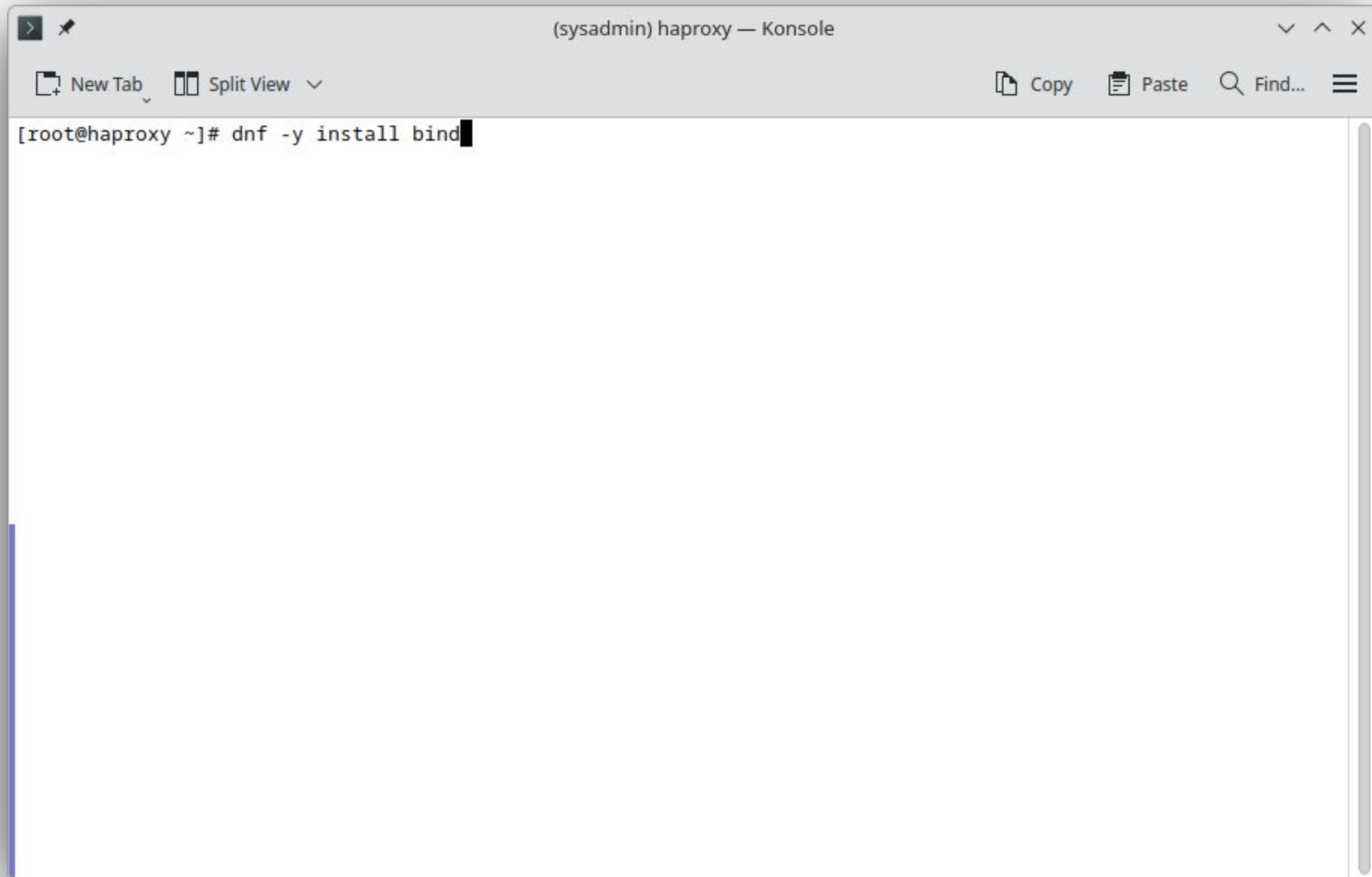
Install bind

Enable and start the service

Configure the configs and zone files

Open up the dns service in the firewall

Test resolution



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal content shows the command `[root@haproxy ~]# dnf -y install bind` with a cursor at the end of the line.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# dnf -y install bind
```

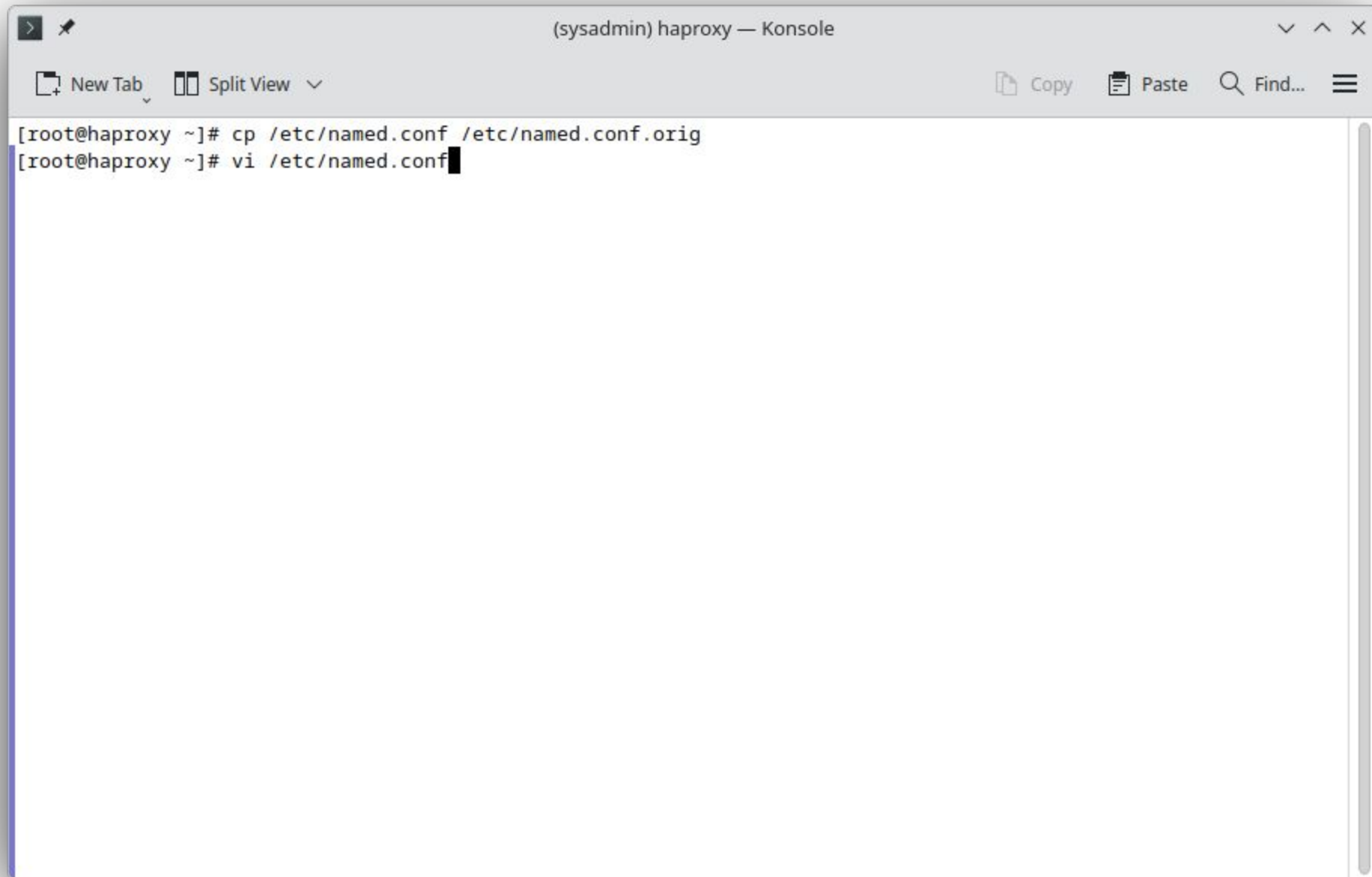
```
(sysadmin) haproxy — Konsole

New Tab Split View Copy Paste Find...

Total download size: 488 k
Installed size: 1.3 M
Downloading Packages:
(1/2): bind-dnssec-utils-9.18.33-10.el10_1.2.x86_64.rpm 2.0 MB/s | 151 kB 00:00
(2/2): bind-9.18.33-10.el10_1.2.x86_64.rpm 4.1 MB/s | 337 kB 00:00
-----
Total 5.8 MB/s | 488 kB 00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 1/1
  Installing     : bind-dnssec-utils-32:9.18.33-10.el10_1.2.x86_64 1/2
  Running scriptlet: bind-32:9.18.33-10.el10_1.2.x86_64 2/2
  Installing     : bind-32:9.18.33-10.el10_1.2.x86_64 2/2
  Running scriptlet: bind-32:9.18.33-10.el10_1.2.x86_64 2/2
Installed products updated.

Installed:
  bind-32:9.18.33-10.el10_1.2.x86_64 bind-dnssec-utils-32:9.18.33-10.el10_1.2.x86_64

Complete!
[root@haproxy ~]# systemctl enable named --now
Created symlink '/etc/systemd/system/multi-user.target.wants/named.service' -> '/usr/lib/systemd/system/named.service'.
[root@haproxy ~]#
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# cp /etc/named.conf /etc/named.conf.orig
[root@haproxy ~]# vi /etc/named.conf
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
//
// named.conf
//
// Provided by Red Hat bind package to configure the ISC BIND named(8) DNS
// server as a caching only nameserver (as a localhost DNS resolver only).
//
// See /usr/share/doc/bind*/sample/ for example named configuration files.
//
options {
    listen-on port 53 { 10.31.100.64; };
    listen-on-v6 port 53 { ::1; };
    directory "/var/named";
    dump-file "/var/named/data/cache_dump.db";
    statistics-file "/var/named/data/named_stats.txt";
    memstatistics-file "/var/named/data/named_mem_stats.txt";
    secroots-file "/var/named/data/named.secroots";
    recursing-file "/var/named/data/named.recursing";
    allow-query { any; };

    /*
    - If you are building an AUTHORITATIVE DNS server, do NOT enable recursion.
    - If you are building a RECURSIVE (caching) DNS server, you need to enable
      recursion.
    - If your recursive DNS server has a public IP address, you MUST enable access
      control to limit queries to your legitimate users. Failing to do so will
      cause your server to become part of large scale DNS amplification
    */
}
```

20,0-1

Top

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
dnssec-validation yes;

managed-keys-directory "/var/named/dynamic";
geoip-directory "/usr/share/GeoIP";

pid-file "/run/named/named.pid";
session-keyfile "/run/named/session.key";

/* https://fedoraproject.org/wiki/Changes/CryptoPolicy */
include "/etc/crypto-policies/back-ends/bind.config";
};

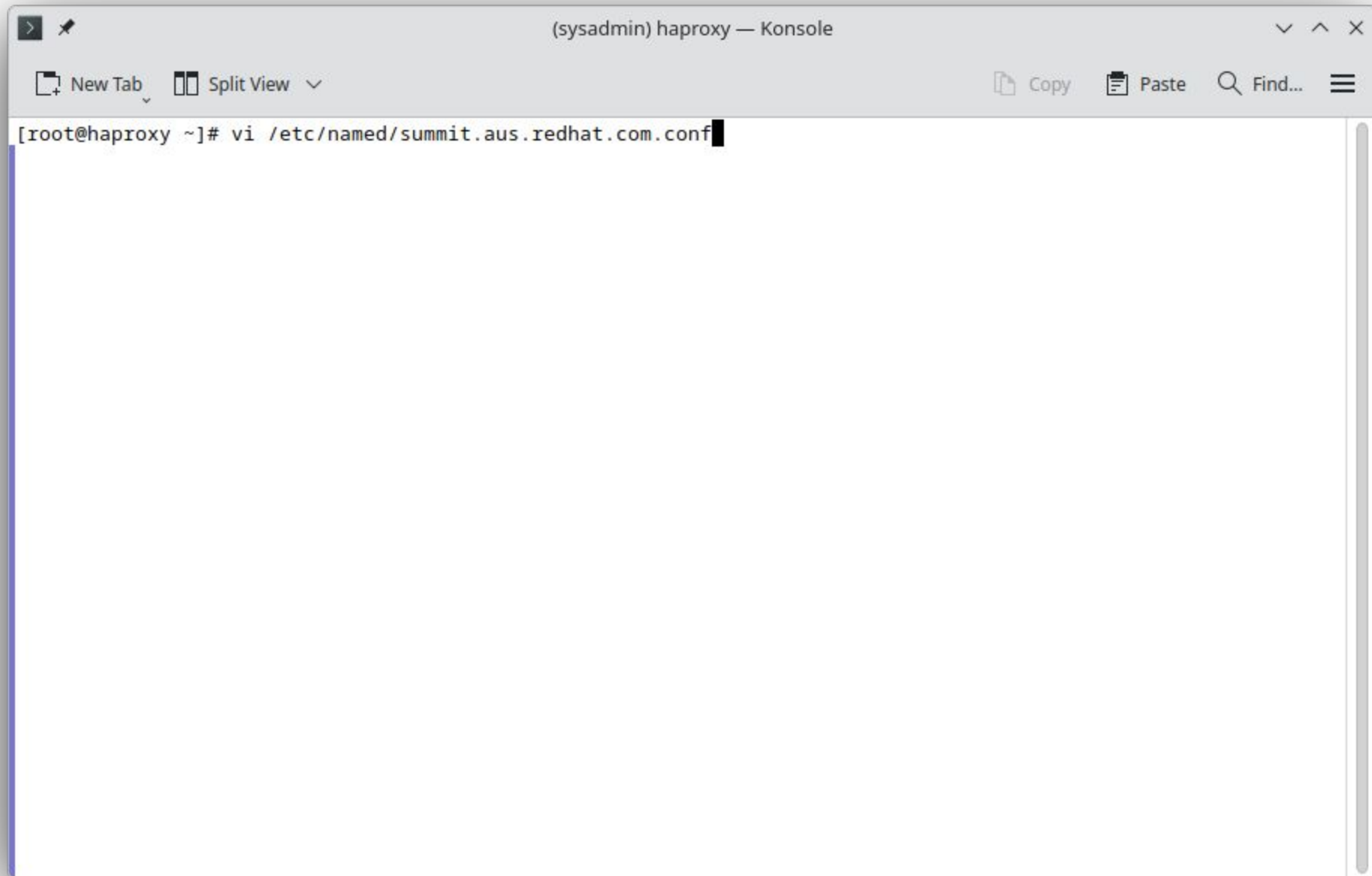
logging {
    channel default_debug {
        file "data/named.run";
        severity dynamic;
    };
};

zone "." IN {
    type hint;
    file "named.ca";
};

include "/etc/named.rfc1912.zones";
include "/etc/named.root.key";
include "/etc/named/*.conf";

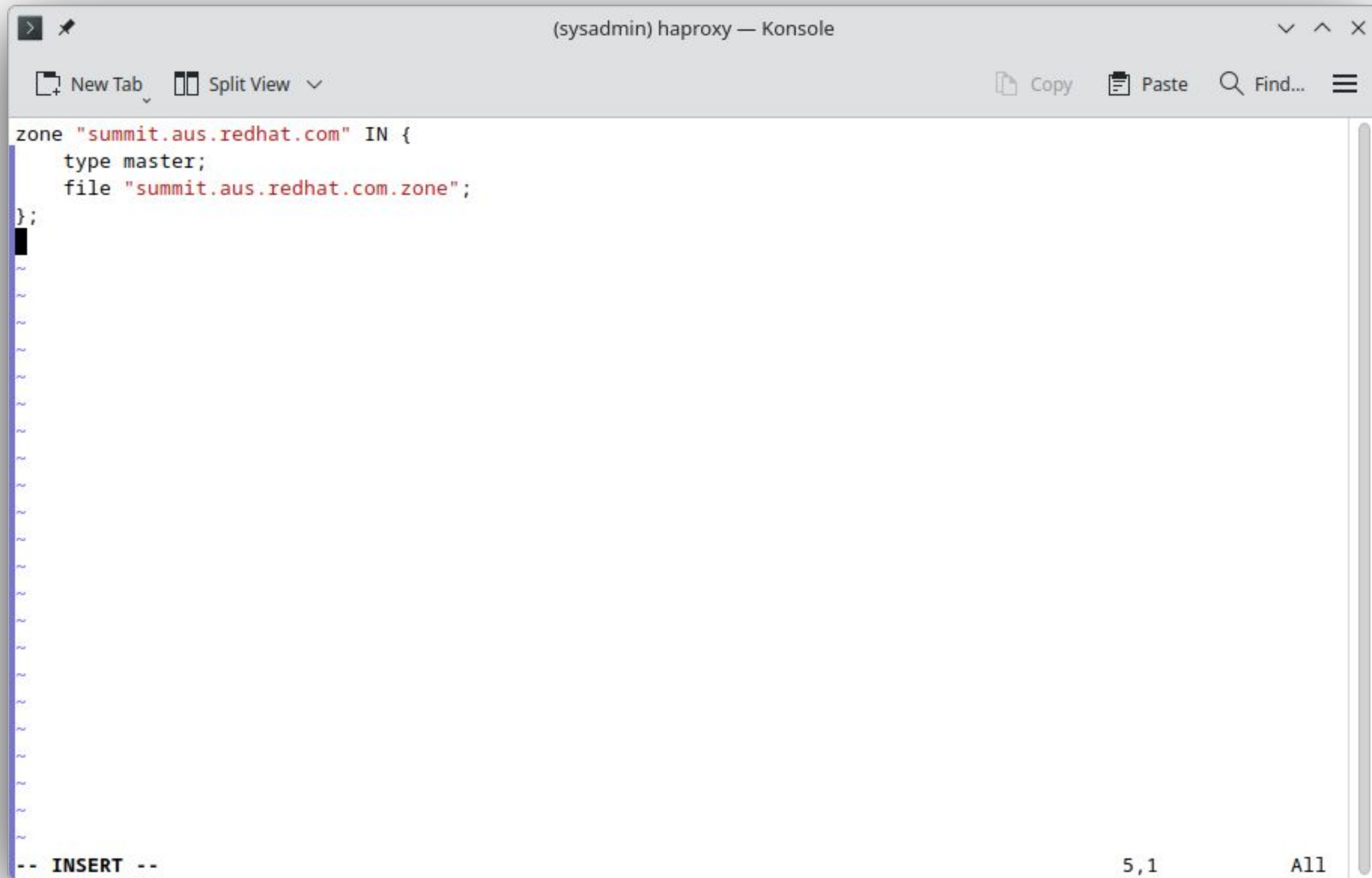
59,1 Bot
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# diff /etc/named.conf /etc/named.conf.orig
11c11
<     listen-on port 53 { 10.31.100.64; };
---
>     listen-on port 53 { 127.0.0.1; };
19c19
<     allow-query    { any; };
---
>     allow-query    { localhost; };
59c59
< include "/etc/named/*.conf";
---
>
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" is shown. The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal prompt is "[root@haproxy ~]#". The command "vi /etc/named/summit.aus.redhat.com.conf" has been entered, and the cursor is at the end of the command line.

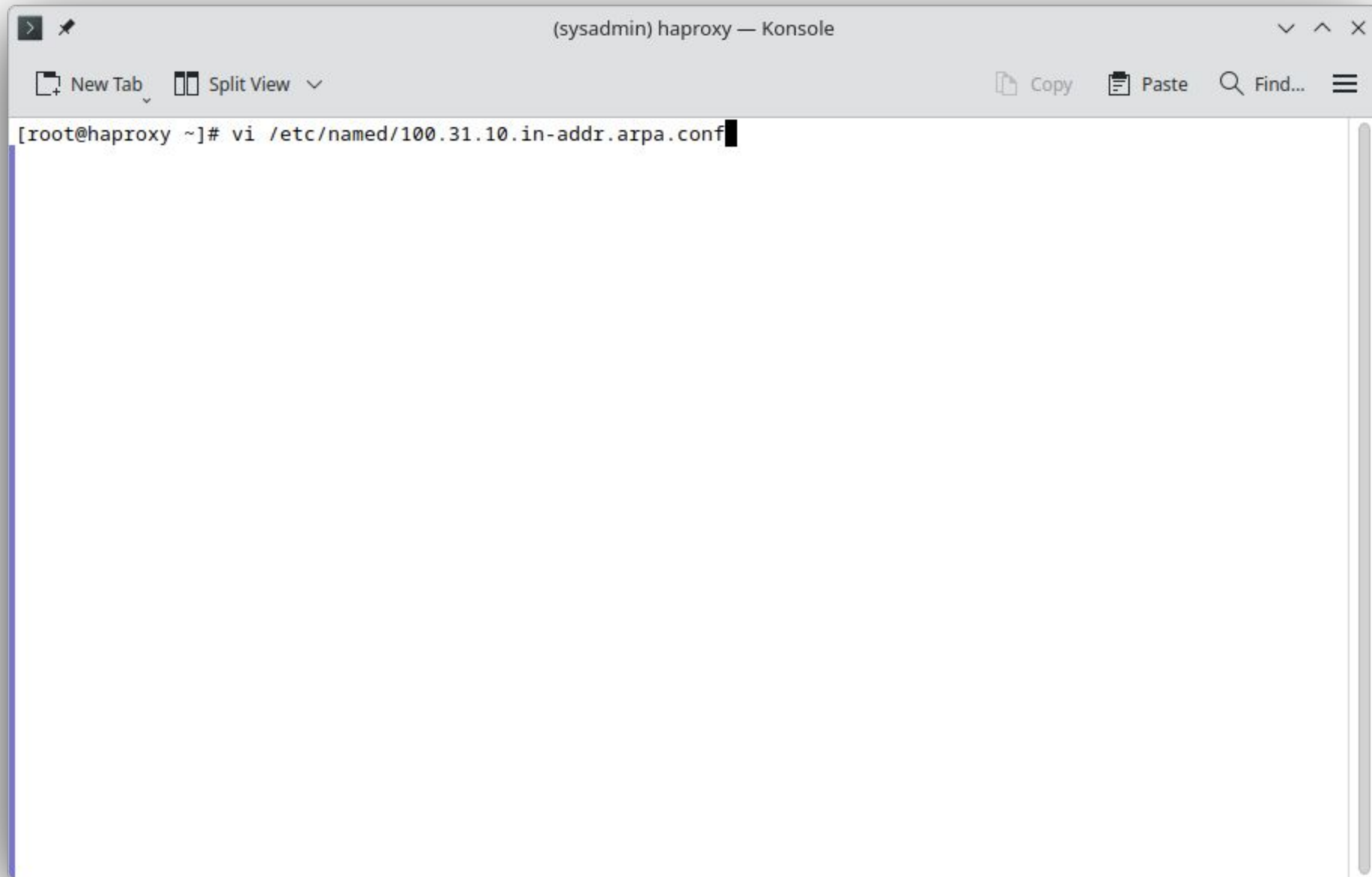
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /etc/named/summit.aus.redhat.com.conf
```

The screenshot shows a terminal window with a title bar that reads "(sysadmin) haproxy — Konsole". The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The main content area displays a configuration snippet for a zone named "summit.aus.redhat.com". The snippet is as follows:

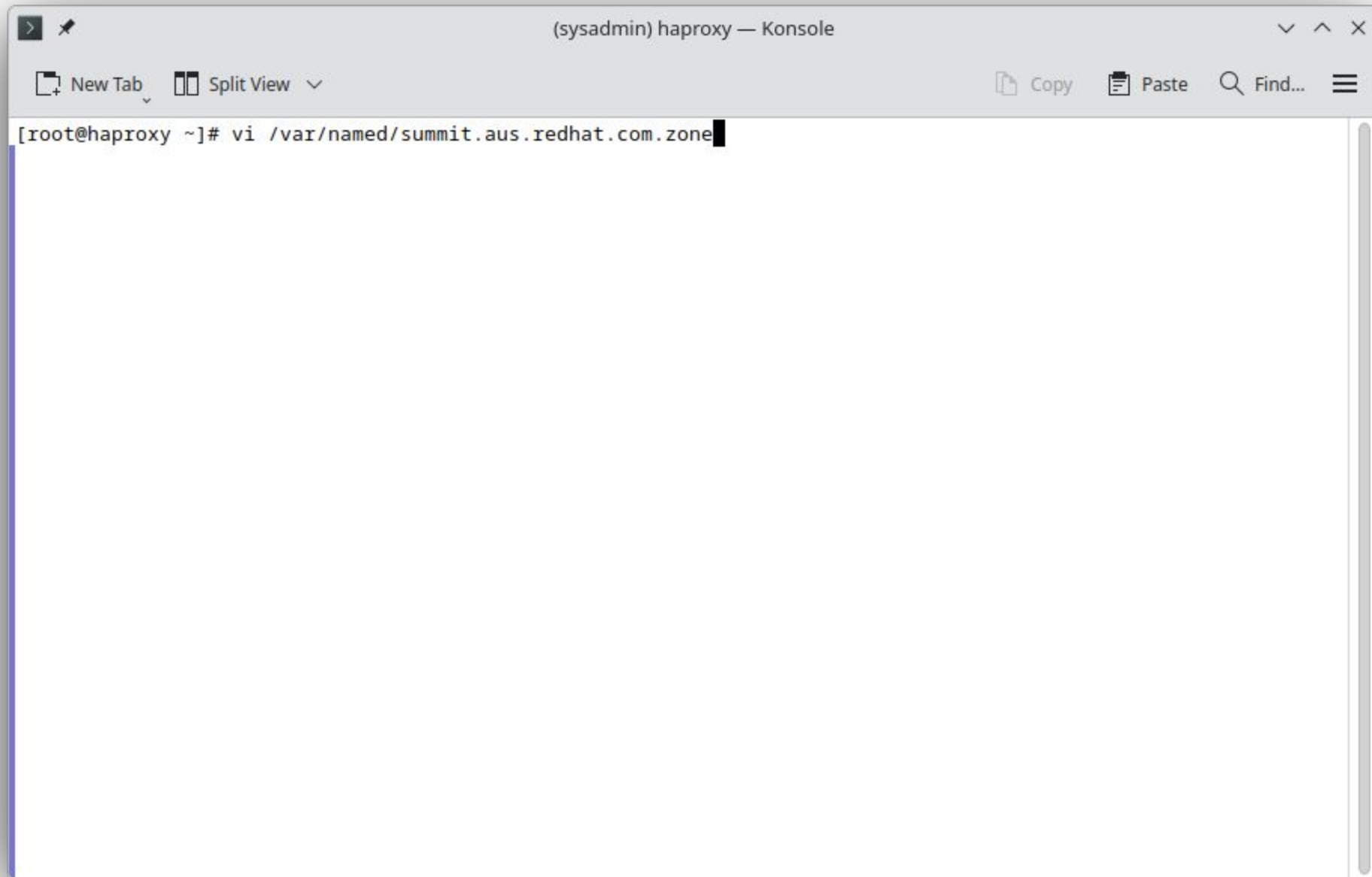
```
zone "summit.aus.redhat.com" IN {  
    type master;  
    file "summit.aus.redhat.com.zone";  
};
```

The cursor is positioned at the end of the first line of the zone definition. The status bar at the bottom of the window shows "-- INSERT --" on the left, "5,1" in the center, and "All" on the right.



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The interface includes a toolbar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The command prompt shows the user is root at haproxy, and the command being executed is `vi /etc/named/100.31.10.in-addr.arpa.conf`. The terminal area is mostly empty, with a vertical scrollbar on the right.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /etc/named/100.31.10.in-addr.arpa.conf
```

A terminal window titled "(sysadmin) haproxy — Konsole" is shown. The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal prompt is "[root@haproxy ~]#". The command entered is "vi /var/named/summit.aus.redhat.com.zone", and the cursor is at the end of the command.

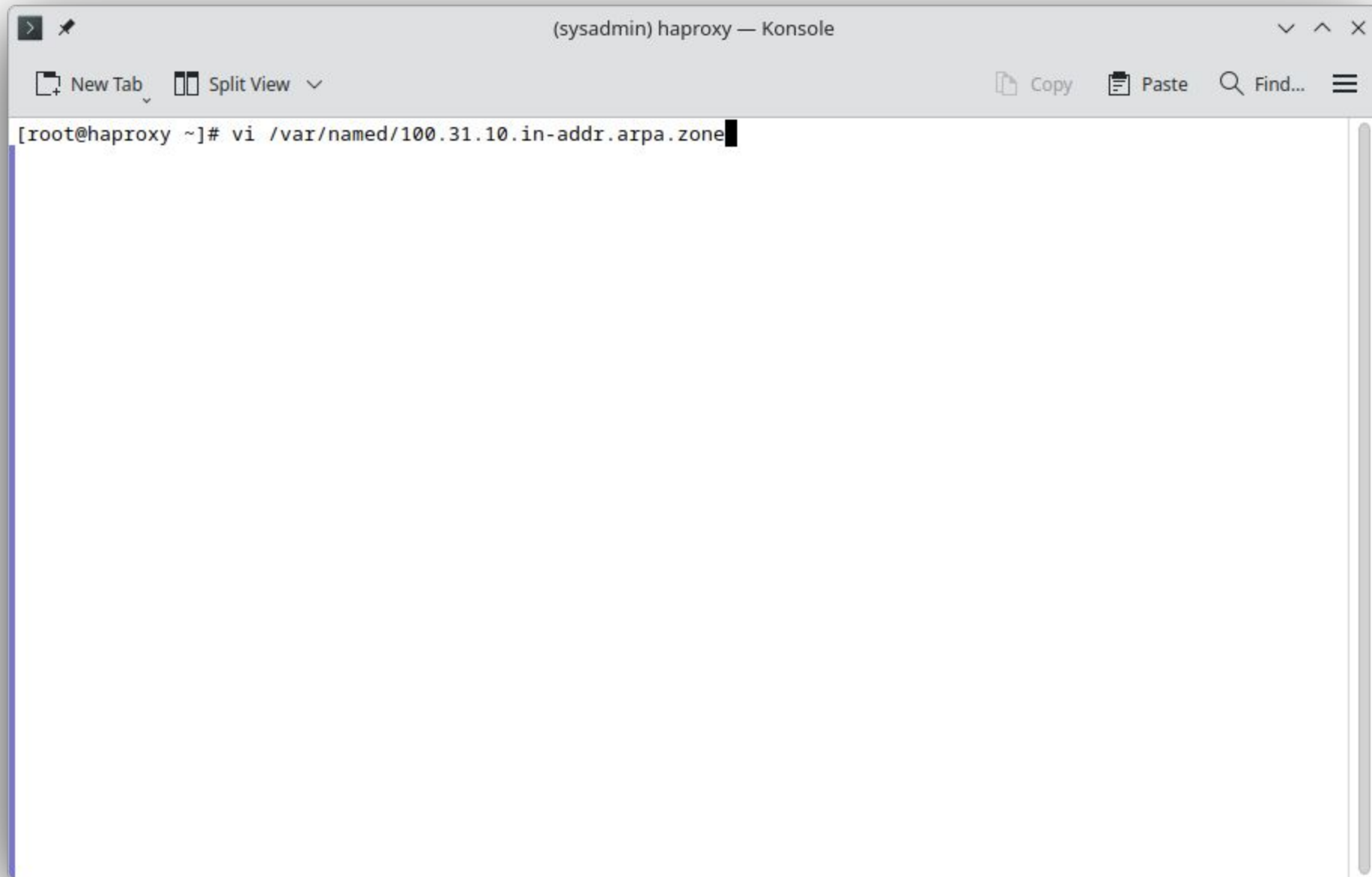
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /var/named/summit.aus.redhat.com.zone
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
$TTL 86400
@ IN SOA haproxy.aus.redhat.com. root.aus.redhat.com. (
    2026051001 ; Serial
    3600       ; Refresh
    1800       ; Retry
    604800     ; Expire
    86400      ; Minimum
)

@ IN NS haproxy.aus.redhat.com.

bootstrap IN A 10.31.100.23
haproxy   IN A 10.31.100.64
ocp1      IN A 10.31.100.65
ocp2      IN A 10.31.100.66
ocp3      IN A 10.31.100.67
ocp4      IN A 10.31.100.68
ocp5      IN A 10.31.100.69
ocp6      IN A 10.31.100.144
api       IN A 10.31.100.64
api-int   IN A 10.31.100.64
*.apps    IN A 10.31.100.64

"/var/named/summit.aus.redhat.com.zone" 22L, 530B 22,0-1 All
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal interface includes a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The command prompt shows a root user at the haproxy machine, and the command entered is `vi /var/named/100.31.10.in-addr.arpa.zone`. The terminal area is mostly empty, with a vertical scrollbar on the right side.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /var/named/100.31.10.in-addr.arpa.zone
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
$TTL 86400
@ IN SOA haproxy.aus.redhat.com. root.aus.redhat.com. (
    2026051001 ; Serial
    3600       ; Refresh
    1800       ; Retry
    604800     ; Expire
    86400 )    ; Minimum

@ IN NS nuc8.aus.redhat.com.

23 PTR bootstrap.summit.aus.redhat.com.
64 PTR haproxy.summit.aus.redhat.com.
65 PTR ocp1.summit.aus.redhat.com.
66 PTR ocp2.summit.aus.redhat.com.
67 PTR ocp3.summit.aus.redhat.com.
68 PTR ocp4.summit.aus.redhat.com.
69 PTR ocp5.summit.aus.redhat.com.
144 PTR ocp6.summit.aus.redhat.com.

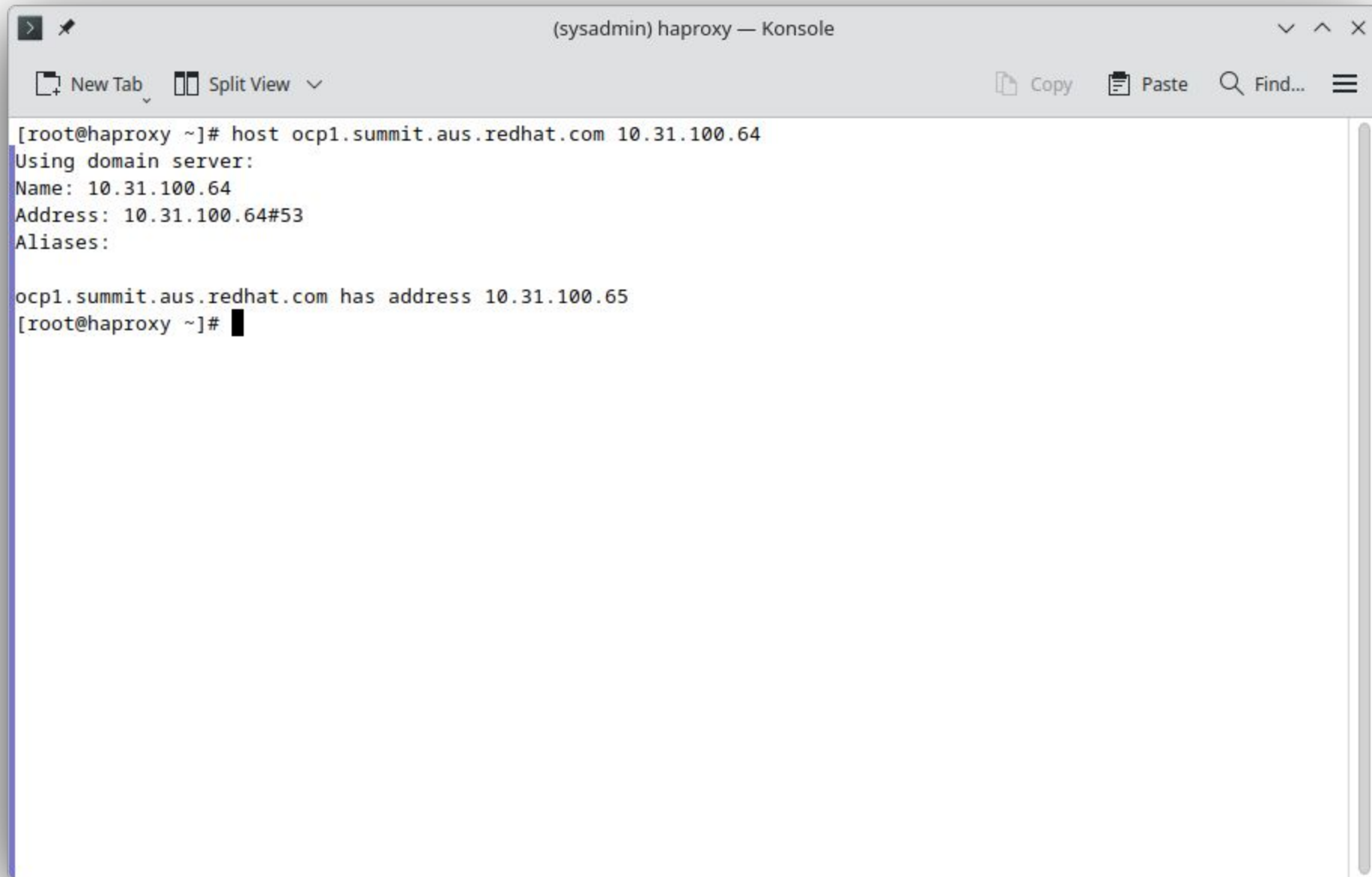
-- INSERT --
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# systemctl restart named
[root@haproxy ~]# systemctl status named
• named.service - Berkeley Internet Name Domain (DNS)
  Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: disabled)
  Active: active (running) since Sun 2026-05-10 10:05:45 UTC; 7s ago
  Invocation: 773f8e61b1194799b840c122b0c9877b
  Process: 3383 ExecStartPre=/bin/bash -c if [ ! "$DISABLE_ZONE_CHECKING" == "yes" ]; then /usr/bin/named-c>
  Process: 3386 ExecStart=/usr/sbin/named -u named -c ${NAMEDCONF} $OPTIONS (code=exited, status=0/SUCCESS)
  Main PID: 3387 (named)
  Tasks: 10 (limit: 23120)
  Memory: 7M (peak: 7.8M)
  CPU: 175ms
  CGroup: /system.slice/named.service
          └─3387 /usr/sbin/named -u named -c /etc/named.conf

May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns1.p02.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p01.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns1.p02.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p01.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p02.nsone.net/A/IN': >
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p01.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns3.p01.nsone.net/A/IN': >
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p02.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns1.p02.nsone.net/AAAA/IN>
May 10 10:05:45 haproxy.aus.redhat.com named[3387]: network unreachable resolving 'dns2.p02.nsone.net/A/IN': >
[root@haproxy ~]#
```



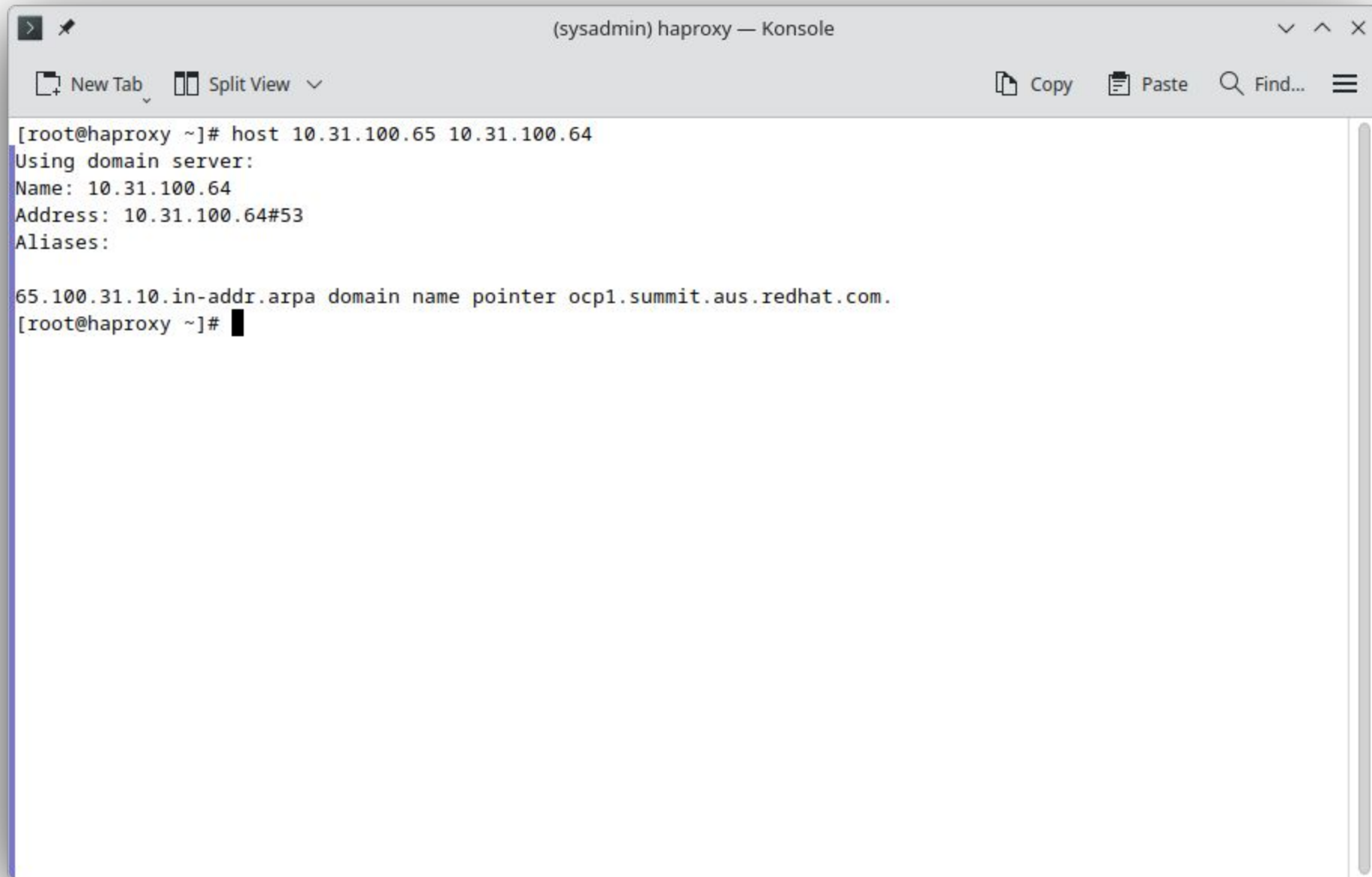
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# firewall-cmd --zone=internal --add-service=dns --permanent
success
[root@haproxy ~]# firewall-cmd --reload
success
[root@haproxy ~]# firewall-cmd --zone=internal --list-all
internal (active)
  target: default
  ingress-priority: 0
  egress-priority: 0
  icmp-block-inversion: no
  interfaces: enp7s0
  sources:
  services: cockpit dhcp dhcpv6-client dns mdns samba-client ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows a command to look up the IP address of "ocp1.summit.aus.redhat.com". The output indicates it used a domain server and found the address 10.31.100.65. The terminal interface includes a top bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. A vertical scrollbar is on the right side of the terminal area.

```
[root@haproxy ~]# host ocp1.summit.aus.redhat.com 10.31.100.64
Using domain server:
Name: 10.31.100.64
Address: 10.31.100.64#53
Aliases:

ocp1.summit.aus.redhat.com has address 10.31.100.65
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal output shows a successful DNS lookup for the IP address 10.31.100.64.

```
[root@haproxy ~]# host 10.31.100.65 10.31.100.64
Using domain server:
Name: 10.31.100.64
Address: 10.31.100.64#53
Aliases:

65.100.31.10.in-addr.arpa domain name pointer ocp1.summit.aus.redhat.com.
[root@haproxy ~]#
```

DNS

Configure DNS for the private network

Test the wildcard resolution, as well!

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# host myapp.apps.summit.aus.redhat.com 10.31.100.64
Using domain server:
Name: 10.31.100.64
Address: 10.31.100.64#53
Aliases:

myapp.apps.summit.aus.redhat.com has address 10.31.100.64
[root@haproxy ~]# host otherapp.apps.summit.aus.redhat.com 10.31.100.64
Using domain server:
Name: 10.31.100.64
Address: 10.31.100.64#53
Aliases:

otherapp.apps.summit.aus.redhat.com has address 10.31.100.64
[root@haproxy ~]# host whatever.apps.summit.aus.redhat.com 10.31.100.64
Using domain server:
Name: 10.31.100.64
Address: 10.31.100.64#53
Aliases:

whatever.apps.summit.aus.redhat.com has address 10.31.100.64
[root@haproxy ~]#
```

HAProxy



Red Hat

HAProxy Service

Install and configure

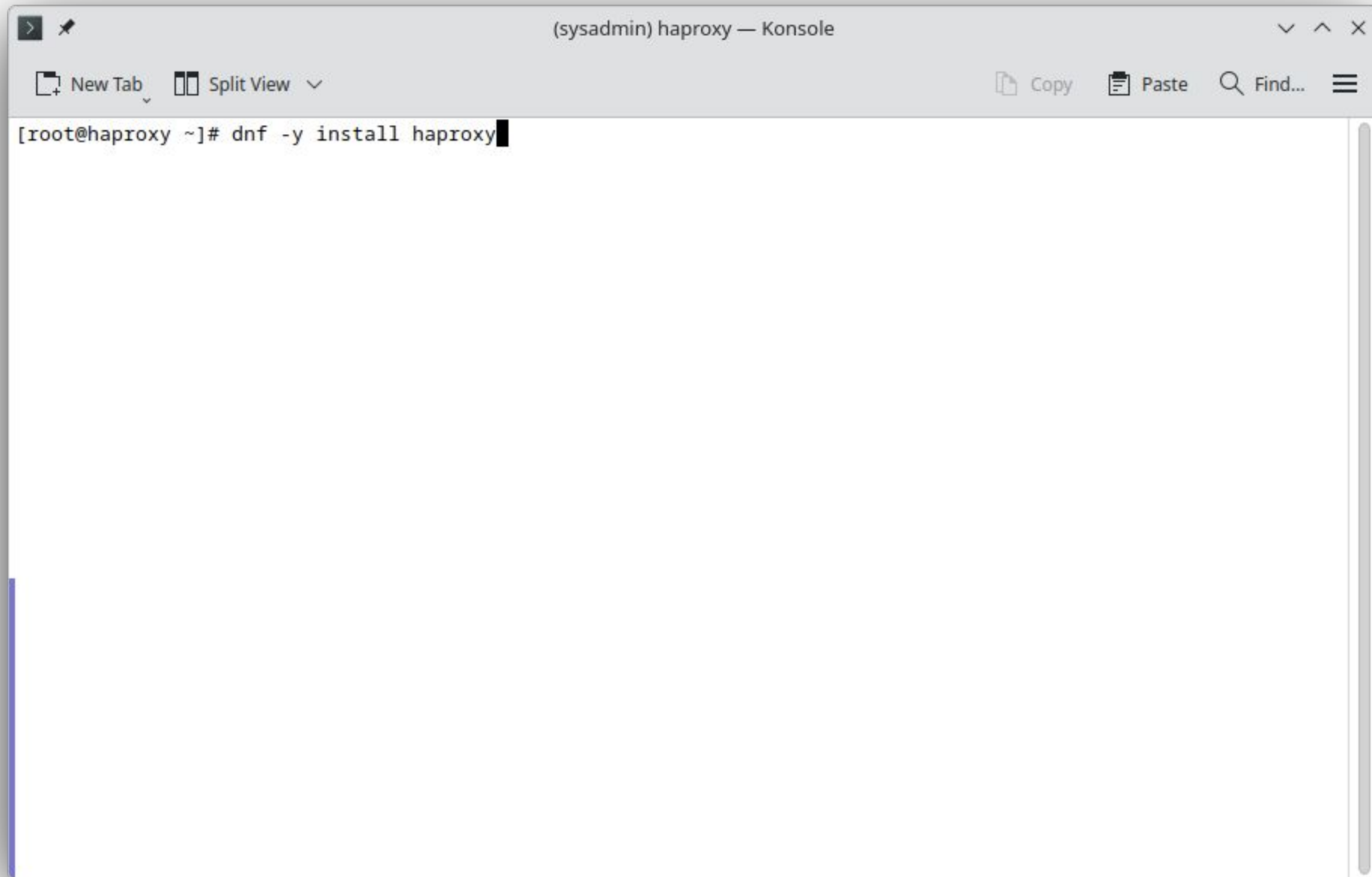
Install the package

Tell SELinux that the proxy can connect

Enable and start the service

Configure the proxy

Test



A terminal window titled "(sysadmin) haproxy — Konsole" is shown. The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal content shows the command `[root@haproxy ~]# dnf -y install haproxy` being entered at the prompt. The cursor is at the end of the command.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# dnf -y install haproxy
```



```
(sysadmin) haproxy — Konsole

New Tab Split View Copy Paste Find...

=====
Install 1 Package

Total download size: 2.6 M
Installed size: 8.0 M
Downloading Packages:
haproxy-3.0.5-4.el10_1.1.x86_64.rpm                25 MB/s | 2.6 MB    00:00
-----
Total                                              24 MB/s | 2.6 MB    00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      :                                1/1
  Running scriptlet: haproxy-3.0.5-4.el10_1.1.x86_64 1/1
  Installing      : haproxy-3.0.5-4.el10_1.1.x86_64 1/1
  Running scriptlet: haproxy-3.0.5-4.el10_1.1.x86_64 1/1
Installed products updated.

Installed:
  haproxy-3.0.5-4.el10_1.1.x86_64

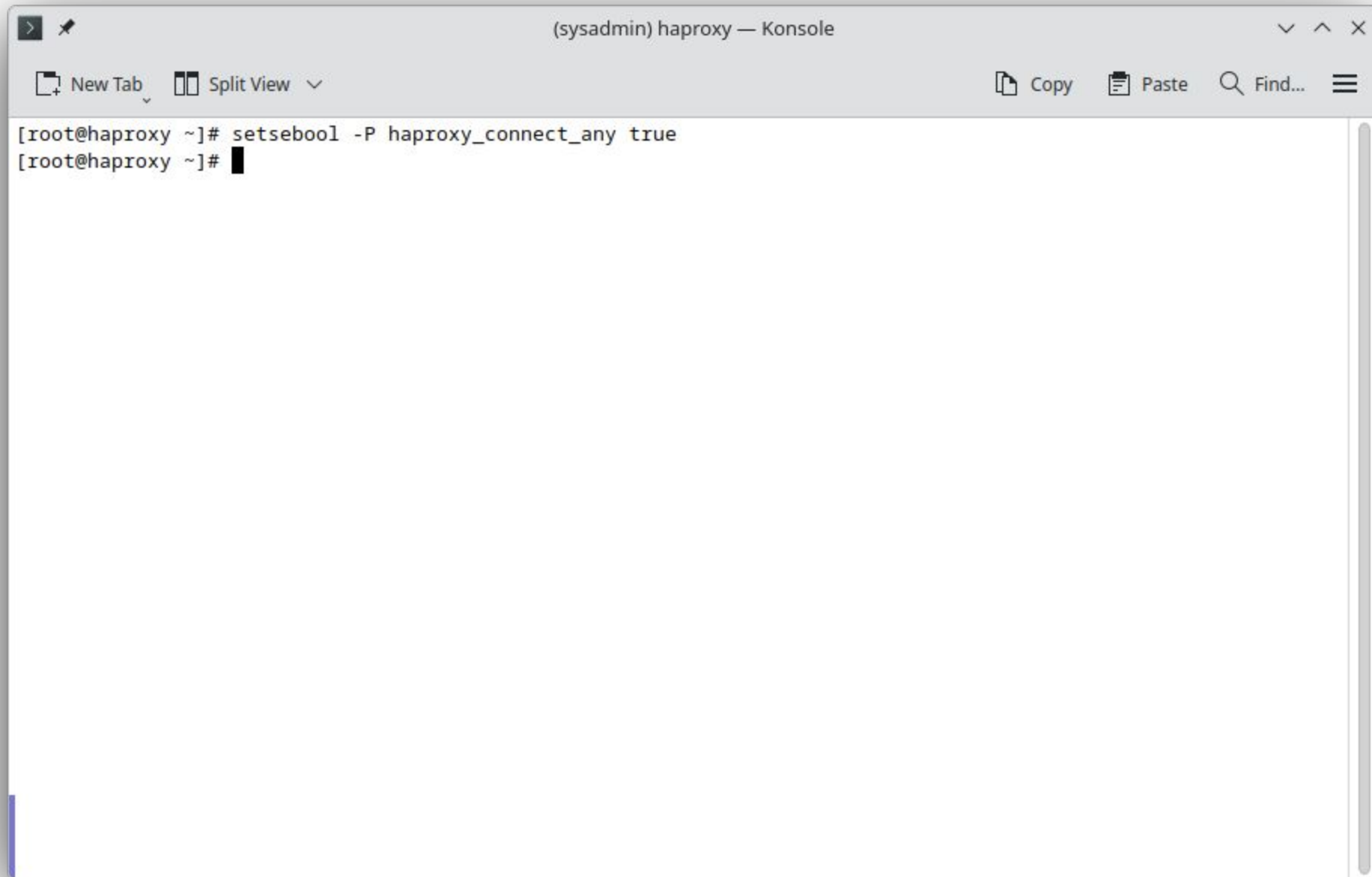
Complete!
[root@haproxy ~]# systemctl enable haproxy.service --now
Created symlink '/etc/systemd/system/multi-user.target.wants/haproxy.service' -> '/usr/lib/systemd/system/haproxy.service'.
[root@haproxy ~]#
```

HAProxy Service

Install and configure

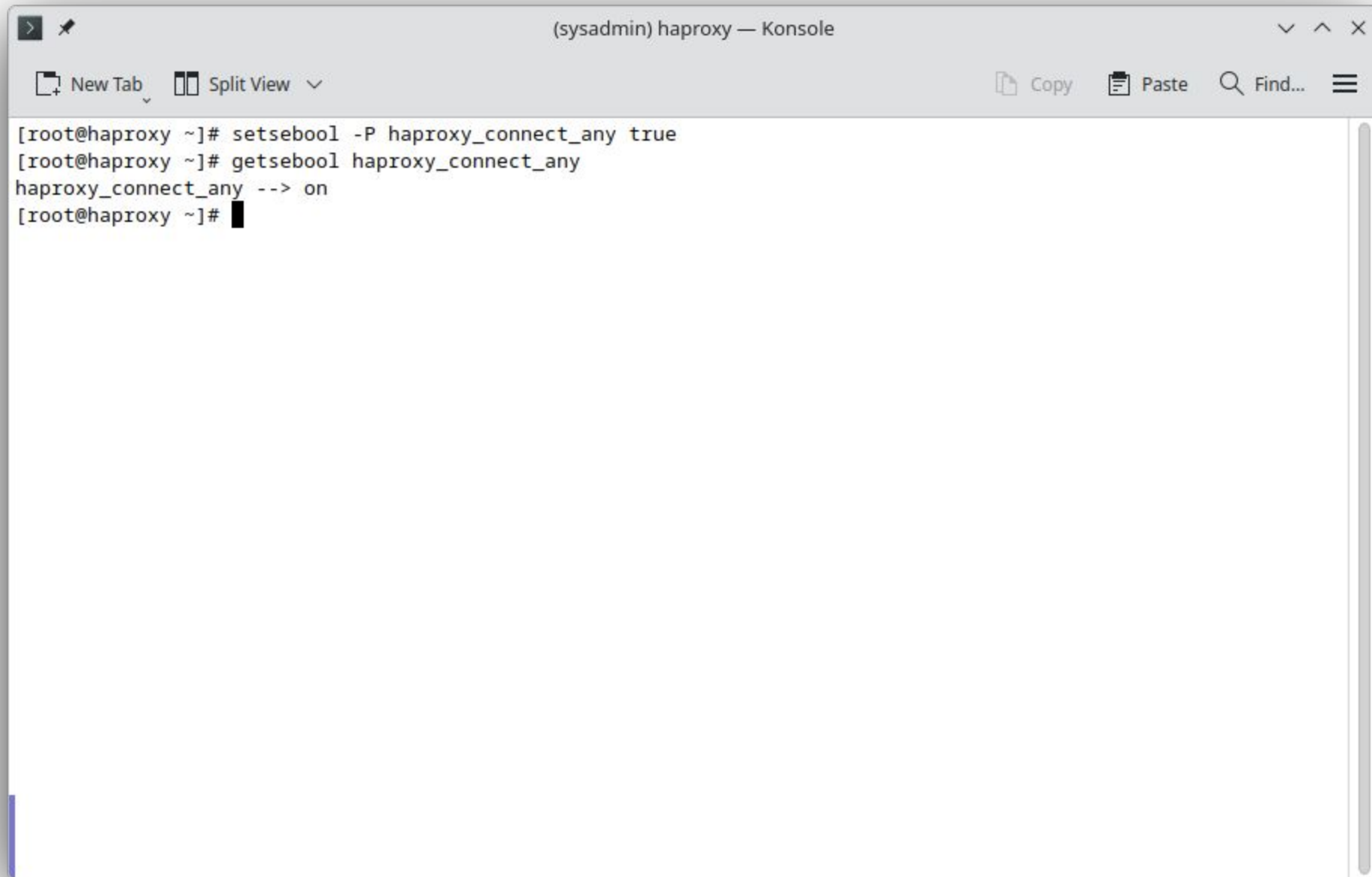
Use setsebool to tell SELinux that it's OK that the proxy software is running

```
setsebool -P haproxy_connect_any true
```



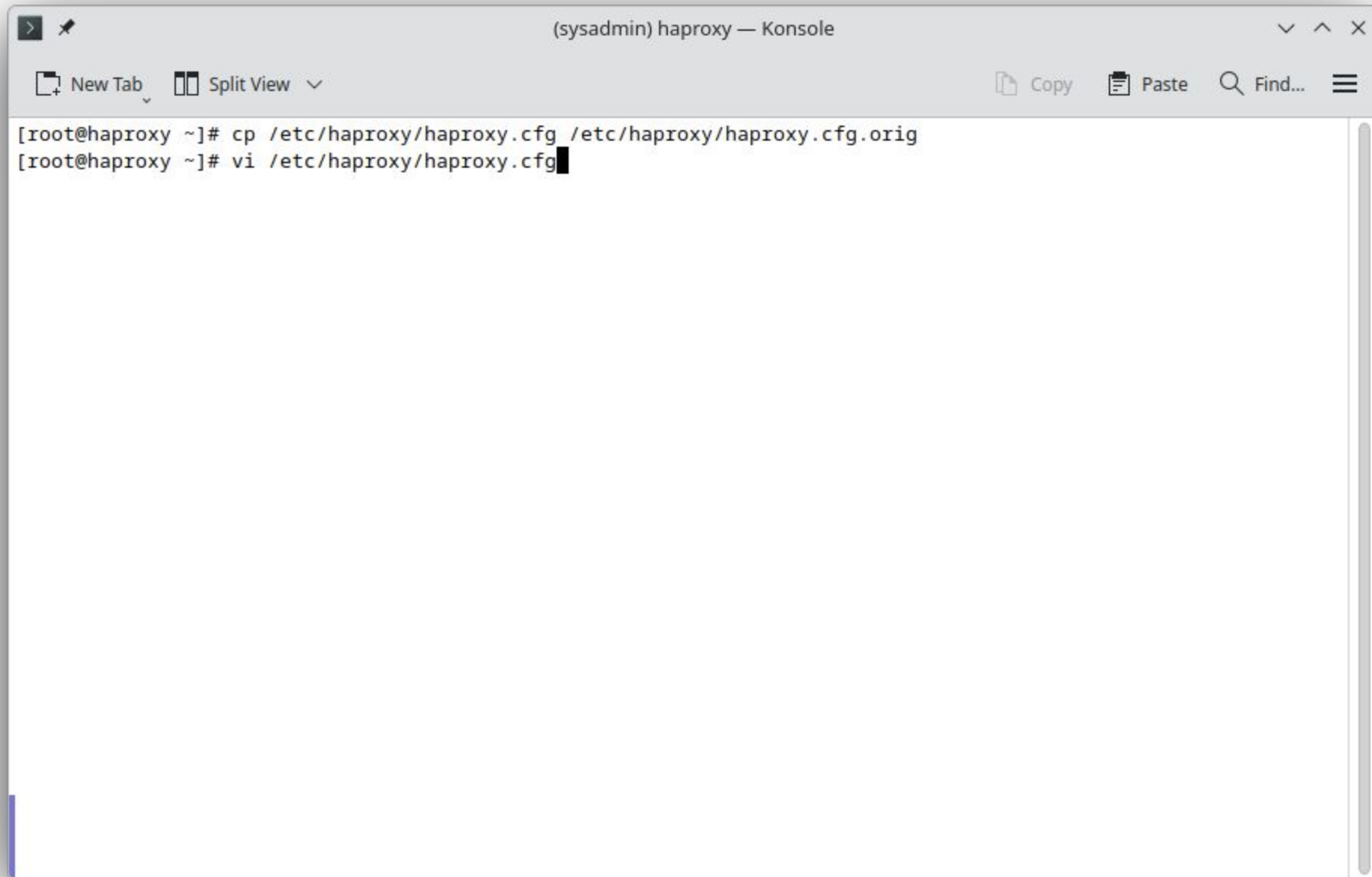
A terminal window titled "(sysadmin) haproxy — Konsole". The window has a toolbar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal content shows a root user at the haproxy machine executing the command "setsebool -P haproxy_connect_any true". The prompt returns to the root user.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# setsebool -P haproxy_connect_any true
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal shows the following commands and output:

```
[root@haproxy ~]# setsebool -P haproxy_connect_any true
[root@haproxy ~]# getsebool haproxy_connect_any
haproxy_connect_any --> on
[root@haproxy ~]#
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# cp /etc/haproxy/haproxy.cfg /etc/haproxy/haproxy.cfg.orig
[root@haproxy ~]# vi /etc/haproxy/haproxy.cfg
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
daemon

# turn on stats unix socket
stats socket /var/lib/haproxy/stats

#-----
# common defaults that all the 'listen' and 'backend' sections will
# use if not designated in their block
#-----
defaults
    log                global
    mode               http
    option              httplog
    option              dontlognull
    option http-server-close
    option redispatch
    option forwardfor   except 127.0.0.0/8
    retries             3
    maxconn             20000
    timeout http-request 10000ms
    timeout http-keep-alive 10000ms
    timeout check       10000ms
    timeout connect     40000ms
    timeout client      300000ms
    timeout server      300000ms
    timeout queue       50000ms

-- INSERT --
```

36,1

Bot

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
# common defaults that all the 'listen' and 'backend' sections will
# use if not designated in their block
#-----
defaults
    log                global
    mode               http
    option              httplog
    option              dontlognull
    option http-server-close
    option redispatch
    option forwardfor   except 127.0.0.0/8
    retries             3
    maxconn             20000
    timeout http-request 10000ms
    timeout http-keep-alive 10000ms
    timeout check        10000ms
    timeout connect      40000ms
    timeout client        300000ms
    timeout server        300000ms
    timeout queue         50000ms

# Enable HAProxy stats
listen stats
    bind :9000
    stats uri /stats
    stats refresh 10000ms

-- INSERT --
```

42,1

Bot


```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
option httplog
option dontlognull
option http-server-close
option redispatch
option forwardfor except 127.0.0.0/8
retries 3
maxconn 20000
timeout http-request 10000ms
timeout http-keep-alive 10000ms
timeout check 10000ms
timeout connect 40000ms
timeout client 300000ms
timeout server 300000ms
timeout queue 50000ms

# Enable HAProxy stats
listen stats
  bind :9000
  stats uri /stats
  stats refresh 10000ms

# Kube API Server
frontend k8s_api_frontend
  bind :6443
  default_backend k8s_api_backend
  mode tcp

-- INSERT --
```

48,1

Bot

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
timeout check 10000ms
timeout connect 40000ms
timeout client 300000ms
timeout server 300000ms
timeout queue 50000ms

# Enable HAProxy stats
listen stats
  bind :9000
  stats uri /stats
  stats refresh 10000ms

# Kube API Server
frontend k8s_api_frontend
  bind :6443
  default_backend k8s_api_backend
  mode tcp

backend k8s_api_backend
  mode tcp
  balance source
  server bootstrap 10.31.100.23:6443 check
  server ocp1 10.31.100.65:6443 check
  server ocp2 10.31.100.66:6443 check
  server ocp3 10.31.100.67:6443 check

-- INSERT --
```

56,1

Bot

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
listen stats
    bind :9000
    stats uri /stats
    stats refresh 10000ms

# Kube API Server
frontend k8s_api_frontend
    bind :6443
    default_backend k8s_api_backend
    mode tcp

backend k8s_api_backend
    mode tcp
    balance source
    server bootstrap 10.31.100.23:6443 check
    server ocp1 10.31.100.65:6443 check
    server ocp2 10.31.100.66:6443 check
    server ocp3 10.31.100.67:6443 check

# OCP Machine Config Server
frontend ocp_machine_config_server_frontend
    mode tcp
    bind :22623
    default_backend ocp_machine_config_server_backend

-- INSERT --
63,1 Bot
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
balance source
server bootstrap 10.31.100.23:6443 check
server ocp1 10.31.100.65:6443 check
server ocp2 10.31.100.66:6443 check
server ocp3 10.31.100.67:6443 check

# OCP Machine Config Server
frontend ocp_machine_config_server_frontend
    mode tcp
    bind :22623
    default_backend ocp_machine_config_server_backend

backend ocp_machine_config_server_backend
    mode tcp
    balance source
    server bootstrap 192.168.22.200:22623 check
    server ocp1 10.31.100.65:22623 check
    server ocp2 10.31.100.66:22623 check
    server ocp3 10.31.100.67:22623 check

-- INSERT --
71,1 Bot
```

```
(sysadmin) haproxy — Konsole

New Tab Split View Copy Paste Find...

balance source
server bootstrap 10.31.100.23:6443 check
server ocp1 10.31.100.65:6443 check
server ocp2 10.31.100.66:6443 check
server ocp3 10.31.100.67:6443 check

# OCP Machine Config Server
frontend ocp_machine_config_server_frontend
    mode tcp
    bind :22623
    default_backend ocp_machine_config_server_backend

backend ocp_machine_config_server_backend
    mode tcp
    balance source
    server bootstrap 192.168.22.200:22623 check
    server ocp1 10.31.100.65:22623 check
    server ocp2 10.31.100.66:22623 check
    server ocp3 10.31.100.67:22623 check

# OCP Ingress - layer 4 tcp mode for each. Ingress Controller will handle layer 7.
frontend ocp_http_ingress_frontend
    bind :80
    default_backend ocp_http_ingress_backend
    mode tcp

-- INSERT --
```

76,1

Bot

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
frontend ocp_machine_config_server_frontend
    mode tcp
    bind :22623
    default_backend ocp_machine_config_server_backend

backend ocp_machine_config_server_backend
    mode tcp
    balance source
    server bootstrap 192.168.22.200:22623 check
    server ocp1 10.31.100.65:22623 check
    server ocp2 10.31.100.66:22623 check
    server ocp3 10.31.100.67:22623 check

# OCP Ingress - layer 4 tcp mode for each. Ingress Controller will handle layer 7.
frontend ocp_http_ingress_frontend
    bind :80
    default_backend ocp_http_ingress_backend
    mode tcp

backend ocp_http_ingress_backend
    balance source
    mode tcp
    server ocp4 10.31.100.68:80 check
    server ocp5 10.31.100.69:80 check
    server ocp6 10.31.100.144:80 check

-- INSERT -- 84,1 Bot
```



```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
backend ocp_machine_config_server_backend
    mode tcp
    balance source
    server bootstrap 192.168.22.200:22623 check
    server ocp1 10.31.100.65:22623 check
    server ocp2 10.31.100.66:22623 check
    server ocp3 10.31.100.67:22623 check

# OCP Ingress - layer 4 tcp mode for each. Ingress Controller will handle layer 7.
frontend ocp_http_ingress_frontend
    bind :80
    default_backend ocp_http_ingress_backend
    mode tcp

backend ocp_http_ingress_backend
    balance source
    mode tcp
    server ocp4 10.31.100.68:80 check
    server ocp5 10.31.100.69:80 check
    server ocp6 10.31.100.144:80 check

frontend ocp_https_ingress_frontend
    bind *:443
    default_backend ocp_https_ingress_backend
    mode tcp

-- INSERT -- 88,1 Bot
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
# OCP Ingress - layer 4 tcp mode for each. Ingress Controller will handle layer 7.
frontend ocp_http_ingress_frontend
    bind :80
    default_backend ocp_http_ingress_backend
    mode tcp

backend ocp_http_ingress_backend
    balance source
    mode tcp
    server ocp4 10.31.100.68:80 check
    server ocp5 10.31.100.69:80 check
    server ocp6 10.31.100.144:80 check

frontend ocp_https_ingress_frontend
    bind *:443
    default_backend ocp_https_ingress_backend
    mode tcp

backend ocp_https_ingress_backend
    mode tcp
    balance source
    server ocp4 10.31.100.68:443 check
    server ocp5 10.31.100.69:443 check
    server ocp6 10.31.100.144:443 check

-- INSERT --
96,1 Bot
```


HAProxy Service

Install and configure

Open the firewall ports

```
firewall-cmd --zone=internal --add-service=http --permanent
firewall-cmd --zone=external --add-service=http --permanent
firewall-cmd --zone=internal --add-service=https --permanent
firewall-cmd --zone=external --add-service=https --permanent
firewall-cmd --zone=internal --add-port=6443/tcp --permanent
firewall-cmd --zone=external --add-port=6443/tcp --permanent
firewall-cmd --zone=internal --add-port=22623/tcp --permanent
firewall-cmd --zone=external --add-port=9000/tcp --permanent
firewall-cmd --reload
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# systemctl restart haproxy.service
[root@haproxy ~]# systemctl status haproxy.service
• haproxy.service - HAProxy Load Balancer
  Loaded: loaded (/usr/lib/systemd/system/haproxy.service; enabled; preset: disabled)
  Active: active (running) since Sun 2026-05-10 10:53:01 UTC; 6s ago
  Invocation: 4757dabf62ba4cbcb492032870017b0
  Process: 4293 ExecStartPre=/usr/sbin/haproxy -f $CONFIG -f $CFGDIR -c -q $OPTIONS (code=exited, status=0/>>
  Main PID: 4296 (haproxy)
  Status: "Ready."
  Tasks: 5 (limit: 23120)
  Memory: 10M (peak: 10.8M)
  CPU: 118ms
  CGroup: /system.slice/haproxy.service
          └─4296 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.>
            4298 /usr/sbin/haproxy -Ws -f /etc/haproxy/haproxy.cfg -f /etc/haproxy/conf.d -p /run/haproxy.>

May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_machine_config_server_ba>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [ALERT] (4298) : backend 'ocp_machine_config_server_>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_http_ingress_backend/ocp>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_http_ingress_backend/ocp>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_http_ingress_backend/ocp>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [ALERT] (4298) : backend 'ocp_http_ingress_backend' >
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_https_ingress_backend/oc>
May 10 10:53:04 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_https_ingress_backend/oc>
May 10 10:53:05 haproxy.aus.redhat.com haproxy[4298]: [WARNING] (4298) : Server ocp_https_ingress_backend/oc>
May 10 10:53:05 haproxy.aus.redhat.com haproxy[4298]: [ALERT] (4298) : backend 'ocp_https_ingress_backend'>
[root@haproxy ~]#
```

HAProxy version 3.0.5-8e879a5, released 2024/09/19

Statistics Report for pid 4298

> General process information

pid = 4298 (process #1, nbproc = 1, nbthread = 4)
uptime = 0d 0h03m05s; warnings = 26
system limits: memmax = unlimited; ulimit-n = 40053
maxsock = 40053; maxconn = 20000; reached = 0; maxpipes = 0
current conns = 2; current pipes = 0/0; conn rate = 2/sec; bit rate = 1.903 kbps
Running tasks: 0/42 (0 niced); idle = 100 %

active UP
active UP, going down
active DOWN, going up
active or backup DOWN
active or backup DOWN for maintenance (MAINT)
active or backup SOFT STOPPED for maintenance
backup UP
backup UP, going down
backup DOWN, going up
not checked
Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

Display option:

- Scope :
- Hide 'DOWN' servers
- Disable refresh
- Refresh now
- CSV export
- JSON export (schema)

External resources:

- Primary site
- Updates (v3.0)
- Online manual

stats

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server									
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
Frontend				2	6	-	2	4	20 000	27			840	104 916	0	0	23					OPEN									
Backend	0	0		0	0		0	0	2 000	0	0	0s	840	104 916	0	0		0	0	0	0	3m5s UP		0/0	0	0		0			

k8s_apl_frontend

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server									
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
Frontend				0	0	-	0	0	20 000	0			0	0	0	0	0					OPEN									

k8s_apl_backend

	Queue			Session rate			Sessions					Bytes		Denied		Errors			Warnings		Server										
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
bootstrap	0	0	-	0	0		0	0		-	0	0	?	0	0	0		0	0	0	0	0	3m5s DOWN	L4CON in 0ms	1/1	Y	-	<u>1</u>	1	3m5s	-
ocp1	0	0	-	0	0		0	0		-	0	0	?	0	0	0		0	0	0	0	0	3m3s DOWN	* L4TOUT in 2002ms	1/1	Y	-	<u>1</u>	1	3m3s	-
ocp2	0	0	-	0	0		0	0		-	0	0	?	0	0	0		0	0	0	0	0	3m3s DOWN	* L4TOUT in 2002ms	1/1	Y	-	<u>1</u>	1	3m3s	-
ocp3	0	0	-	0	0		0	0		-	0	0	?	0	0	0		0	0	0	0	0	3m3s DOWN	* L4TOUT in 2000ms	1/1	Y	-	<u>1</u>	1	3m3s	-
Backend	0	0		0	0		0	0		2 000	0	0	?	0	0	0		0	0	0	0	0	3m3s DOWN		0/0	0	0		1	3m3s	

ocp_machine_config_server_frontend

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server									
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
Frontend				0	0	-	0	0	20 000	0			0	0	0	0	0					OPEN									

ocp_machine_config_server_backend

	Queue			Session rate			Sessions						Bytes		Denied		Errors			Warnings		Server									
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwntme	Thrtle	
bootstrap	0	0	-	0	0		0	0		0	0	?	0	0		0		0	0	0	0	3m3s DOWN	* L4TOUT in 2001ms	1/1	Y	-	<u>1</u>	1	3m3s	-	
ocp1	0	0	-	0	0		0	0		0	0	?	0	0		0		0	0	0	0	3m3s DOWN	* L4TOUT in 2002ms	1/1	Y	-	<u>1</u>	1	3m3s	-	
ocp2	0	0	-	0	0		0	0		0	0	?	0	0		0		0	0	0	0	3m2s DOWN	* L4TOUT in 2001ms	1/1	Y	-	<u>1</u>	1	3m2s	-	
ocp3	0	0	-	0	0		0	0		0	0	?	0	0		0		0	0	0	0	3m2s DOWN	* L4TOUT in 2001ms	1/1	Y	-	<u>1</u>	1	3m2s	-	
Backend	0	0		0	0		0	0	2 000	0	0	?	0	0	0	0		0	0	0	0	3m2s DOWN		0/0	0	0		1	3m2s		

NFS Server



Red Hat

NFS Server

Storage for the OpenShift registry

Since the HAProxy machine is available to the private network, I added a 200Gb drive, formatted it, and am sharing it via NFS

haproxy.summit.aus.redhat.com on QEMU/KVM: armitage

File Virtual Machine View Send Key

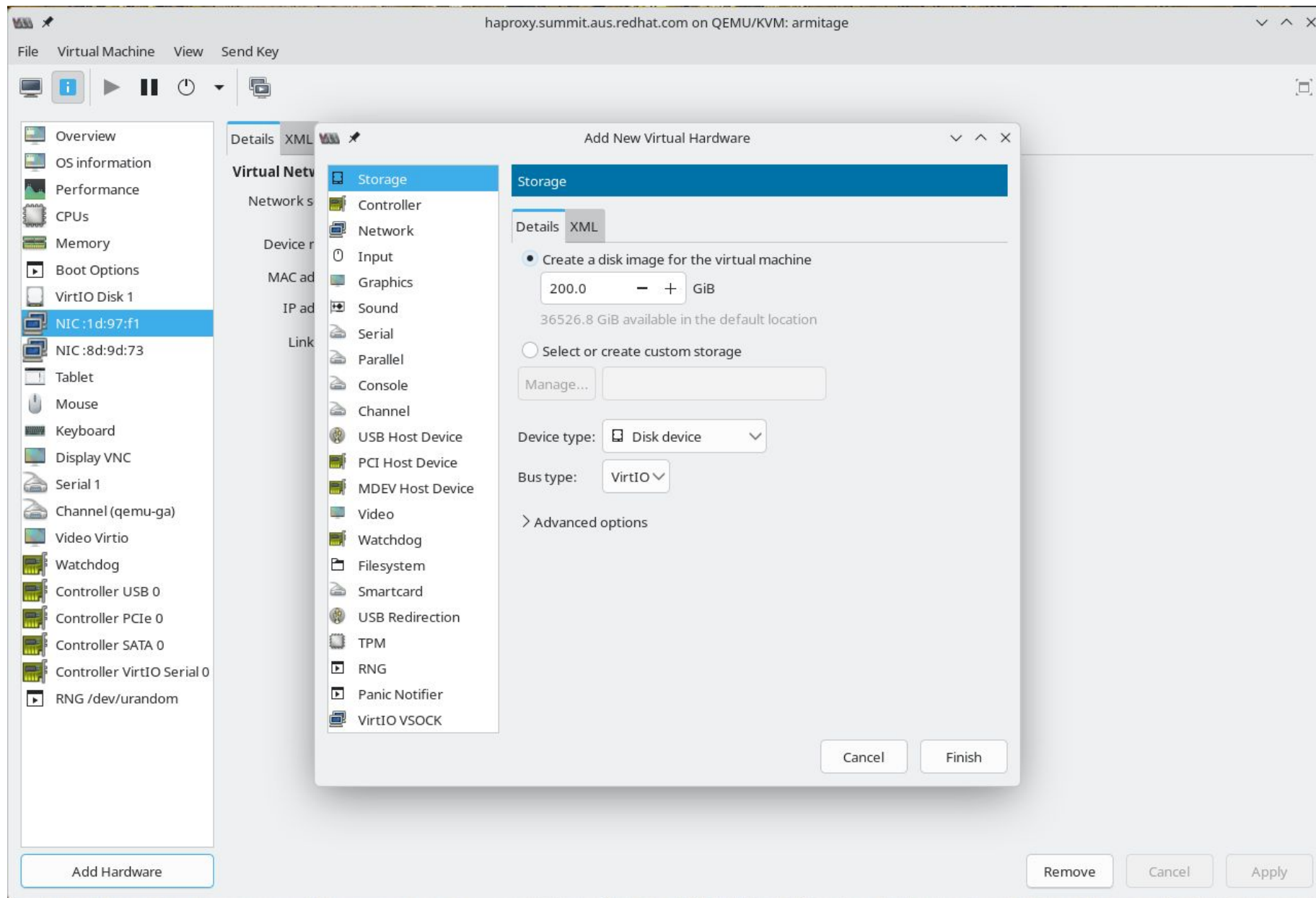
Red Hat Enterprise Linux 10.1 (Coughlan)
Kernel 6.12.0-124.39.1.el10_1.x86_64 on x86_64

Activate the web console with: `systemctl enable --now cockpit.socket`

haproxy login: root
Password:
Last login: Sun May 10 09:18:12 on tty1
[root@haproxy ~]# lsblk

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
vda	252:0	0	40G	0	disk	
├─vda1	252:1	0	1M	0	part	
├─vda2	252:2	0	1G	0	part	/boot
├─vda3	252:3	0	39G	0	part	
└─rhel_haproxy-root	253:0	0	35.1G	0	lvm	/
└─rhel_haproxy-swap	253:1	0	3.9G	0	lvm	[SWAP]

[root@haproxy ~]#



haproxy.summit.aus.redhat.com on QEMU/KVM: armitage

File Virtual Machine View Send Key

Red Hat Enterprise Linux 10.1 (Coughlan)
Kernel 6.12.0-124.39.1.el10_1.x86_64 on x86_64

Activate the web console with: `systemctl enable --now cockpit.socket`

haproxy login: root
Password:
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NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
vda	252:0	0	40G	0	disk	
├─vda1	252:1	0	1M	0	part	
├─vda2	252:2	0	1G	0	part	/boot
├─vda3	252:3	0	39G	0	part	
└─rhel_haproxy-root	253:0	0	35.1G	0	lvm	/
└─rhel_haproxy-swap	253:1	0	3.9G	0	lvm	[SWAP]

[root@haproxy ~]# lsblk

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
vda	252:0	0	40G	0	disk	
├─vda1	252:1	0	1M	0	part	
├─vda2	252:2	0	1G	0	part	/boot
├─vda3	252:3	0	39G	0	part	
└─rhel_haproxy-root	253:0	0	35.1G	0	lvm	/
└─rhel_haproxy-swap	253:1	0	3.9G	0	lvm	[SWAP]
vdb	252:16	0	200G	0	disk	

[root@haproxy ~]# _


```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# fdisk /dev/vdb

Welcome to fdisk (util-linux 2.40.2).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS (MBR) disklabel with disk identifier 0x2b8533b0.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-419430399, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-419430399, default 419430399):

Created a new partition 1 of type 'Linux' and of size 200 GiB.

Command (m for help): t
Selected partition 1
Hex code or alias (type L to list all): 8e
Changed type of partition 'Linux' to 'Linux LVM'.

Command (m for help):
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# pvcreate /dev/vdb1
Physical volume "/dev/vdb1" successfully created.
[root@haproxy ~]# vgcreate nfsvol /dev/vdb1
Volume group "nfsvol" successfully created
[root@haproxy ~]# lvcreate -l 100%FREE -n nfslv nfsvol
Logical volume "nfslv" created.
[root@haproxy ~]# lvs
LV VG Attr LSize Pool Origin Data% Meta% Move Log Cpy%Sync Convert
nfslv nfsvol -wi-a----- <200.00g
root rhel_haproxy -wi-ao----- 35.05g
swap rhel_haproxy -wi-ao----- <3.95g
[root@haproxy ~]# mkfs.xfs /dev/nfsvol/nfslv
meta-data=/dev/nfsvol/nfslv isize=512 agcount=4, agsize=13106944 blks
= sectsz=512 attr=2, projid32bit=1
= crc=1 finobt=1, sparse=1, rmapbt=1
= reflink=1 bigtime=1 inobtcount=1 nnext64=1
= exchange=0
data = bsize=4096 blocks=52427776, imaxpct=25
= sunit=0 swidth=0 blks
naming =version 2 bsize=4096 ascii-ci=0, ftype=1, parent=0
log =internal log bsize=4096 blocks=25599, version=2
= sectsz=512 sunit=0 blks, lazy-count=1
realtime =none extsz=4096 blocks=0, rtextents=0
Discarding blocks...Done.
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# mkdir /var/export
[root@haproxy ~]# mount /dev/nfsvol/nfslv /var/export
[root@haproxy ~]# tail -1 /etc/mtab
/dev/mapper/nfsvol-nfslv /var/export xfs rw,seclabel,relatime,attr2,inode64,logbufs=8,logbsize=32k,noquota 0 0
[root@haproxy ~]# tail -1 /etc/mtab >> /etc/fstab
[root@haproxy ~]# umount /var/export
[root@haproxy ~]# mount -a
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[root@haproxy ~]# systemctl daemon-reload
[root@haproxy ~]# umount /var/export
[root@haproxy ~]# mount -a
[root@haproxy ~]# df -h | grep export
/dev/mapper/nfsvol-nfslv      200G  3.9G  197G   2% /var/export
[root@haproxy ~]#
```

NFS Server

Storage for the OpenShift registry

Now we install nfs-utils and configure the export

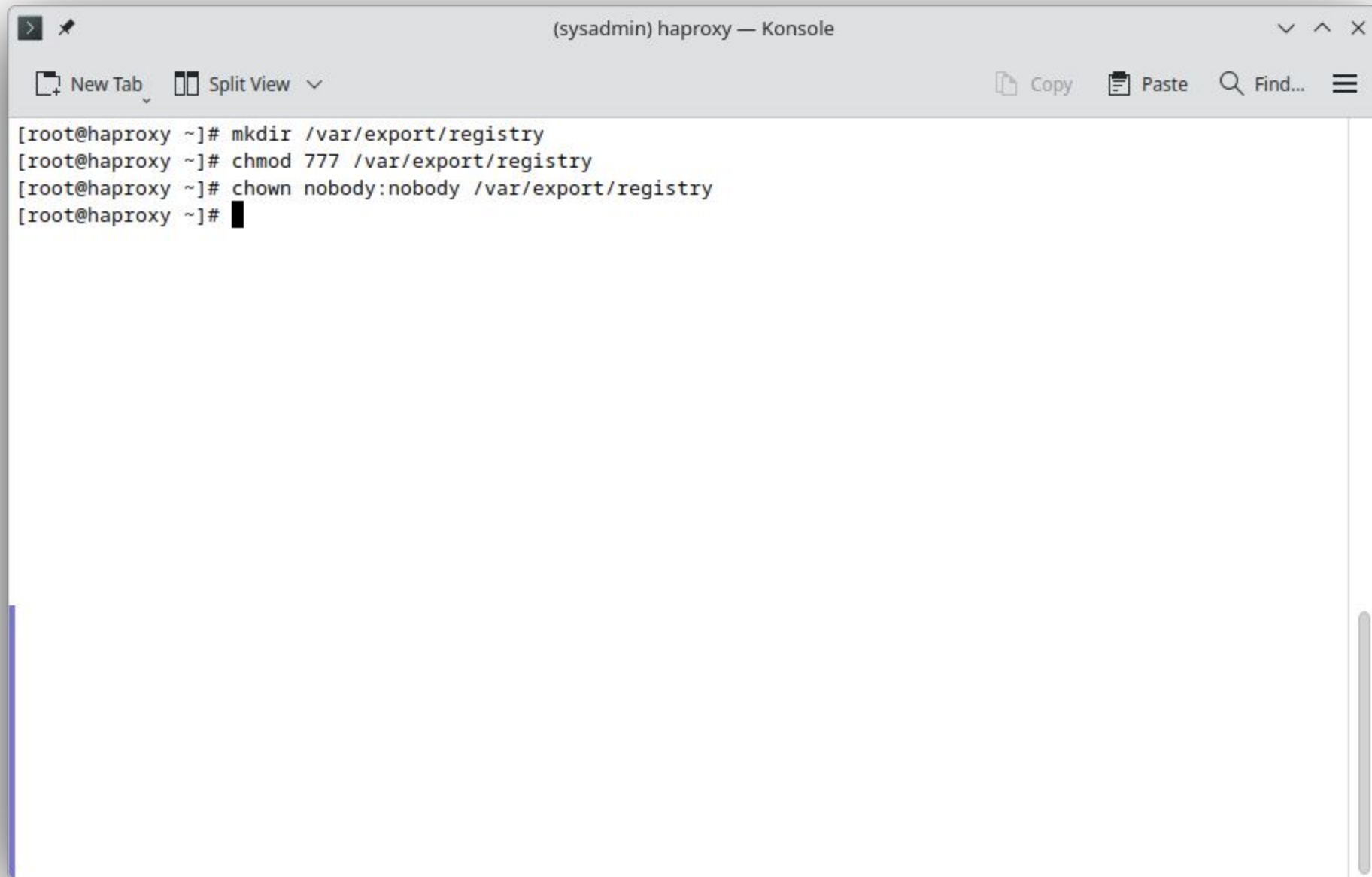
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# dnf -y install nfs-utils
Updating Subscription Management repositories.

This system has release set to 10 and it receives updates only for this release.

Red Hat Enterprise Linux 10 for x86_64 - BaseOS (RPMs) 51 kB/s | 4.1 kB 00:00
Red Hat Satellite Client 6 for RHEL 10 x86_64 (RPMs) 48 kB/s | 3.8 kB 00:00
Red Hat CodeReady Linux Builder for RHEL 10 x86_64 (RPMs) 52 kB/s | 4.0 kB 00:00
Red Hat Enterprise Linux 10 for x86_64 - AppStream (RPMs) 53 kB/s | 4.1 kB 00:00
EPEL 10 x86_64 53 kB/s | 4.0 kB 00:00
Dependencies resolved.

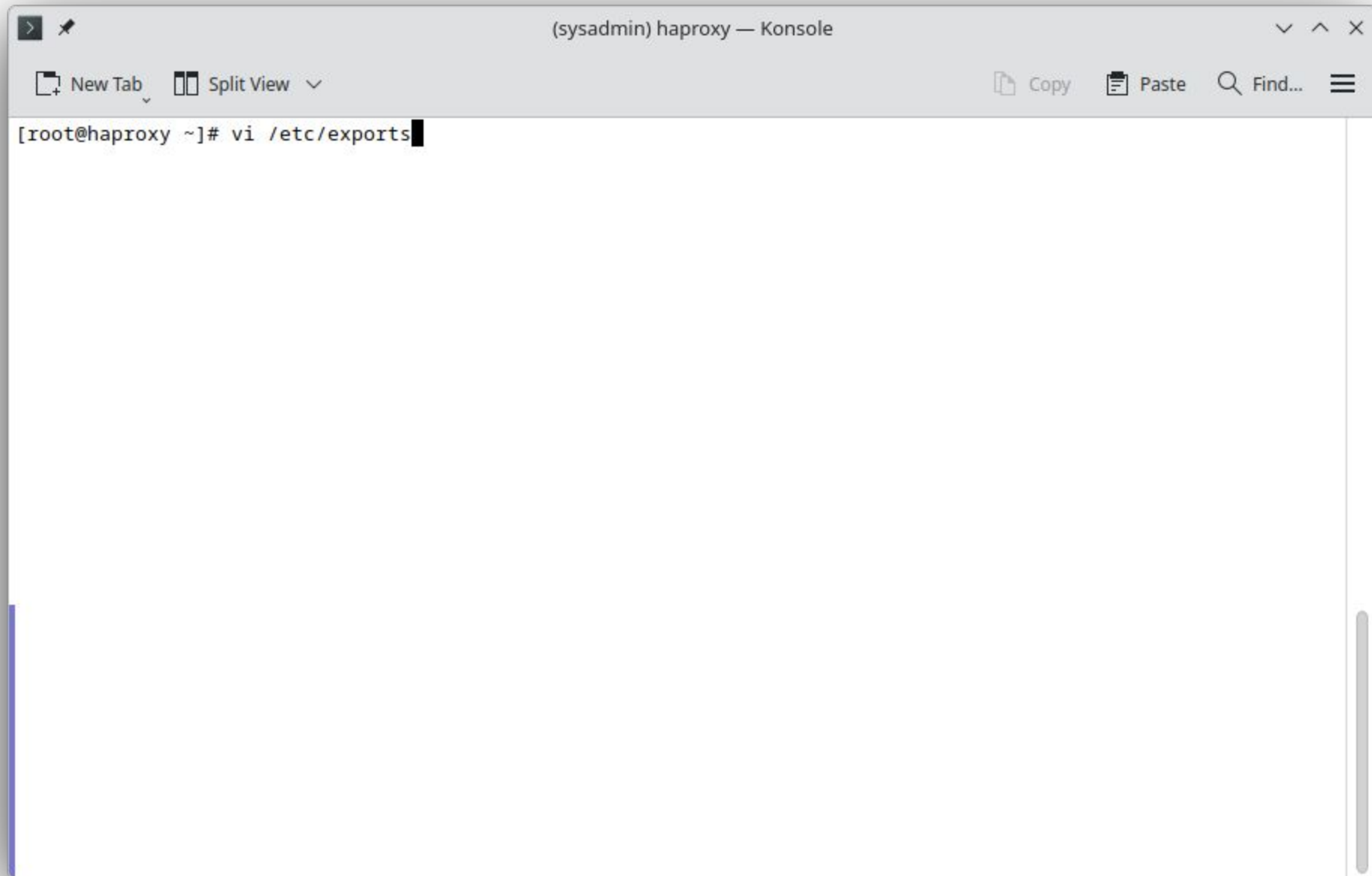
=====
Package Architecture Version Repository Size
=====
Installing:
nfs-utils x86_64 1:2.8.3-0.el10 rhel-10-for-x86_64-baseos-rpms 485 k
Installing dependencies:
gssproxy x86_64 0.9.2-10.el10 rhel-10-for-x86_64-baseos-rpms 118 k
libev x86_64 4.33-14.el10 rhel-10-for-x86_64-baseos-rpms 56 k
libnfsidmap x86_64 1:2.8.3-0.el10 rhel-10-for-x86_64-baseos-rpms 63 k
libverto-libev x86_64 0.3.2-10.el10 rhel-10-for-x86_64-baseos-rpms 15 k
rpcbind x86_64 1.2.7-3.el10 rhel-10-for-x86_64-baseos-rpms 63 k
sssd-nfs-idmap x86_64 2.11.1-2.el10_1.1 rhel-10-for-x86_64-baseos-rpms 36 k

Transaction Summary
=====
Install 7 Packages
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows a sequence of commands to create a directory and set permissions:

```
[root@haproxy ~]# mkdir /var/export/registry
[root@haproxy ~]# chmod 777 /var/export/registry
[root@haproxy ~]# chown nobody:nobody /var/export/registry
[root@haproxy ~]#
```

A terminal window titled "(sysadmin) haproxy — Konsole" is shown. The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal content shows a root user at the haproxy machine, with the command `vi /etc/exports` entered and the cursor at the end of the line.

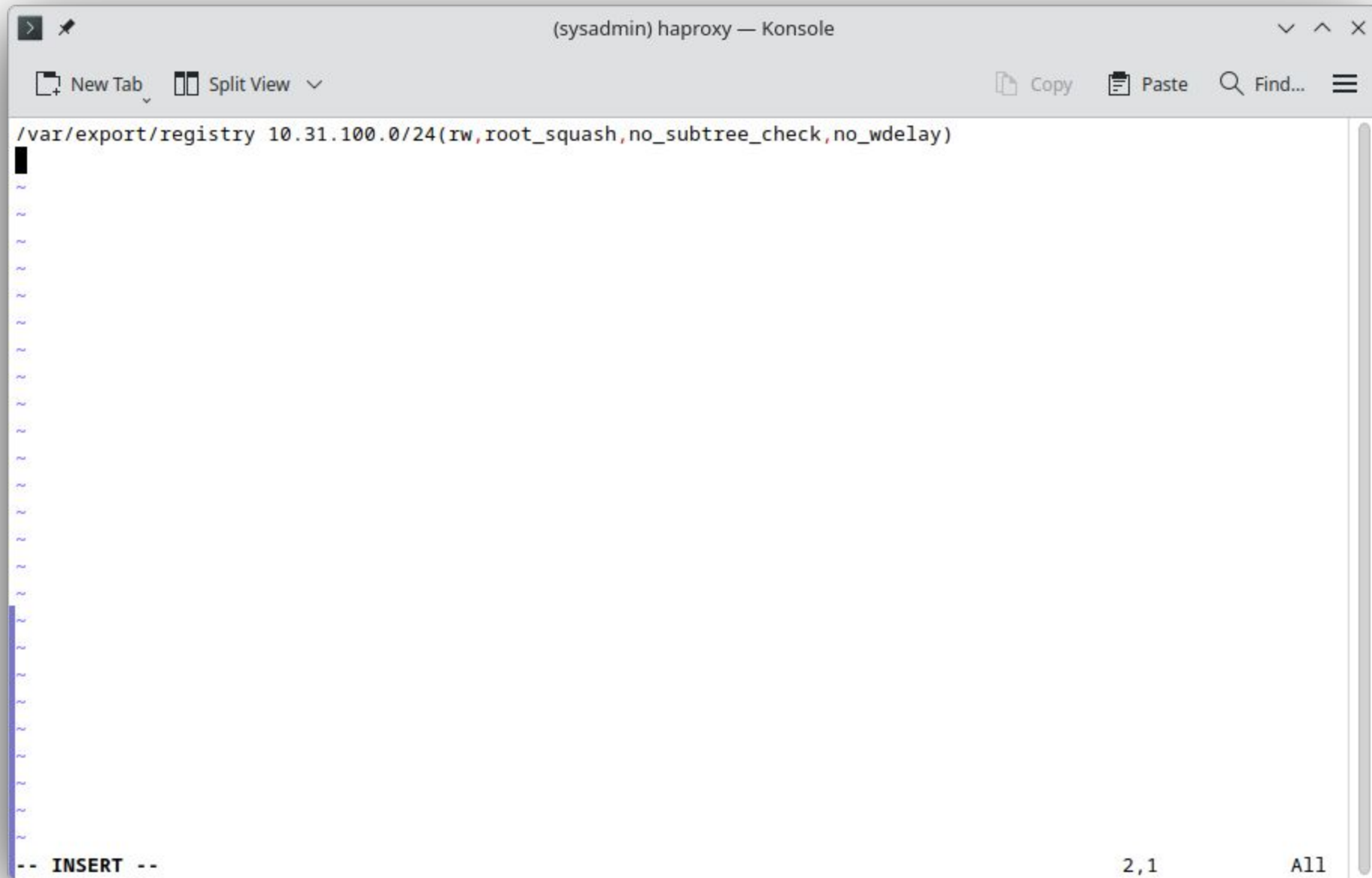
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /etc/exports
```

NFS Server

Storage for the OpenShift registry

The export needs to be shared with specific settings

```
rw,root_squash,no_subtree_check,no_wdelay
```

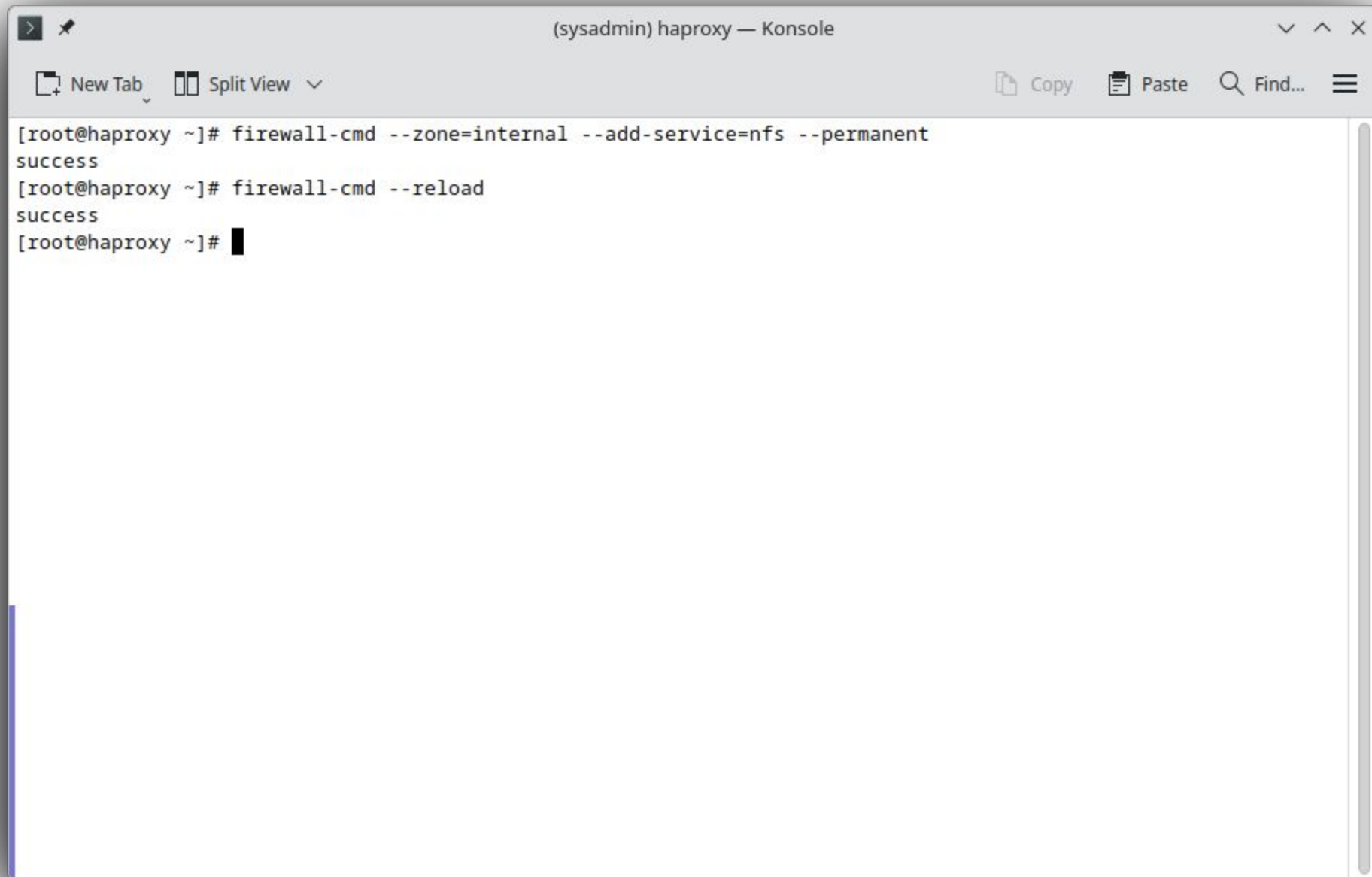



```
(sysadmin) haproxy — Konsole
```

New Tab Split View Copy Paste Find... ≡

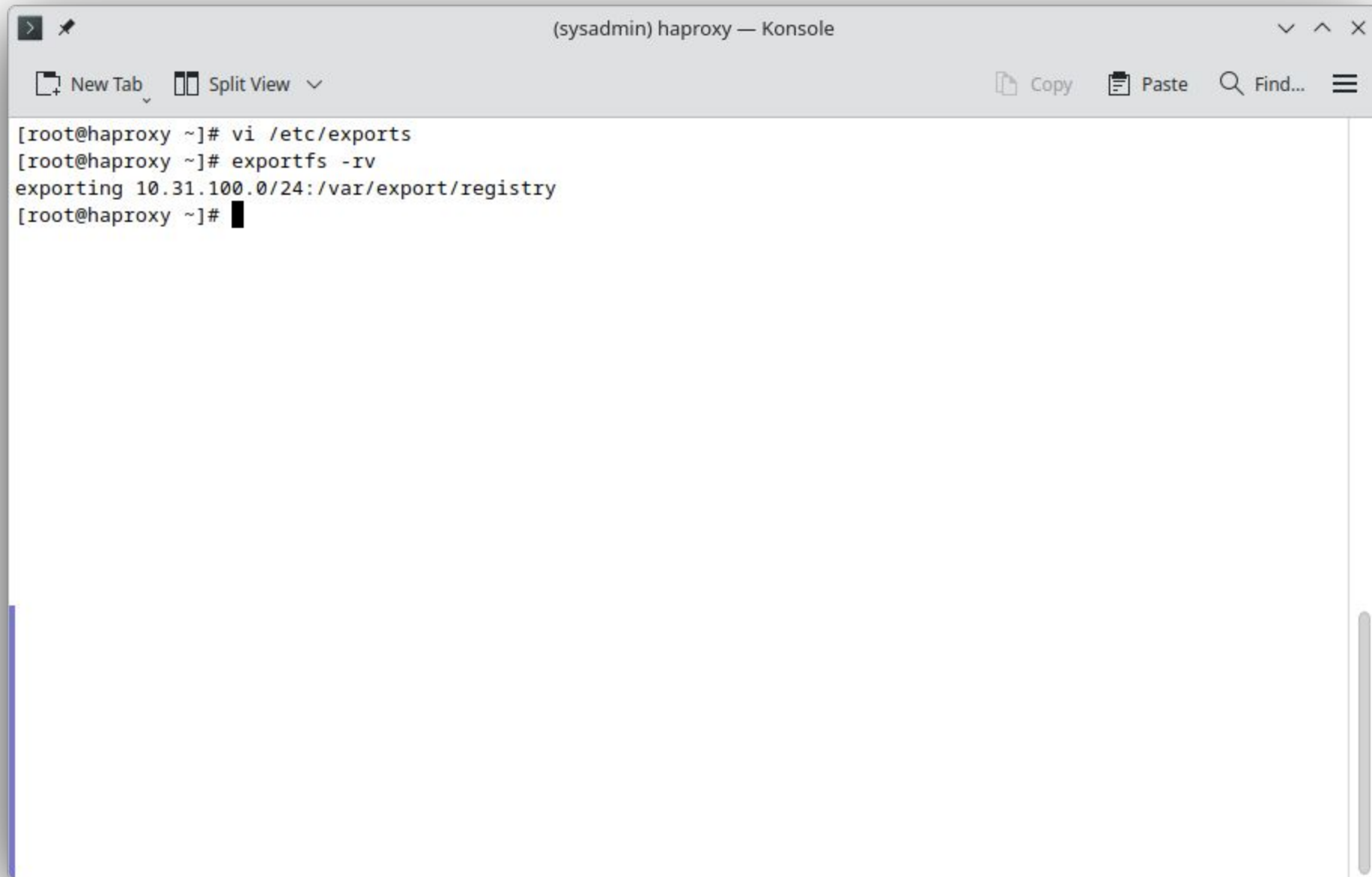
```
/var/export/registry 10.31.100.0/24(rw,root_squash,no_subtree_check,no_wdelay)
```

-- INSERT -- 2,1 All



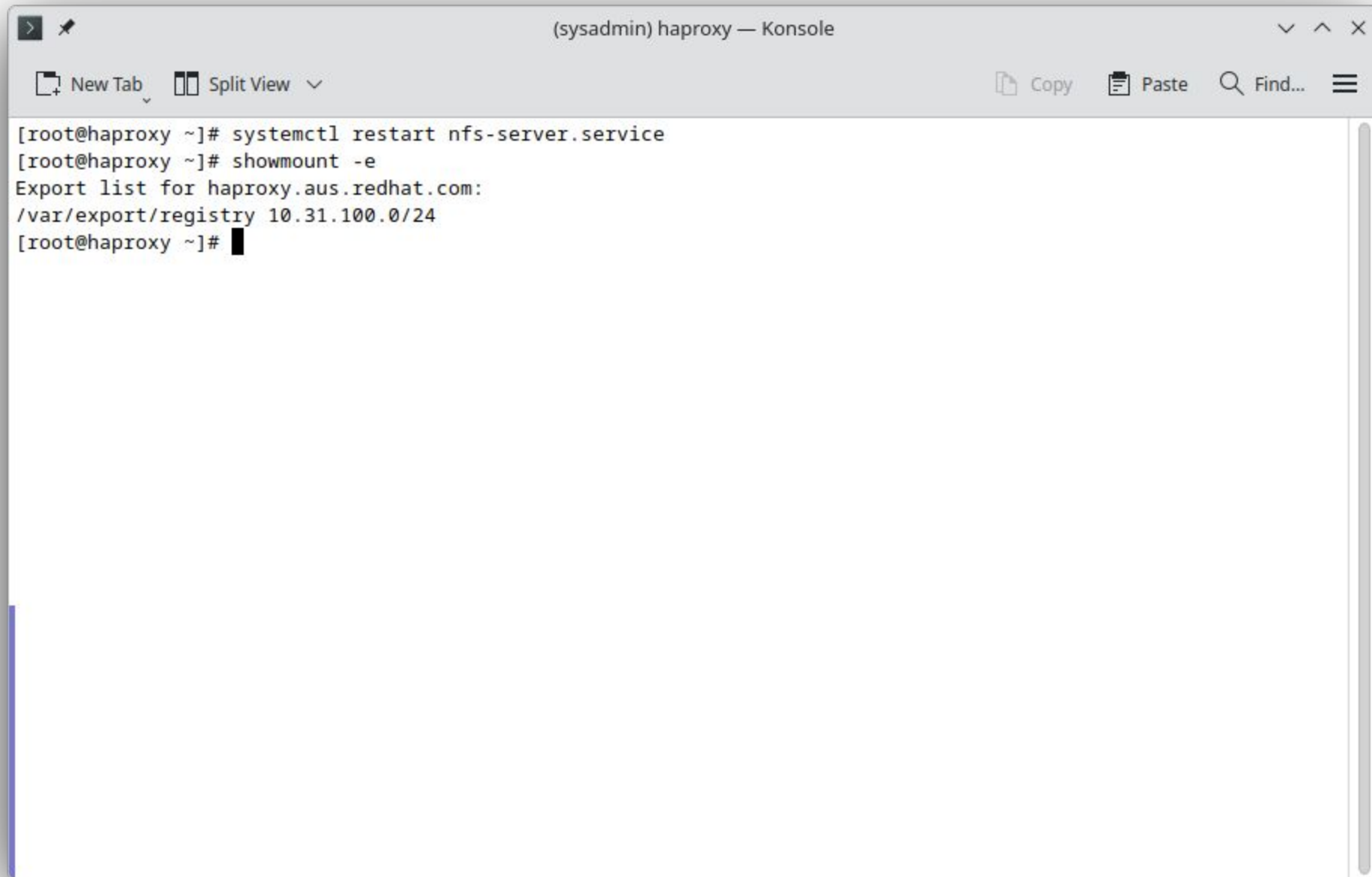
A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal interface includes a top bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The command history shows the user adding a permanent firewall rule for NFS and reloading the firewall.

```
[root@haproxy ~]# firewall-cmd --zone=internal --add-service=nfs --permanent
success
[root@haproxy ~]# firewall-cmd --reload
success
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal shows the following commands and output:

```
[root@haproxy ~]# vi /etc/exports
[root@haproxy ~]# exportfs -rv
exporting 10.31.100.0/24:/var/export/registry
[root@haproxy ~]#
```



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The terminal shows the execution of two commands: `systemctl restart nfs-server.service` and `showmount -e`. The output of the second command displays the export list for `haproxy.aus.redhat.com`, showing `/var/export/registry` exported to `10.31.100.0/24`. The prompt `[root@haproxy ~]#` is visible at the end of the output line.

```
[root@haproxy ~]# systemctl restart nfs-server.service
[root@haproxy ~]# showmount -e
Export list for haproxy.aus.redhat.com:
/var/export/registry 10.31.100.0/24
[root@haproxy ~]#
```

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# mkdir /mnt/nfs
[root@haproxy ~]# mount 10.31.100.64:/var/export/registry /mnt/nfs/
[root@haproxy ~]# mount | grep registry
10.31.100.64:/var/export/registry on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsz=524288,wsz=524288,namlen=255,hard,fatal_nerrors=none,proto=tcp,timeo=600,retrans=2,sec=sys,clientaddr=10.31.100.64,local_lock=none,addr=10.31.100.64)
[root@haproxy ~]# df -h /mnt/nfs/
Filesystem                Size      Used Avail Use% Mounted on
10.31.100.64:/var/export/registry 200G   3.9G   197G   2% /mnt/nfs
[root@haproxy ~]# umount /mnt/nfs/
[root@haproxy ~]# df -h /mnt/nfs/
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/rhel_haproxy-root 35G    2.6G    33G   8% /
[root@haproxy ~]#
```

Download Client and Installer

Client and Installer

Get them from <https://console.redhat.com/openshift>

- You can also get the upstream tools from the OKD project

Downloads - Resources | OpenShift - Google Chrome

console.redhat.com/openshift/downloads

Red Hat | Personal | Rover Apps | Red Hat Support | I.T. New Hire Hub | I.T. Toolbox | Red Hat External

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode

Red Hat Hybrid Cloud Console

Thomas Cameron

OpenShift

Overview

Fleet management

Cloud console settings

Resources

Learning Resources

Downloads

Releases

Assisted Installer

OpenShift > Resources > Downloads

Downloads

All categories > Expand all

Command-line interface (CLI) tools

Download command line tools to manage and work with OpenShift from your terminal.

	Name	OS type	Architecture type	
>	OpenShift command-line interface (oc)	Linux	x86_64	Download
>	OpenShift Cluster Manager API command-line interface (ocm) Developer Preview	Linux	x86_64	Download
>	Red Hat OpenShift Service on AWS command-line interface (rosa)	Linux	x86_64	Download
>	Knative command-line interface for OpenShift Serverless (kn)			

Feedback

Downloads - Resources | x +

console.redhat.com/openshift/downloads

Red Hat Personal Rover Apps Red Hat Support I.T. New Hire Hub I.T. Toolbox Red Hat External

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode

Red Hat Hybrid Cloud Console

OpenShift

Overview

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Learning Resources

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Releases

Assisted Installer

OpenShift > Resources > Downloads

OpenShift installation

Install OpenShift based on your infrastructure. For the installer matching your infrastructure type, select the operating system and architecture on which you wish to run the installer. Then follow the steps provided within your infrastructure's tab on the create cluster page to install an OpenShift cluster.

	Name	OS type	Architecture type	
>	OpenShift for x86_64 Installer	Linux	x86_64	Download
>	OpenShift for ARM Installer	Linux	x86_64	Download
>	OpenShift for IBM Z (s390x) Installer	Linux	x86_64	Download
>	OpenShift for Power Installer	Linux	x86_64	Download
>	OpenShift Installer with multi-architecture compute machines	Linux	x86_64	Download
>	OpenShift Local (crc)	Linux	x86_64	Download

Feedback

Client and Installer

Get them from <https://console.redhat.com/openshift>

- Note that this is on a workstation machine

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ curl -O https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable/openshift-client-linux.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 42.4M 100 42.4M 0 0 44.7M 0 --:--:-- --:--:-- --:--:-- 44.8M
[sysadmin@workstation ~]$ curl -O https://mirror.openshift.com/pub/openshift-v4/x86_64/clients/ocp/stable/openshift-install-linux.tar.gz
% Total % Received % Xferd Average Speed Time Time Time Current
Dload Upload Total Spent Left Speed
100 431M 100 431M 0 0 43.9M 0 0:00:09 0:00:09 --:--:-- 38.3M
[sysadmin@workstation ~]$ tar zxvf openshift-client-linux.tar.gz
README.md
oc
kubectl
[sysadmin@workstation ~]$ tar zxvf openshift-install-linux.tar.gz
README.md
openshift-install
[sysadmin@workstation ~]$ ls
kubectl oc openshift-client-linux.tar.gz openshift-install openshift-install-linux.tar.gz README.md
[sysadmin@workstation ~]$
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ sudo install oc /usr/local/bin/
[sysadmin@workstation ~]$ sudo install kubectl /usr/local/bin/
[sysadmin@workstation ~]$ sudo install openshift-install /usr/local/bin/
[sysadmin@workstation ~]$ ls -l /usr/local/bin/
total 947216
-rwxr-xr-x. 1 root root 140315144 May 11 10:48 kubectl
-rwxr-xr-x. 1 root root 140315144 May 11 10:48 oc
-rwxr-xr-x. 1 root root 689311928 May 11 10:48 openshift-install
[sysadmin@workstation ~]$
```

Start the Installation

Get You Pull Secret

This registers your cluster to Red Hat

- <https://console.redhat.com/openshift/install/pull-secret>

Install OpenShift 4 | Pull Secret - Google Chrome

tcameron - Red Hat Summit x Install OpenShift 4 | Pull Se x Hybrid Cloud Console Hom x +

console.redhat.com/openshift/install/pull-secret

You're in Hybrid Cloud Console production mode. To see new pre-production features, turn on Preview mode

Red Hat Hybrid Cloud Console

OpenShift

Overview

Fleet management >

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OpenShift

Downloads > Pull secret

Install OpenShift Container Platform 4

Pull secret

Download or copy your pull secret. You'll be prompted for this information during installation.

[Download pull secret](#) [Copy pull secret](#)

Red Hat collects a limited amount of telemetry data. By installing OpenShift Container Platform 4, you accept our data collection policy. [Learn more](#) about the data we collect.

Feedback

Create an SSH Key

This is the key you can use to ssh into your OpenShift nodes

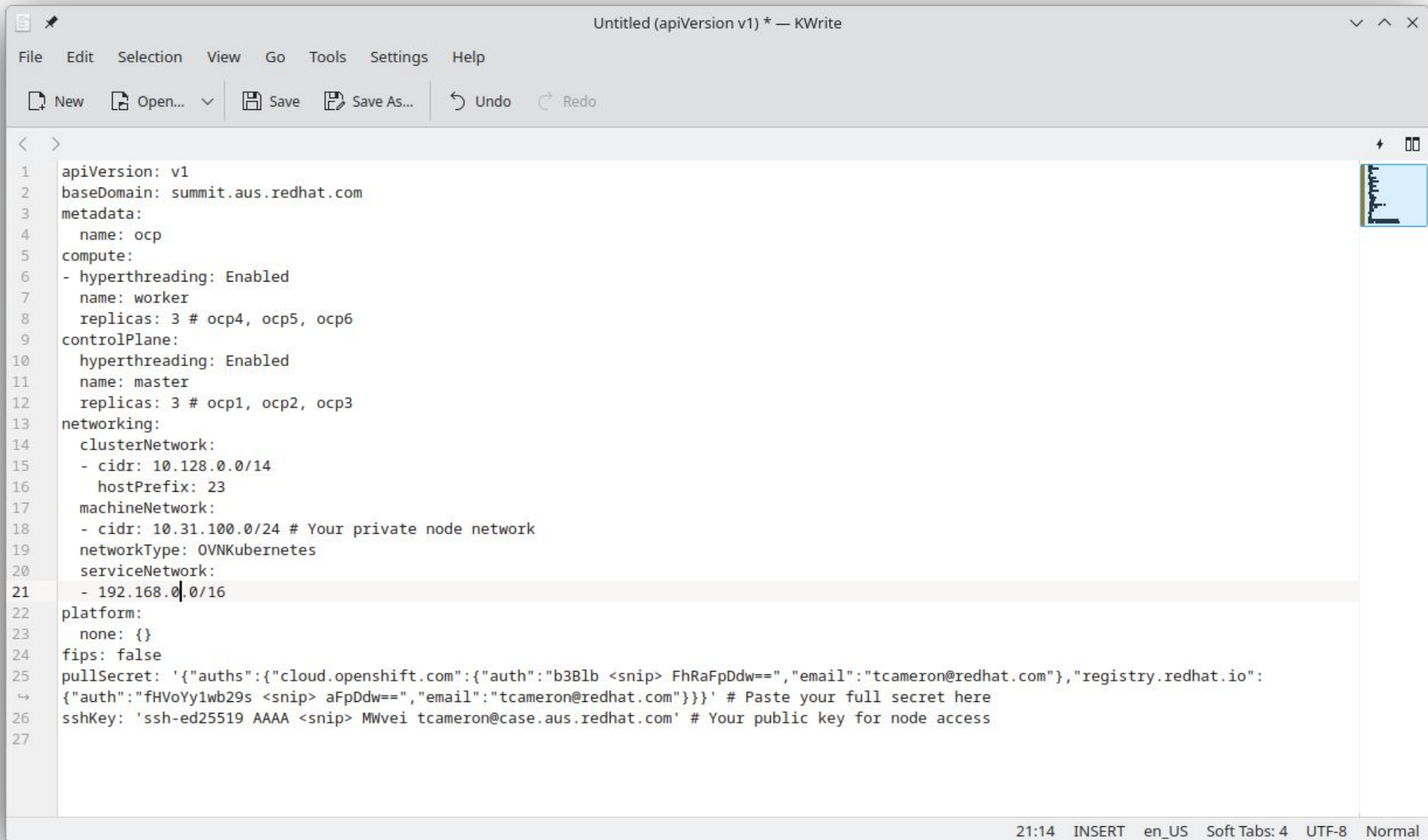
- I use ed25519 to create my key


```
(sysadmin) bootstrap — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@bootstrap ~]$ ssh-keygen -t ed25519
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/sysadmin/.ssh/id_ed25519):
Enter passphrase for "/home/sysadmin/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/sysadmin/.ssh/id_ed25519
Your public key has been saved in /home/sysadmin/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:6qAQBfDrwLiUiIGKS78PXV4pdTAwaa6hUFAJCEF5jwI sysadmin@bootstrap.aus.redhat.com
The key's randomart image is:
+--[ED25519 256]--+
|X==..  ooo      |
|E+ +   o. o     |
|+ = o o  . .    |
|B=.o o  .. o    |
|O=+ . ooSo     |
|o*....o.o      |
|+ .o....       |
| . .oo        |
| .....       |
+-----[SHA256]-----+
[sysadmin@bootstrap ~]$
```

Create Your Installation

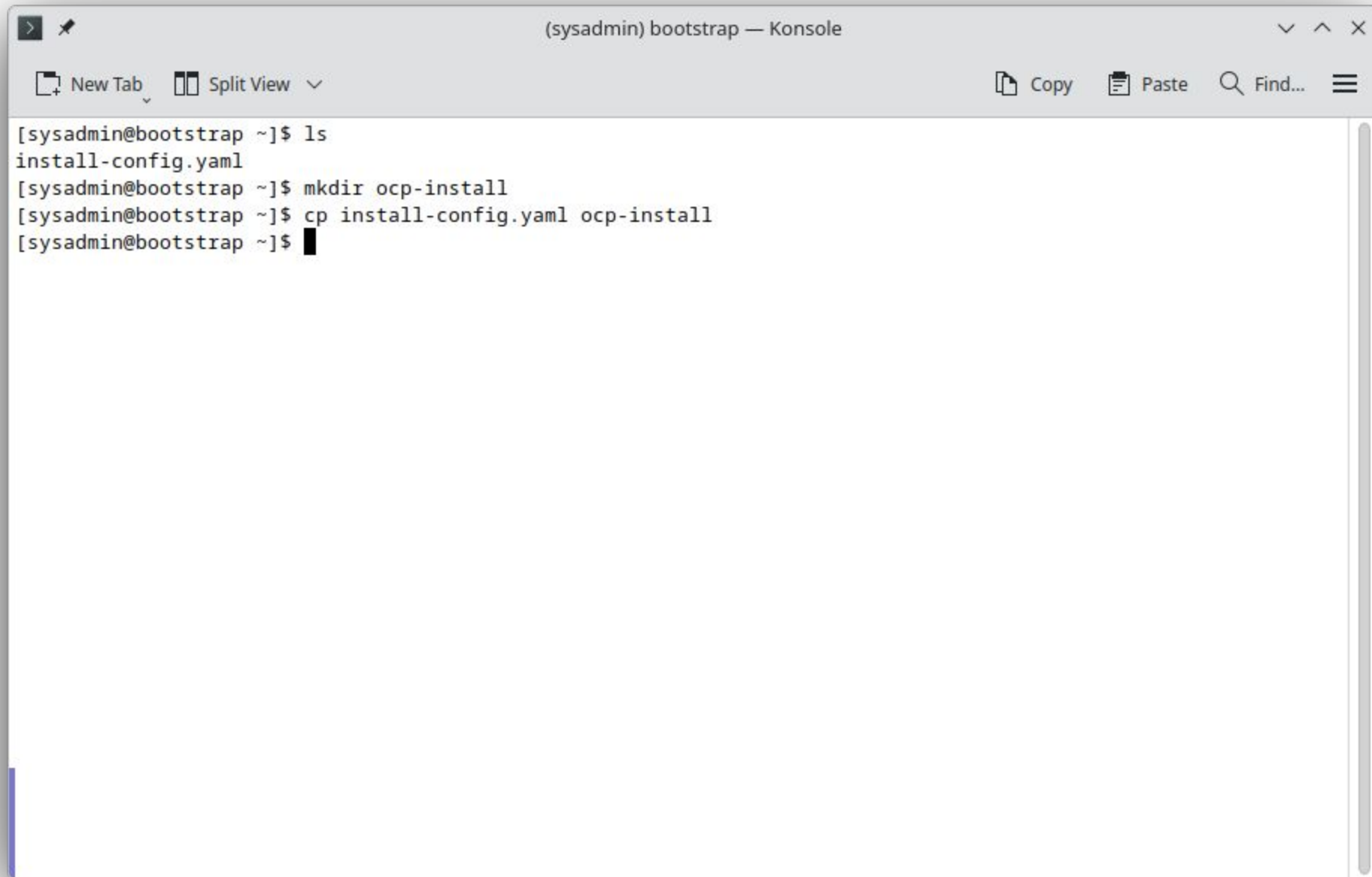
You can create an `install-config.yaml`

- I used a really simple one. It's in my github repo
- **NOTE:** when you run the installer, it will nuke your `install-config.yaml` file! COPY it, don't MOVE it to your installation directory!

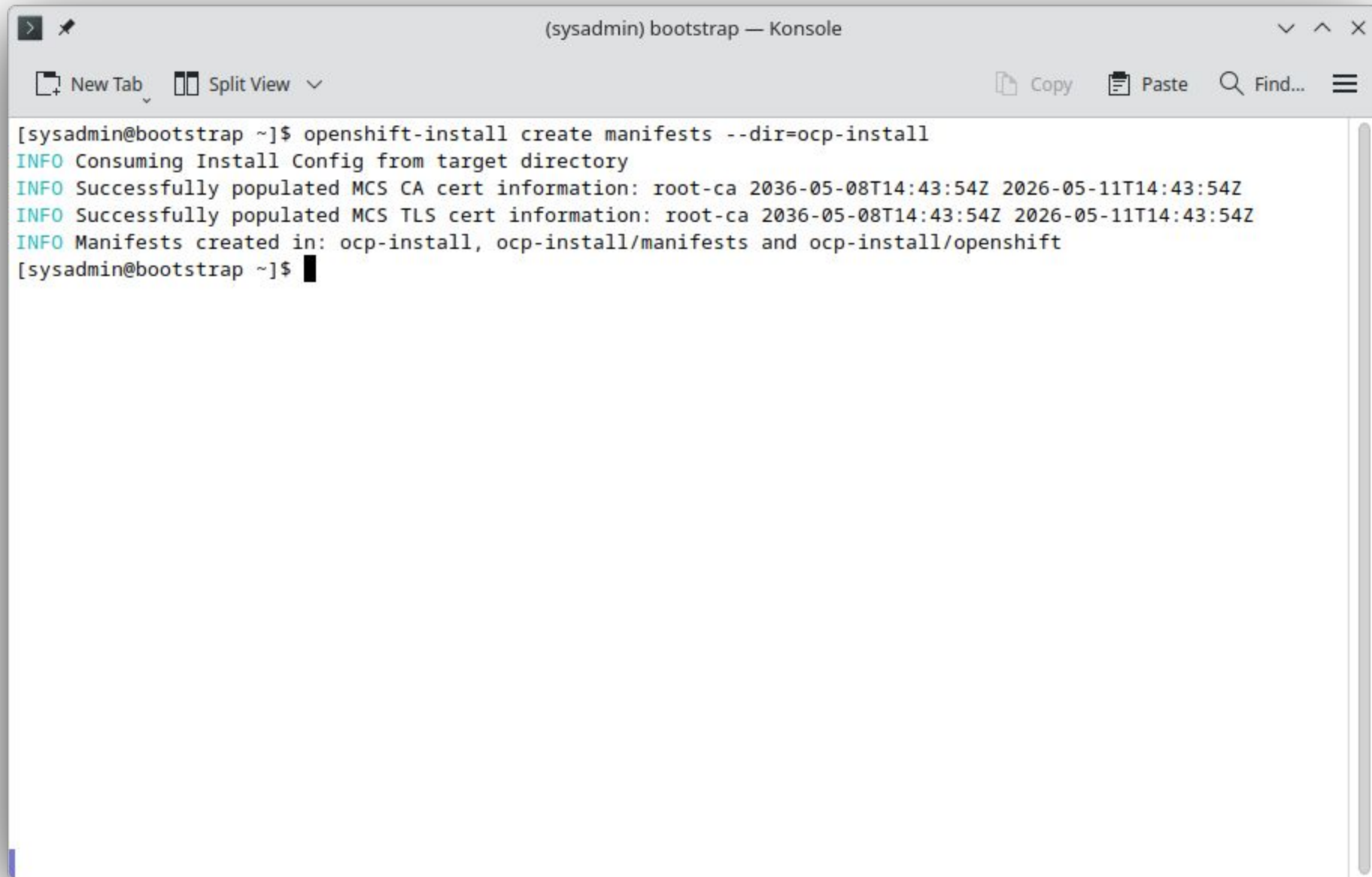


The screenshot shows a KWrite text editor window titled "Untitled (apiVersion v1) * — KWrite". The window contains a YAML configuration file for an OpenShift installation. The configuration includes fields for apiVersion, baseDomain, metadata, compute, controlPlane, networking, platform, fips, pullSecret, and sshKey. The networking section is highlighted with a light yellow background. The status bar at the bottom shows the time as 21:14, the mode as INSERT, the language as en_US, the number of soft tabs as 4, the encoding as UTF-8, and the font style as Normal.

```
1  apiVersion: v1
2  baseDomain: summit.aus.redhat.com
3  metadata:
4    name: ocp
5  compute:
6    - hyperthreading: Enabled
7      name: worker
8      replicas: 3 # ocp4, ocp5, ocp6
9  controlPlane:
10    hyperthreading: Enabled
11    name: master
12    replicas: 3 # ocp1, ocp2, ocp3
13  networking:
14    clusterNetwork:
15      - cidr: 10.128.0.0/14
16        hostPrefix: 23
17    machineNetwork:
18      - cidr: 10.31.100.0/24 # Your private node network
19    networkType: OVNKubernetes
20    serviceNetwork:
21      - 192.168.0.0/16
22  platform:
23    none: {}
24  fips: false
25  pullSecret: '{"auths":{"cloud.openshift.com":{"auth":"b3B1b <snip> FhRaFpDdw==","email":"tcameron@redhat.com"},"registry.redhat.io":
26    {"auth":"fHVoYy1wb29s <snip> aFpDdw==","email":"tcameron@redhat.com"}}}' # Paste your full secret here
27  sshKey: 'ssh-ed25519 AAAA <snip> MWvei tcameron@case.aus.redhat.com' # Your public key for node access
```

A terminal window titled "(sysadmin) bootstrap — Konsole" with a toolbar containing "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal shows the following commands and output:

```
[sysadmin@bootstrap ~]$ ls
install-config.yaml
[sysadmin@bootstrap ~]$ mkdir ocp-install
[sysadmin@bootstrap ~]$ cp install-config.yaml ocp-install
[sysadmin@bootstrap ~]$
```



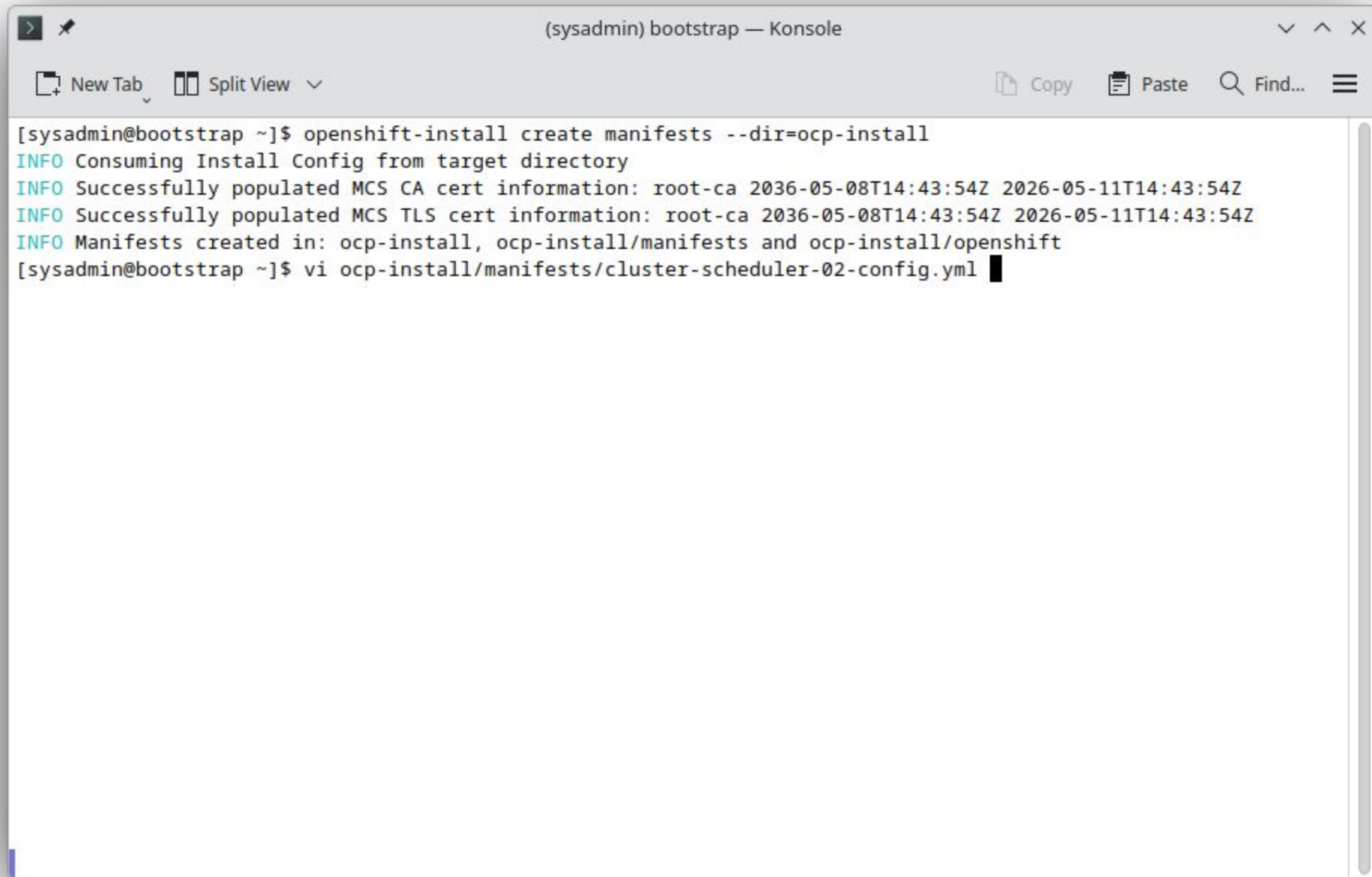
A terminal window titled "(sysadmin) bootstrap — Konsole" with standard window controls. The terminal shows the execution of the command `openshift-install create manifests --dir=ocp-install`. The output consists of four lines of informational messages, each starting with "INFO" in cyan. The messages indicate that the install config was consumed, MCS CA and TLS certificates were populated with specific dates, and manifests were created in the specified directory and its subdirectories. The prompt returns to the user after the command completes.

```
[sysadmin@bootstrap ~]$ openshift-install create manifests --dir=ocp-install
INFO Consuming Install Config from target directory
INFO Successfully populated MCS CA cert information: root-ca 2036-05-08T14:43:54Z 2026-05-11T14:43:54Z
INFO Successfully populated MCS TLS cert information: root-ca 2036-05-08T14:43:54Z 2026-05-11T14:43:54Z
INFO Manifests created in: ocp-install, ocp-install/manifests and ocp-install/openshift
[sysadmin@bootstrap ~]$
```

Set the Control Plane to Schedulable

By default, they are not

- Edit the manifests/cluster-scheduler-02-config.yml file



A terminal window titled "(sysadmin) bootstrap — Konsole" with standard window controls. The terminal shows the execution of the `openshift-install create manifests --dir=ocp-install` command. The output consists of four informational messages: consuming the install config, populating MCS CA and TLS cert information with specific timestamps, and creating manifests in the `ocp-install` directory and its subdirectories. The prompt then changes to `vi ocp-install/manifests/cluster-scheduler-02-config.yml`, where the cursor is positioned at the end of the line.

```
(sysadmin) bootstrap — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@bootstrap ~]$ openshift-install create manifests --dir=ocp-install
INFO Consuming Install Config from target directory
INFO Successfully populated MCS CA cert information: root-ca 2036-05-08T14:43:54Z 2026-05-11T14:43:54Z
INFO Successfully populated MCS TLS cert information: root-ca 2036-05-08T14:43:54Z 2026-05-11T14:43:54Z
INFO Manifests created in: ocp-install, ocp-install/manifests and ocp-install/openshift
[sysadmin@bootstrap ~]$ vi ocp-install/manifests/cluster-scheduler-02-config.yml
```

```
(sysadmin) bootstrap — Konsole

apiVersion: config.openshift.io/v1
kind: Scheduler
metadata:
  name: cluster
spec:
  mastersSchedulable: false
  policy:
    name: ""
  profileCustomizations:
    dynamicResourceAllocation: ""
status: {}

"ocp-install/manifests/cluster-scheduler-02-config.yml" 11L, 204B
```


[illegible]

```
(sysadmin) bootstrap — Konsole

New Tab Split View Copy Paste Find...

[sysadmin@bootstrap ~]$ tree ocp-install/
ocp-install/
├── 000_capi-namespace.yaml
├── manifests
│   ├── cluster-config.yaml
│   ├── cluster-dns-02-config.yml
│   ├── cluster-infrastructure-02-config.yml
│   ├── cluster-ingress-02-config.yml
│   ├── cluster-network-02-config.yml
│   ├── cluster-proxy-01-config.yaml
│   ├── cluster-scheduler-02-config.yml
│   ├── cvo-overrides.yaml
│   ├── kube-cloud-config.yaml
│   ├── kube-system-configmap-root-ca.yaml
│   ├── machine-config-server-ca-configmap.yaml
│   ├── machine-config-server-ca-secret.yaml
│   ├── machine-config-server-tls-secret.yaml
│   └── openshift-config-secret-pull-secret.yaml
└── openshift
    ├── 99_feature-gate.yaml
    ├── 99_kubeadmin-password-secret.yaml
    ├── 99_openshift-cluster-api_master-user-data-secret.yaml
    ├── 99_openshift-cluster-api_worker-user-data-secret.yaml
    ├── 99_openshift-machineconfig_99-master-ssh.yaml
    ├── 99_openshift-machineconfig_99-worker-ssh.yaml
    └── openshift-install-manifests.yaml

3 directories, 22 files
```

Create The Ignition Files

These will be used by the installer

- `openshift-install create ignition-configs --dir=ocp-install`
- Note that it deletes and creates stuff!

```
(sysadmin) bootstrap — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@bootstrap ~]$ openshift-install create ignition-configs --dir=ocp-install
INFO Consuming Openshift Manifests from target directory
INFO Consuming Common Manifests from target directory
INFO Consuming OpenShift Install (Manifests) from target directory
INFO Consuming Worker Machines from target directory
INFO Consuming Master Machines from target directory
INFO Ignition-Configs created in: ocp-install and ocp-install/auth
[sysadmin@bootstrap ~]$ tree ocp-install/
ocp-install/
├── 000_capi-namespace.yaml
├── auth
│   ├── kubeadmin-password
│   └── kubeconfig
├── bootstrap.ign
├── master.ign
├── metadata.json
└── worker.ign

2 directories, 7 files
[sysadmin@bootstrap ~]$
```

Fix The Permissions

When you copy the files, they may not be world-readable

- Check the permissions on the files

```
(sysadmin) bootstrap — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@bootstrap ~]$ ls -l ocp-install/
total 308
-rw-r-----. 1 sysadmin sysadmin      0 May 11 14:43 000_capi-namespace.yaml
drwxr-x---. 2 sysadmin sysadmin    50 May 11 15:02 auth
-rw-r-----. 1 sysadmin sysadmin 300113 May 11 15:02 bootstrap.ign
-rw-r-----. 1 sysadmin sysadmin  1727 May 11 15:02 master.ign
-rw-r-----. 1 sysadmin sysadmin   134 May 11 15:02 metadata.json
-rw-r-----. 1 sysadmin sysadmin  1727 May 11 15:02 worker.ign
[sysadmin@bootstrap ~]$ chmod 644 ocp-install/*.ign
[sysadmin@bootstrap ~]$ chmod 644 ocp-install/*.json
[sysadmin@bootstrap ~]$ chmod 755 ocp-install/auth/
[sysadmin@bootstrap ~]$ ls -l ocp-install/
total 308
-rw-r-----. 1 sysadmin sysadmin      0 May 11 14:43 000_capi-namespace.yaml
drwxr-xr-x. 2 sysadmin sysadmin    50 May 11 15:02 auth
-rw-r--r--. 1 sysadmin sysadmin 300113 May 11 15:02 bootstrap.ign
-rw-r--r--. 1 sysadmin sysadmin  1727 May 11 15:02 master.ign
-rw-r--r--. 1 sysadmin sysadmin   134 May 11 15:02 metadata.json
-rw-r--r--. 1 sysadmin sysadmin  1727 May 11 15:02 worker.ign
[sysadmin@bootstrap ~]$
```

Share The Installation Files

I put them on the workstation machine

- It is connected to both networks
- I installed httpd and started it
- I copied the installation content to `/var/www/html`

Index of /ocp-install

Index of /ocp-install - Google Chrome

workstation.aus.redhat.com/ocp-install/

☆

📁

👤

⋮

📁 ILO

📁 Personal

Index of /ocp-install

	Name	Last modified	Size	Description
📁	Parent Directory		-	
📄	000_capi-namespace.yaml	2026-05-11 10:51	0	
📁	auth/	2026-05-11 10:51	-	
📄	bootstrap.ign	2026-05-11 10:51	293K	
📄	master.ign	2026-05-11 10:51	1.7K	
📄	metadata.json	2026-05-11 10:51	134	
📄	worker.ign	2026-05-11 10:51	1.7K	

Download The CoreOS ISO

You can get them from

<https://mirror.openshift.com/pub/openshift-v4/dependencies/rhcos/latest/>

- I downloaded it to my hypervisor

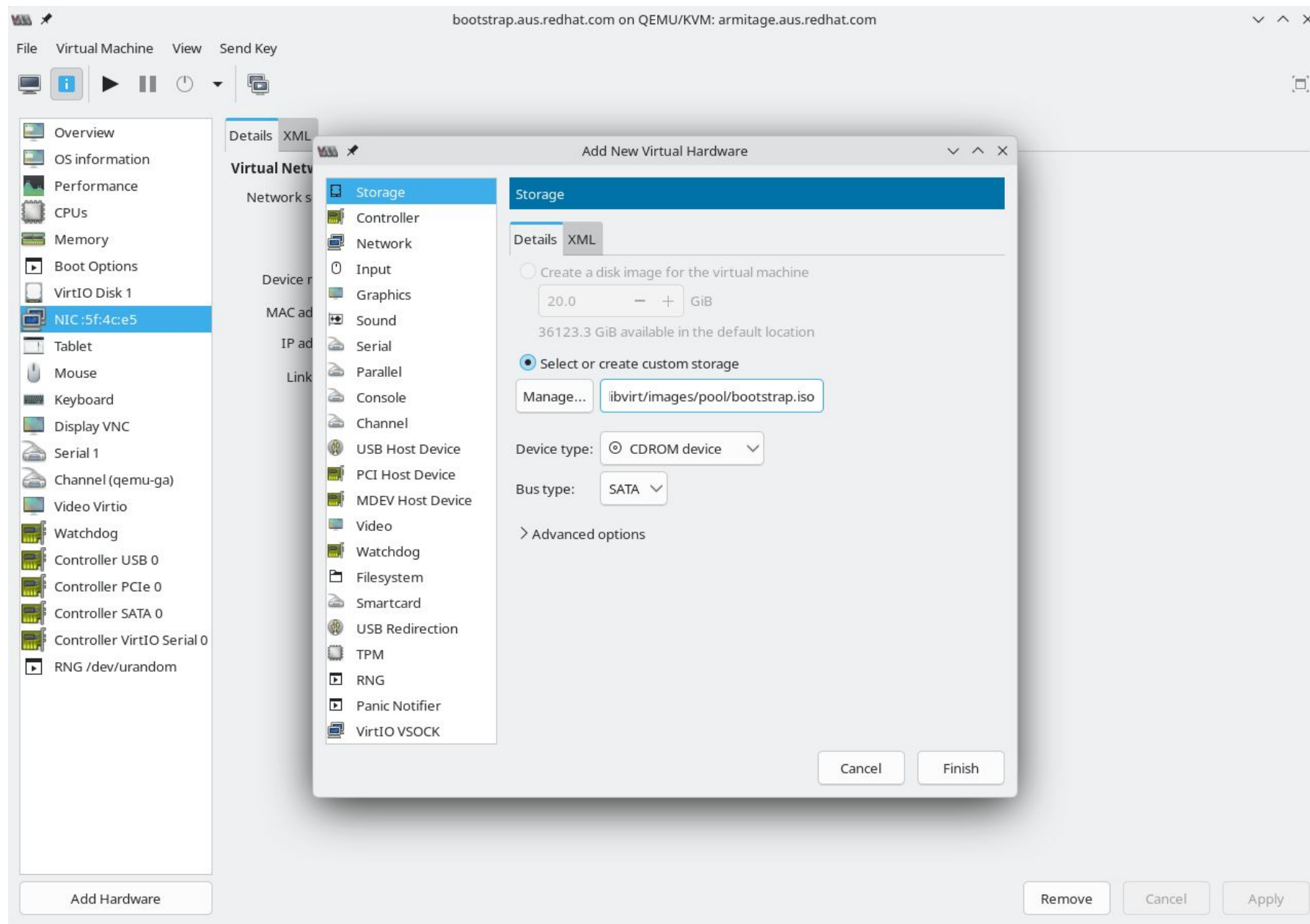
mirror.openshift.com/pub/openshift-v4/dependencies/rhcos/latest/

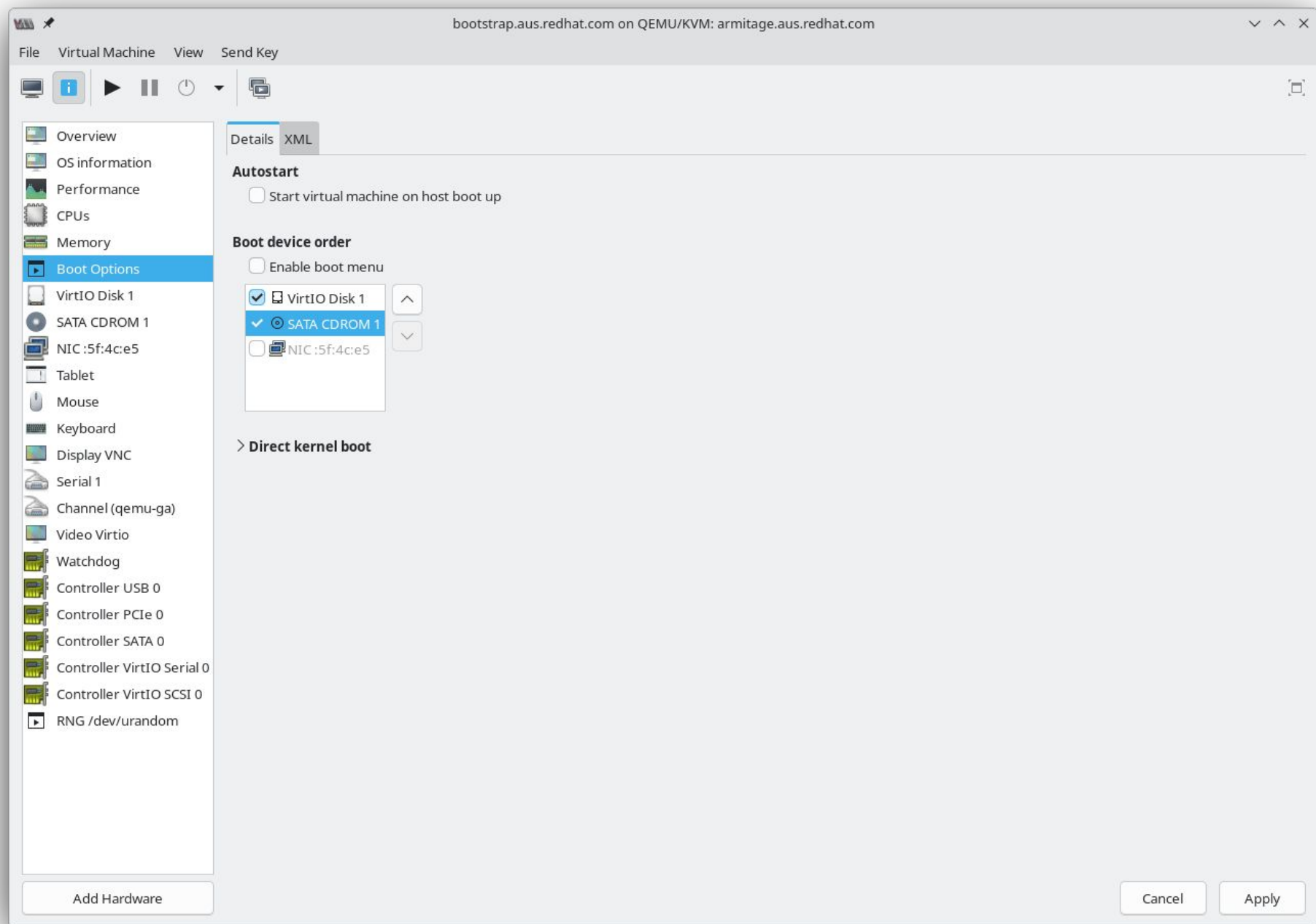
Name	Size	Modified
rhcos-4.21.0-x86_64-aws.x86_64.vmdk.gz	1 GB	Fri Feb 20 03:08:21 2026
rhcos-4.21.0-x86_64-azure.x86_64.vhd.gz	1 GB	Fri Feb 20 03:08:21 2026
rhcos-4.21.0-x86_64-azurestack.x86_64.vhd.gz	1 GB	Fri Feb 20 03:08:21 2026
rhcos-4.21.0-x86_64-gcp.x86_64.tar.gz	1 GB	Fri Feb 20 03:08:21 2026
rhcos-4.21.0-x86_64-ibmcloud.x86_64.qcow2.gz	1 GB	Fri Feb 20 03:08:21 2026
rhcos-4.21.0-x86_64-kubevirt.x86_64.ociarhive	1 GB	Fri Feb 20 03:08:36 2026
rhcos-4.21.0-x86_64-live-initramfs.x86_64.img	102 MB	Fri Feb 20 03:08:36 2026
rhcos-4.21.0-x86_64-live-iso.x86_64.iso	1 GB	Fri Feb 20 03:08:36 2026
rhcos-4.21.0-x86_64-live-kernel.x86_64	14 MB	Fri Feb 20 03:08:37 2026
rhcos-4.21.0-x86_64-live-rootfs.x86_64.img	1 GB	Fri Feb 20 03:08:37 2026
rhcos-4.21.0-x86_64-metal.x86_64.raw.gz	1 GB	Fri Feb 20 03:08:37 2026
rhcos-4.21.0-x86_64-metal4k.x86_64.raw.gz	1 GB	Fri Feb 20 03:08:37 2026
rhcos-4.21.0-x86_64-nutanix.x86_64.qcow2	1 GB	Fri Feb 20 03:08:51 2026
rhcos-4.21.0-x86_64-openstack.x86_64.qcow2.gz	1 GB	Fri Feb 20 03:08:52 2026
rhcos-4.21.0-x86_64-ostree.x86_64-manifest.json	17 KB	Fri Feb 20 03:08:52 2026
rhcos-4.21.0-x86_64-ostree.x86_64.ociarhive	1 GB	Fri Feb 20 03:08:52 2026
rhcos-4.21.0-x86_64-qemu.x86_64.qcow2.gz	1 GB	Fri Feb 20 03:08:53 2026
rhcos-4.21.0-x86_64-vmware.x86_64.ova	1 GB	Fri Feb 20 03:08:53 2026
rhcos-aws.x86_64.vmdk.gz	1 GB	Fri Feb 20 03:09:07 2026
rhcos-azure.x86_64.vhd.gz	1 GB	Fri Feb 20 03:09:07 2026
rhcos-azurestack.x86_64.vhd.gz	1 GB	Fri Feb 20 03:09:08 2026
rhcos-gcp.x86_64.tar.gz	1 GB	Fri Feb 20 03:09:09 2026
rhcos-ibmcloud.x86_64.qcow2.gz	1 GB	Fri Feb 20 03:09:09 2026
rhcos-ld.txt	15 bytes	Fri Feb 20 03:09:22 2026
rhcos-installer-initramfs.x86_64.img	102 MB	Fri Feb 20 03:09:22 2026
rhcos-installer-iso.x86_64.iso	1 GB	Fri Feb 20 03:09:22 2026
rhcos-installer-kernel.x86_64	14 MB	Fri Feb 20 03:09:23 2026
rhcos-installer-rootfs.x86_64.img	1 GB	Fri Feb 20 03:09:23 2026
rhcos-kubevirt.x86_64.ociarhive	1 GB	Fri Feb 20 03:09:24 2026
rhcos-live-initramfs.x86_64.img	102 MB	Fri Feb 20 03:09:24 2026
rhcos-live-iso.x86_64.iso	1 GB	Fri Feb 20 03:09:24 2026
rhcos-live-kernel.x86_64	14 MB	Fri Feb 20 03:09:25 2026
rhcos-live-rootfs.x86_64.img	1 GB	Fri Feb 20 03:09:25 2026
rhcos-metal.x86_64.raw.gz	1 GB	Fri Feb 20 03:09:38 2026

Mount the ISO On Each Machine

I created links from the download for each machine

- Mount it and make set the machine to boot off it





```
bootstrap.aus.redhat.com on QEMU/KVM: armitage.aus.redhat.com
File Virtual Machine View Send Key

Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Plow)
Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64

SSH host key: SHA256:7FuCrQTayQt1b5a6i5hW0IacKm12n1kHRZ6TpmFDhFM (ECDSA)
SSH host key: SHA256:jSxkQHov+Seg08u3C8LS0iCFE7fXNjCgiBUF5Txin5U (ED25519)
SSH host key: SHA256:39VD0xb1Fmnb2jdLTcPC1sKoY2/3+0avo/w+kjQAI0E (RSA)
enp7s0: 10.31.100.23 fe80::4ce6:8454:3416:a99
Ignition: ran on 2026/05/11 16:38:10 UTC (this boot)
Ignition: no config provided by user
bootstrap login: core (automatic login)

#####
Welcome to the CoreOS live environment. This system is running completely
from memory, making it a good candidate for hardware discovery and
installing persistently to disk. Here is an example of running an install
to disk via coreos-installer:

sudo coreos-installer install /dev/sda \
    --ignition-url https://example.com/example.ign

You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have
that configuration persist into the installed system by passing the
'--copy-network' argument to 'coreos-installer install'. Please run
'coreos-installer install --help' for more information on the possible
install options.
#####

[core@bootstrap ~]$ sudo coreos-installer install /dev/vda -I http://10.31.100.12/ocp-install/bootstrap.ign --insecure-ignition_
```

ocp1.summit.redhat.com File Virtual Machine View Send Key Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Plow) Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64 SSH host key: SHA256:•Hn2hsUdy1122tndNz0ct•GicAJ7InttLn56Lm08s (ECDSA) SSH host key: SHA256:Phao9•WU1Iq6Ck4uuhJLwqal16w4bu1YMT4phe3kMu (ED25519) SSH host key: SHA256:Kkkfn5KXP15kx1EH•8IuzgrtRj2Y88DQgd1r03zoca/E (RSA) emp1s0: 10.31.100.65 fe80::1d76:5b2d:19d:7292 Ignition: ran on 2026-05-11 16:46:29 UTC (this boot) Ignition: no config provided by user ocp1 login: core (automatic login) ===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp1 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install</pre>	ocp2.summit.aus.redhat.com on QEMU/KVM File Virtual Machine View Send Key Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Plow) Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64 SSH host key: SHA256:LfKZuFACs24tuEE2SpSynn1fxLSSEjW98W9s066seto (ECDSA) SSH host key: SHA256:6u2DF1spH1WpTdmmNLShCOhcKt6S18jPlfa2WU•2X8 (ED25519) SSH host key: SHA256:8t4uKzN3uafUDaawUWGBpcSDU4t9cRDmu9QR18HeS0aQ (RSA) emp1s0: 10.31.100.66 fe80::f83c:70bc:1bf2:b404 Ignition: ran on 2026-05-11 16:46:31 UTC (this boot) Ignition: no config provided by user ocp2 login: core (automatic login) ===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp2 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install</pre>	ocp3.summit.aus.redhat.com on QEMU/KVM: molly.aus.redhat.com File Virtual Machine View Send Key Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Plow) Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64 SSH host key: SHA256:5s082U/2Fpz7K•h5qp6Uq65mMsd1E5qU9aJVTUo2GpY (ECDSA) SSH host key: SHA256:21Y•XT1owFc022C10hrQ•U7HGRD01H2aBBKS5Jg•Cc (ED25519) SSH host key: SHA256:8jtuA5QnaS2UXTu/LrF17uPBM0pkL6••yALUtoGpWn (RSA) emp1s0: 10.31.100.67 fe80::4ed3:68dd:c5df:718f Ignition: ran on 2026-05-11 16:48:08 UTC (this boot) Ignition: no config provided by user ocp3 login: core (automatic login) ===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp3 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install/master.ign --insecure-ignition</pre>
===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp4 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install</pre>	===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp5 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install</pre>	===== Welcome to the CoreOS live environment. This system is running completely from memory, making it a good candidate for hardware discovery and installing persistently to disk. Here is an example of running an install to disk via coreos-installer: <pre>sudo coreos-installer install /dev/sda \ --ignition-url https://example.com/example.ign</pre> You may configure networking via 'sudo nmcli' or 'sudo nmtui' and have that configuration persist into the installed system by passing the '--copy-network' argument to 'coreos-installer install'. Please run 'coreos-installer install --help' for more information on the possible install options. ===== <pre>[core@ocp6 ~]\$ sudo coreos-installer install /dev/uda -I http://10.31.100.12/ocp-install/worker.ign --insecure-ignition</pre>

Reboot

Once the initial installation is done

- Reboot the machines once they complete the **initial** installation


```
ocp1.summit.redhat.com
File Virtual Machine View Send Key
Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Flow)
Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64
SSH host key: SHA256:rUUMsdhpipchc3SnDp00UuIT+HhbcKlIE8LassRSdNk (ECDSA)
SSH host key: SHA256:scXMcZa2h5HF9u9KQ05Um6VEpfqkIM4pBuEsyeY8 (ED25519)
SSH host key: SHA256:gK/e0DctPArGHUoFhLKrTU9JuLuEtMgQkzySba/JLSQ (RSA)
emp1s0: 10.31.100.65 fe80::18af:efdf:ef86:ffe6
Ignition: ran on 2026/05/11 17:12:17 UTC (this boot)
Ignition: user-provided config was applied
ocp1 login: Starting rpm-ostree System Management Daemon...
[ OK ] Started rpm-ostree System Management Daemon.
Starting Hold /boot Open for OSTree Finalize Staged Deployment.
[ OK ] Started Hold /boot Open for OSTree Finalize Staged Deployment.
[ OK ] Finished OSTree Finalize Staged Deployment.
```

```
ocp2.summit.aus.redhat.com on QEMU/KVM
File Virtual Machine View Send Key
Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Flow)
Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64
SSH host key: SHA256:KgTx5gJla+9/z+/P667nJwrcH39Im9WPhn085M7bMg (ECDSA)
SSH host key: SHA256:SA/M5Gd2FXBIFeRULAnruKEAmcX7pS+nSmmu2Y2xSEI (ED25519)
SSH host key: SHA256:IBLLrTTSheuKPDs/USCFxlmudZor4BMGsGSu8Xf4x4 (RSA)
emp1s0: 10.31.100.66 fe80::1851:5053:ce15:ebf9
Ignition: ran on 2026/05/11 17:12:19 UTC (this boot)
Ignition: user-provided config was applied
ocp2 login: Starting rpm-ostree System Management Daemon...
[ OK ] Started rpm-ostree System Management Daemon.
```

```
ocp3.summit.aus.redhat.com on QEMU/KVM: molly.aus.redhat.com
File Virtual Machine View Send Key
Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Flow)
Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64
SSH host key: SHA256:xxIM1fKa4fMxq3HUTSz/16+Udo0NZQralB0rfDilaHQ (ECDSA)
SSH host key: SHA256:kPhrQsAdf4tMzJz1a5320htfJbu81CG5ZF1hlox98 (ED25519)
SSH host key: SHA256:0Jun8Q6gPGCuMb88Kcx6Sh+J3pygmDfeE3SEw8JJB50 (RSA)
emp1s0: 10.31.100.67 fe80::4852:2871:de0f:5fd7
Ignition: ran on 2026/05/11 17:12:22 UTC (this boot)
Ignition: user-provided config was applied
ocp3 login: Starting rpm-ostree System Management Daemon...
[ OK ] Started rpm-ostree System Management Daemon.
```

```
bootstrap.aus.redhat.com on QEMU/KVM: armitage.aus.redhat.com
File Virtual Machine View Send Key
Red Hat Enterprise Linux CoreOS 9.6.20251212-1 (Flow)
Kernel 5.14.0-570.74.1.el9_6.x86_64 on an x86_64
emp7s0: 10.31.100.23 fe80::2d07:34ed:11d:c9b0
bootstrap login: [ 109.144237] Warning: Unmaintained driver is detected: nft_compat
```

```
ocp4.summit.aus.redhat.com
File Virtual Machine View Send Key
ig/worker: attempt #22
[ 93.594429] ignition[8721]: GET re
[ *** ] A start job is running for
[ 98.599367] ignition[8721]: GET ht
3/config/worker: attempt #23
[ * ] A start job is running for
[ 103.601992] ignition[8721]: GET ht
3/config/worker: attempt #24
[ *** ] A start job is running for
[ 108.608923] ignition[8721]: GET ht
3/config/worker: attempt #25
[ ** ] A start job is running for
[ 113.615915] ignition[8721]: GET ht
3/config/worker: attempt #26
[ *** ] A start job is running for
[ 118.622552] ignition[8721]: GET ht
3/config/worker: attempt #27
[ *** ] A start job is running for
[ 123.629451] ignition[8721]: GET ht
3/config/worker: attempt #28
[ ** ] A start job is running for
[ 128.632241] ignition[8721]: GET ht
3/config/worker: attempt #29
[ ** ] A start job is running for
```

```
on QEMU/KVM: neuromancer.aus.redhat.com
File Virtual Machine View Send Key
gnition (fetch) (1min 30s / no limit)[ 93
pi-int.ocp.summit.aus.redhat.com:22623/conf
ult: Internal Server Error
gnition (fetch) (1min 35s / no limit)
ps://api-int.ocp.summit.aus.redhat.com:2262
gnition (fetch) (1min 40s / no limit)
ps://api-int.ocp.summit.aus.redhat.com:2262
gnition (fetch) (1min 45s / no limit)
ps://api-int.ocp.summit.aus.redhat.com:2262
gnition (fetch) (1min 50s / no limit)
ps://api-int.ocp.summit.aus.redhat.com:2262
gnition (fetch) (1min 55s / no limit)
[ 118.872972] ignition[8801]: GET https://api-int.ocp.summit.aus.redhat.com:2262
3/config/worker: attempt #27
[ *** ] A start job is running for Ignition (fetch) (2min / no limit)
[ 123.875987] ignition[8801]: GET https://api-int.ocp.summit.aus.redhat.com:2262
3/config/worker: attempt #28
[ ** ] A start job is running for Ignition (fetch) (2min 2s / no limit)
```

Watch the bootstrap

Run this command on your workstation

- `openshift-install wait-for bootstrap-complete --dir=ocp-install --log-level=debug`
- You'll want to take a break for this!

```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

[sysadmin@workstation ~]$ openshift-install wait-for bootstrap-complete --dir=ocp-install --log-level=debug
DEBUG OpenShift Installer 4.21.14
DEBUG Built from commit a8bea03c72112c0f859af3694676da9483baec99
INFO Waiting up to 20m0s (until 12:33PM CDT) for the Kubernetes API at https://api.ocp.summit.aus.redhat.com:6443...
DEBUG Loading Agent Config...
INFO API v1.34.6 up
DEBUG Loading Install Config...
DEBUG Loading SSH Key...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Cluster Name...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Pull Secret...
DEBUG Loading Platform...
DEBUG Using Install Config loaded from state file
INFO Waiting up to 45m0s (until 12:58PM CDT) for bootstrapping to complete...
█
```

```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

[sysadmin@workstation ~]$ openshift-install wait-for bootstrap-complete --dir=ocp-install --log-level=debug
DEBUG OpenShift Installer 4.21.14
DEBUG Built from commit a8bea03c72112c0f859af3694676da9483baec99
INFO Waiting up to 20m0s (until 1:30PM CDT) for the Kubernetes API at https://api.ocp.summit.aus.redhat.com:6443...
DEBUG Loading Agent Config...
INFO API v1.34.6 up
DEBUG Loading Install Config...
DEBUG Loading SSH Key...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Cluster Name...
DEBUG Loading Base Domain...
DEBUG Loading Platform...
DEBUG Loading Pull Secret...
DEBUG Loading Platform...
DEBUG Using Install Config loaded from state file
INFO Waiting up to 45m0s (until 1:55PM CDT) for bootstrapping to complete...
DEBUG Bootstrap status: complete
E0511 13:10:28.120499 2278 reflector.go:205] "Failed to watch" err="Get \"https://api.ocp.summit.aus.redhat.com:6443/api/v1/namespaces/kube-system/configmaps?allowWatchBookmarks=true&fieldSelector=metadata.name%3Dbootstrap&resourceVersion=32902&timeoutSeconds=525&watch=true\": context canceled" logger="UnhandledError" reflector="k8s.io/client-go/tools/watch/informerwatcher.go:162" type="*v1.ConfigMap"
INFO Waiting for the bootstrap etcd member to be removed...
INFO Bootstrap etcd member has been removed
INFO It is now safe to remove the bootstrap resources
INFO Time elapsed: 0s
[sysadmin@workstation ~]$
```



```
pid = 4993 (process #1, nbproc = 1, nbthread = 4)
uptime = 0d 0h00m03s; warnings = 20
system limits: memmax = unlimited; ulimit-n = 40053
maxsock = 40053; maxconn = 20000; reached = 0; maxpipes = 0
current conns = 43; current pipes = 0; conn rate = 11/sec; bit rate = 1.418 Mbps
Running tasks: 0/63 (0 nice), idle = 98 %
```

active UP
active UP, going down
active DOWN, going up
active or backup DOWN
active or backup DOWN for maintenance (MAINT)
active or backup SOFT STOPPED for maintenance

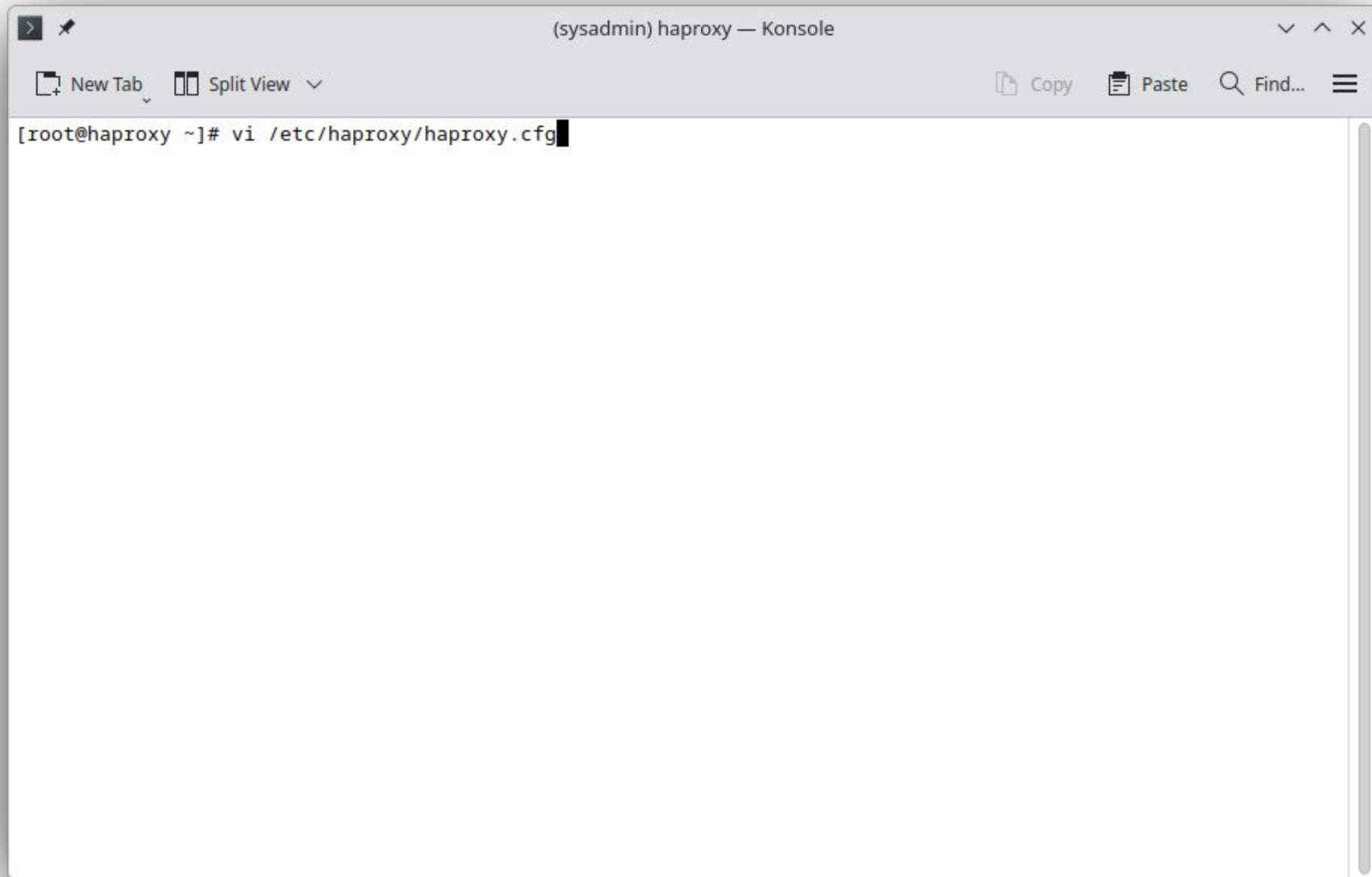
backup UP
backup UP, going down
backup DOWN, going up
not checked

Note: "NOLB"/"DRAIN" = UP with load-balancing disabled

- Scope :
- [Hide 'DOWN' servers](#)
- [Disable refresh](#)
- [Refresh now](#)
- [CSV export](#)
- [JSON export \(schemas\)](#)

- [Primary site](#)
- [Updates \(v3.0\)](#)
- [Online manual](#)

tcp_mips_ingress_backend		Queue		Session rate				Sessions				Bytes				Denied				Errors				Warnings				Server					
	Curr	Max	Limit	Curr	Max	Limit	Curr	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dumtme	Thrtle			
ocq4	0	0	-	1	1	1	0	Max	1	-	2	1	1s	0	0	0	0	0	0	0	1	1	2s DOWN	LACON in Desc	1/1	Y	-	1	1	2s	-		
ocq5	0	0	-	0	0	0	0	0	0	-	0	?	0	0	0	0	0	0	0	0	0	0	2s DOWN	LACON in Desc	1/1	Y	-	1	1	2s	-		
ocq6	0	0	-	1	2	0	0	1	0	5	2	3s	0	0	0	0	0	0	0	3	2	1s DOWN	LACON in Desc	1/1	Y	-	1	1	1s	-			
Backend	0	0	-	0	0	0	0	0	0	3,000	0	0	4.54s	0	0	0	0	0	0	0	0			0	0	0	0	0	0	0	0		



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The toolbar includes "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal text shows a root user at the haproxy host running the command `vi /etc/haproxy/haproxy.cfg`, with the cursor positioned at the end of the command.

```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
[root@haproxy ~]# vi /etc/haproxy/haproxy.cfg
```

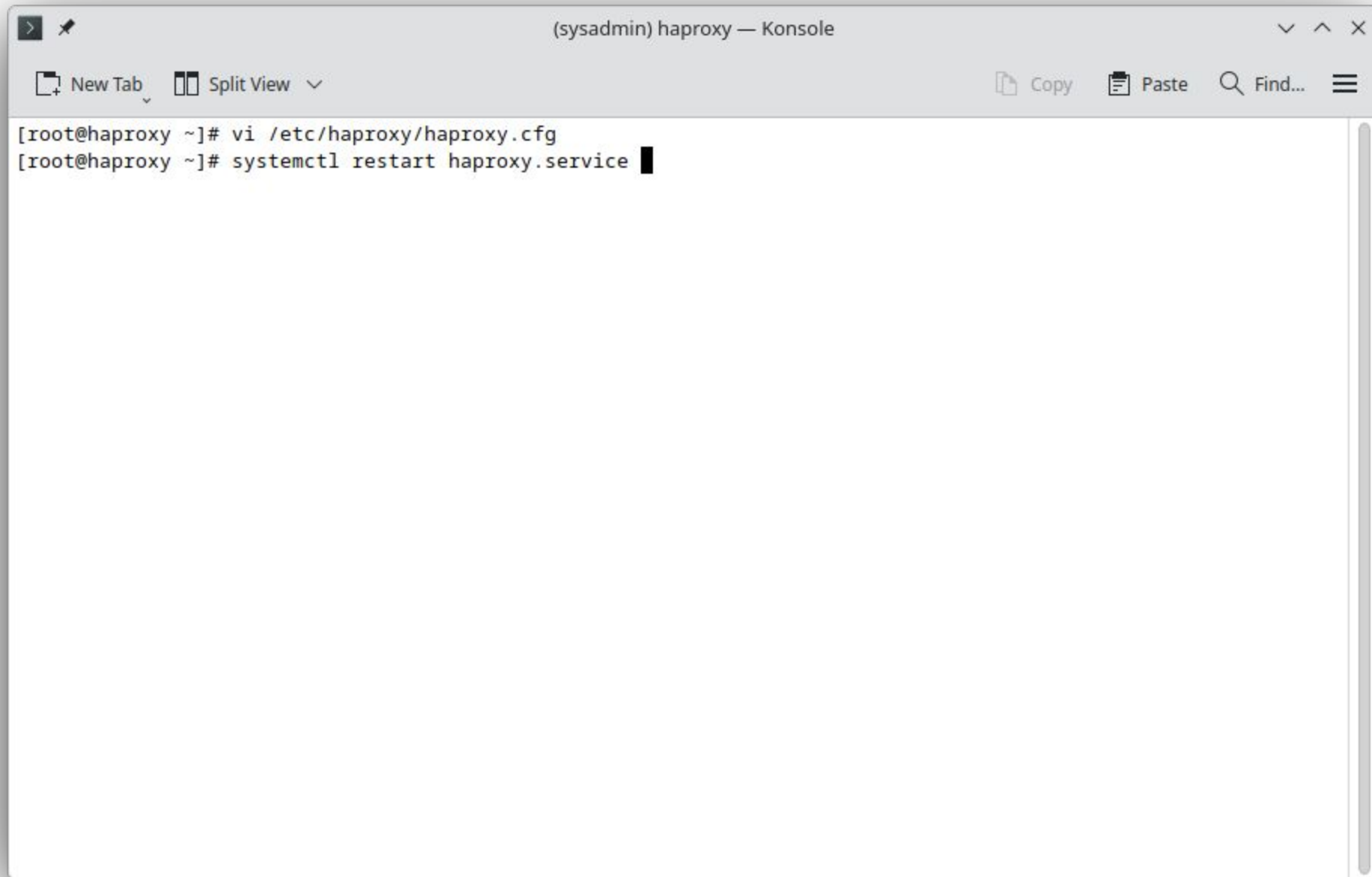
```
(sysadmin) haproxy — Konsole
New Tab Split View Copy Paste Find...
mode tcp
backend k8s_api_backend
  mode tcp
  balance source
#   server    bootstrap 10.31.100.23:6443 check
  server    ocp1 10.31.100.65:6443 check
  server    ocp2 10.31.100.66:6443 check
  server    ocp3 10.31.100.67:6443 check

# OCP Machine Config Server
frontend ocp_machine_config_server_frontend
  mode tcp
  bind :22623
  default_backend ocp_machine_config_server_backend

backend ocp_machine_config_server_backend
  mode tcp
  balance source
#   server    bootstrap 10.31.100.23:22623 check
  server    ocp1 10.31.100.65:22623 check
  server    ocp2 10.31.100.66:22623 check
  server    ocp3 10.31.100.67:22623 check

# OCP Ingress - layer 4 tcp mode for each. Ingress Controller will handle layer 7.
frontend ocp_http_ingress_frontend
  bind :80
```

66,1 66%



A terminal window titled "(sysadmin) haproxy — Konsole" with standard window controls. The toolbar includes "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal content shows two commands executed in a root shell on a machine named haproxy: editing the configuration file and restarting the service.

```
[root@haproxy ~]# vi /etc/haproxy/haproxy.cfg
[root@haproxy ~]# systemctl restart haproxy.service
```


Post Install

Set Up Access

To talk to the cluster, set your KUBECONFIG variable

- `export KUBECONFIG=/home/sysadmin/ocp-install/auth/kubeconfig`
- Then make sure you can “see” the cluster

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ export KUBECONFIG=/home/sysadmin/ocp-install/auth/kubeconfig
[sysadmin@workstation ~]$ oc get nodes
NAME     STATUS    ROLES                                AGE    VERSION
ocp1     Ready     control-plane,master,worker         71m    v1.34.6
ocp2     Ready     control-plane,master,worker         71m    v1.34.6
ocp3     Ready     control-plane,master,worker         71m    v1.34.6
[sysadmin@workstation ~]$ oc whoami
system:admin
[sysadmin@workstation ~]$
```

Approve All The Certificate Signing Requests

As the cluster comes up, it creates a ton of self signed certificates

- So a human needs to approve the requests
- We'll be pretty ham-handed and approve all of them

```
oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | \
xargs --no-run-if-empty oc adm certificate approve
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ oc get csr
NAME                                     AGE    SIGNERNAME                                     REQUESTEDDURATION    CONDITION    REQUESTOR
csr-4nt6x                               45m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-4rwxl                               15m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-54sgr                               60m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-6flbw                               78m    kubernetes.io/kubelet-serving                  <none>               Approved, Issued    system:no
de:ocp3
csr-8fp99                               15m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-crgkl                               75m    kubernetes.io/kube-apiserver-client            24h                 Approved, Issued    system:no
de:ocp2
csr-dc4r5                               75m    kubernetes.io/kube-apiserver-client            24h                 Approved, Issued    system:no
de:ocp1
csr-dg4ml                               75m    kubernetes.io/kube-apiserver-client            24h                 Approved, Issued    system:no
de:ocp3
csr-fhmkq                               14m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-fk5lt                               60m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-gzp7c                               30m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-jkhk8                               78m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Approved, Issued    system:se
viceaccount:openshift-machine-config-operator:node-bootstrapper
csr-lgkbs                               30m    kubernetes.io/kube-apiserver-client-kubelet    <none>               Pending      system:se
```



```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

[sysadmin@workstation ~]$ oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | \
xargs --no-run-if-empty oc adm certificate approve
certificatesigningrequest.certificates.k8s.io/csr-4nt6x approved
certificatesigningrequest.certificates.k8s.io/csr-4rwx1 approved
certificatesigningrequest.certificates.k8s.io/csr-54sgr approved
certificatesigningrequest.certificates.k8s.io/csr-8fp99 approved
certificatesigningrequest.certificates.k8s.io/csr-fhmkq approved
certificatesigningrequest.certificates.k8s.io/csr-fk5lt approved
certificatesigningrequest.certificates.k8s.io/csr-gzp7c approved
certificatesigningrequest.certificates.k8s.io/csr-lgkbs approved
certificatesigningrequest.certificates.k8s.io/csr-mkqwz approved
certificatesigningrequest.certificates.k8s.io/csr-mlwt5 approved
certificatesigningrequest.certificates.k8s.io/csr-rskwb approved
certificatesigningrequest.certificates.k8s.io/csr-sclxj approved
certificatesigningrequest.certificates.k8s.io/csr-vg6kx approved
certificatesigningrequest.certificates.k8s.io/csr-vtmcw approved
certificatesigningrequest.certificates.k8s.io/csr-vv7cp approved
[sysadmin@workstation ~]$
```

Approve All The Certificate Signing Requests

As the cluster comes up, it creates a ton of self signed certificates

- Do this until all are approved


```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

[sysadmin@workstation ~]$ oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | xargs --no-run-if-empty oc adm certificate approve
certificatesigningrequest.certificates.k8s.io/csr-9prt2 approved
certificatesigningrequest.certificates.k8s.io/csr-d8bqn approved
certificatesigningrequest.certificates.k8s.io/csr-n245f approved
[sysadmin@workstation ~]$ oc get csr -o go-template='{{range .items}}{{if not .status}}{{.metadata.name}}{{"\n"}}{{end}}{{end}}' | xargs --no-run-if-empty oc adm certificate approve
[sysadmin@workstation ~]$
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ oc get nodes
NAME      STATUS   ROLES                                AGE      VERSION
ocp1      Ready    control-plane,master,worker         88m      v1.34.6
ocp2      Ready    control-plane,master,worker         88m      v1.34.6
ocp3      Ready    control-plane,master,worker         88m      v1.34.6
ocp4      Ready    worker                              3m54s    v1.34.6
ocp5      Ready    worker                              3m54s    v1.34.6
ocp6      Ready    worker                              3m57s    v1.34.6
[sysadmin@workstation ~]$
```


Statistics Report for pid 5030

```
pid = 5030 (process #1, nbgproc = 1, nbthread = 4)
uptime = 6d 11h0m00s, warnings = 24
system limits: memmax = unlimited, ulim_n = 40051
maxsock = 40051, maxconn = 20000, reached = 0, maxpipes = 0
current cores = 57, current pipes = 0,0; rate = 29sec, bit rate = 148.463 kbps
Running tasks: 0/105 (0 nice'd), idle = 100 %
```

Note: "NOLB"/"DRAIN" = UP with load-balancing disabled.

Mls_api_frontend																															
	Queue			Session rate			Sessions						Bytes				Req		Derrd		Errors		Warnings		Server						
	Cur	Max	Limit	Cur	Max	Limit	Cur	Max	Limit	Total	LoTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Rtrr	Redis	Status	LastChk	Wght	Act	Bck	Chk	Dwn	Dwrtn	Thrtle	
Frontend				0	104	--	63	120	20 000	5 008			26 688 914	247 754 996	0	0	0	0	0			OPEN									

[illegible][illegible]

ocp_https_ingress_frontend																																
	Queue				Session rate			Sessions					Bytes				Denied			Errors			Warnings		Status	Server						
	Cur	Max	Limit		Cur	Max	Limit	Cur	Max	Limit	Total	LbTot	Last	In	Out	Req	Resp	Req	Conn	Resp	Retr	Redir	LastChk	Wght		Act	Bck	Chk	Dwn	Dwnrtn	Thrle	
Frontend					0	9	-	0		5	20 000	3 319		173 148	276 514	0	0	0	0				OPEN									

Set Up NFS

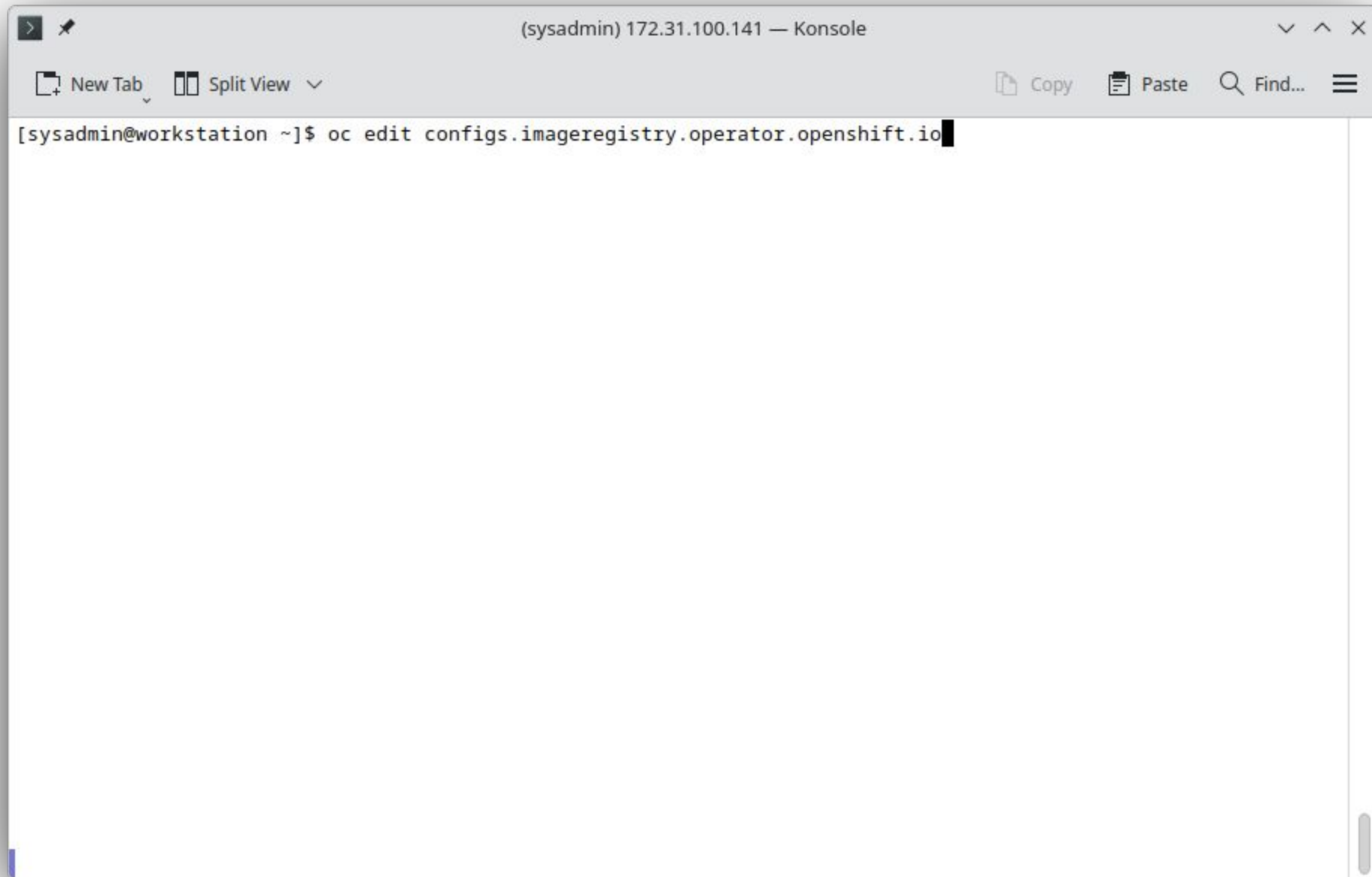


Red Hat

Tell OpenShift about Your Storage

Edit `configs.imageregistry.operator.openshift.io`

- `oc edit configs.imageregistry.operator.openshift.io`



The image shows a Konsole terminal window with the title "(sysadmin) 172.31.100.141 — Konsole". The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal prompt is "[sysadmin@workstation ~]\$". The command entered is "oc edit configs.imageregistry.operator.openshift.io".

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ oc edit configs.imageregistry.operator.openshift.io
```

```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: imageregistry.operator.openshift.io/v1
kind: Config
metadata:
  creationTimestamp: "2026-05-11T17:31:37Z"
  finalizers:
  - imageregistry.operator.openshift.io/finalizer
  generation: 1
  name: cluster
  resourceVersion: "39442"
  uid: e5e2908b-08d1-46c8-9c04-9513997d2282
spec:
  logLevel: Normal
  managementState: Removed
  observedConfig: null
  operatorLogLevel: Normal
  proxy: {}
  replicas: 1
  requests:
    read:
      maxWaitInQueue: 0s
    write:
      maxWaitInQueue: 0s
  rolloutStrategy: RollingUpdate
"/tmp/oc-edit-3989935609.yaml" 88L, 2513B 17,19 Top
```



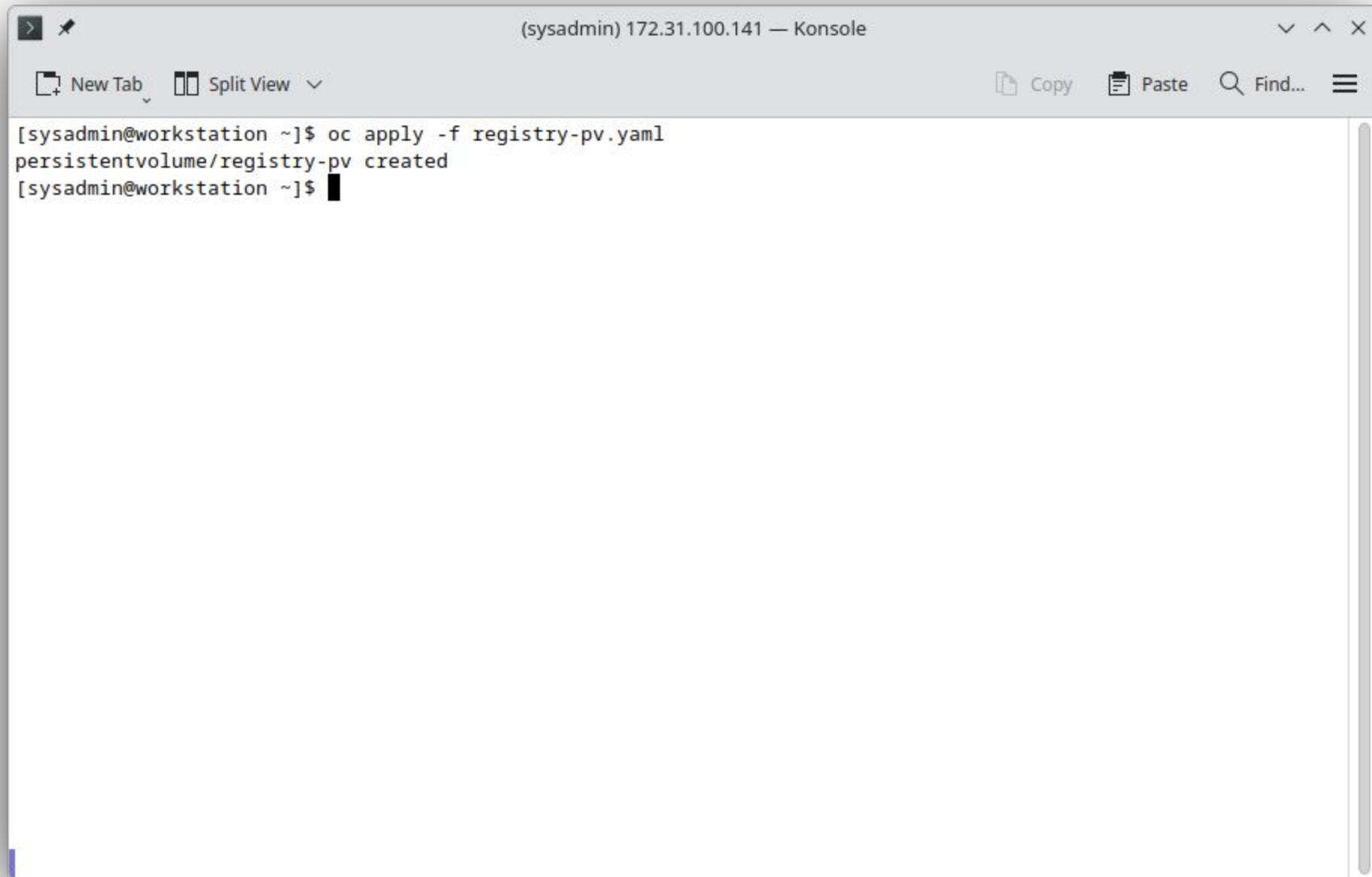
```
(sysadmin) 172.31.100.141 — Konsole

New Tab Split View Copy Paste Find...

# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: imageregistry.operator.openshift.io/v1
kind: Config
metadata:
  creationTimestamp: "2026-05-11T17:31:37Z"
  finalizers:
    - imageregistry.operator.openshift.io/finalizer
  generation: 1
  name: cluster
  resourceVersion: "39442"
  uid: e5e2908b-08d1-46c8-9c04-9513997d2282
spec:
  logLevel: Normal
  managementState: Removed
  observedConfig: null
  operatorLogLevel: Normal
  proxy: {}
  replicas: 1
  requests:
    read:
      maxWaitInQueue: 0s
    write:
      maxWaitInQueue: 0s
  rolloutStrategy: RollingUpdate
"/tmp/oc-edit-3989935609.yaml" 88L, 2513B 17,19 Top
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ oc edit configs.imageregistry.operator.openshift.io
config.imageregistry.operator.openshift.io/cluster edited
[sysadmin@workstation ~]$ oc get pvc -n openshift-image-registry
NAME                                STATUS    VOLUME    CAPACITY    ACCESS MODES    STORAGECLASS    VOLUMEATTRIBUTESCLASS    A
GE
image-registry-storage    Pending                                <unset>                                4
4s
[sysadmin@workstation ~]$
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ cat registry-pv.yaml
apiVersion: v1
kind: PersistentVolume
metadata:
  name: registry-pv
spec:
  accessModes:
    - ReadWriteMany
  capacity:
    storage: 100Gi
  persistentVolumeReclaimPolicy: Retain
  nfs:
    path: /var/export/registry
    server: 10.31.100.64
[sysadmin@workstation ~]$
```



A terminal window titled "(sysadmin) 172.31.100.141 — Konsole". The window has a toolbar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a menu icon. The terminal content shows a command being executed and its output.

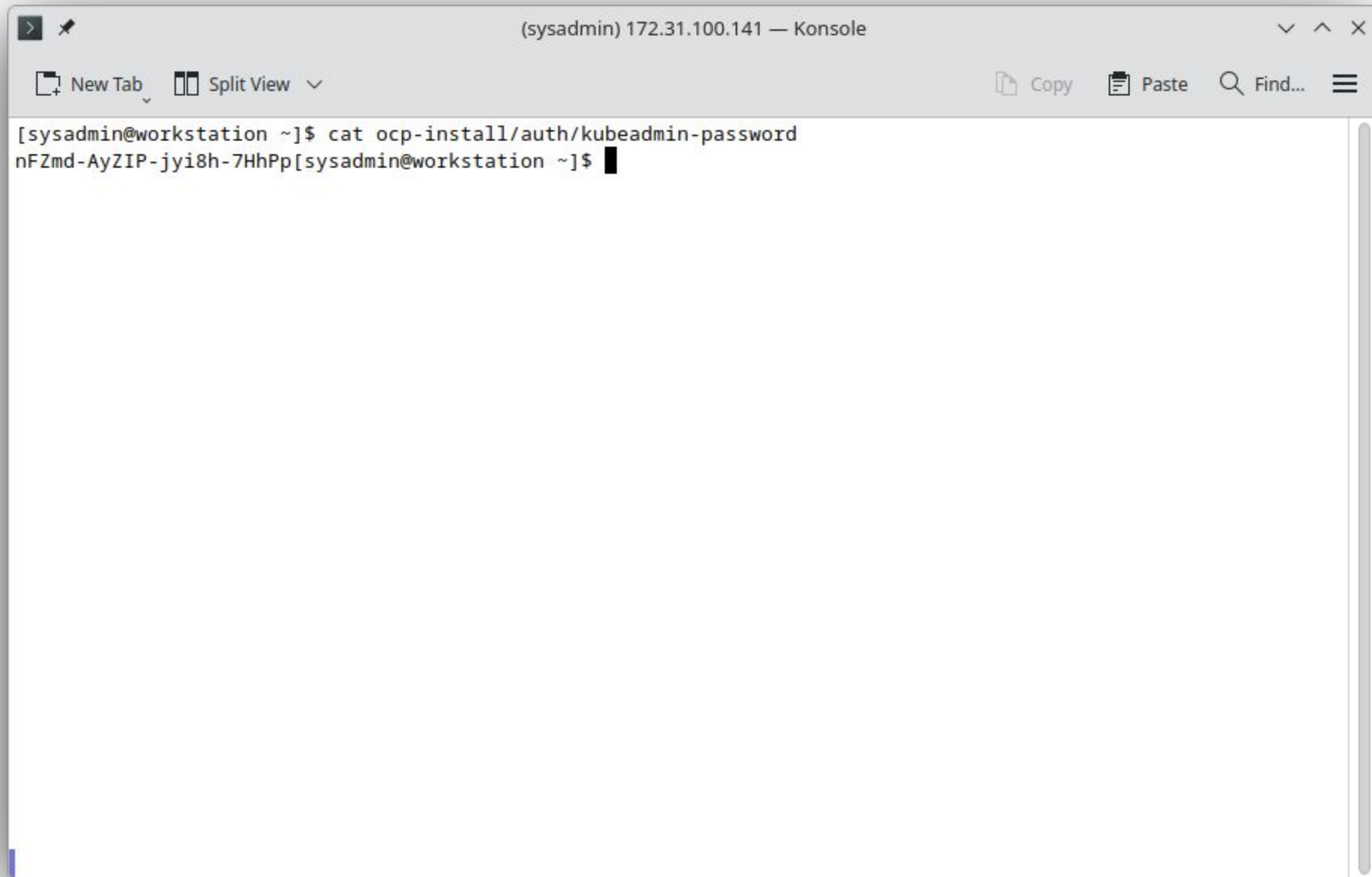
```
[sysadmin@workstation ~]$ oc apply -f registry-pv.yaml
persistentvolume/registry-pv created
[sysadmin@workstation ~]$
```

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ oc get pvc -n openshift-image-registry
NAME                                STATUS  VOLUME      CAPACITY  ACCESS MODES  STORAGECLASS  VOLUMEATTRIBUTESCLASS
image-registry-storage             Bound   registry-pv  100Gi     RWX                                     <unset>
8m21s
[sysadmin@workstation ~]$
```

Log In!



Red Hat



A terminal window titled "(sysadmin) 172.31.100.141 — Konsole". The window has a menu bar with "New Tab", "Split View", "Copy", "Paste", "Find...", and a hamburger menu icon. The terminal content shows a command being executed: `[sysadmin@workstation ~]$ cat ocp-install/auth/kubeadmin-password`. The output of the command is displayed on the next line: `nFZmd-AyZIP-jyi8h-7HhPp`. The prompt `[sysadmin@workstation ~]$` is followed by a cursor.

```
(sysadmin) 172.31.100.141 — Konsole
New Tab Split View Copy Paste Find...
[sysadmin@workstation ~]$ cat ocp-install/auth/kubeadmin-password
nFZmd-AyZIP-jyi8h-7HhPp[sysadmin@workstation ~]$
```

Overview · Cluster · Red Hat OpenShift - Google Chrome

Not secure https://console-openshift-console.apps.ocp.summit.aus.redhat.com/dashboards

Red Hat OpenShift

You are logged in as a temporary administrative user. Update the [cluster OAuth configuration](#) to allow others to log in.

Overview

Cluster

Getting started resources

Set up your cluster

Finish setting up your cluster with recommended configurations.

[Take console tour](#)

[View all steps in documentation](#)

Build with guided documentation

Follow guided documentation to build applications and familiarize yourself with key features.

[Enable the Developer Perspective](#)

[Impersonating the system:admin user](#)

[View all quick starts](#)

Explore new features and capabilities

OpenShift AI

Build, deploy, and manage AI-enabled applications.

Trusted Software Supply Chain

Assess risk, validate integrity, secure artifacts, release safely.

OpenShift Lightspeed

Your personal AI helper.

[See what's new in OpenShift 4.21](#)

Details

Cluster API address

https://api.ocp.summit.aus.redhat.com:6443

Cluster ID

51a310bb-b367-414d-833e-f0981bc999fd

[OpenShift Cluster Manager](#)

Infrastructure provider

None

OpenShift version

4.21.14

Service Level Agreement (SLA)

Self-support, 60 day trial

59 days remaining

[Manage subscription settings](#)

Update channel

stable-4.21

Cluster inventory

[6 Nodes](#)

[Pods](#)

[StorageClasses](#)

[PersistentVolumeClaim](#)

CPU

48 available of 48

Memory

159.5 GiB available of 188 GiB

[28.53 GiB](#)

Filesystem

Network transfer

Pod count

Activity

Ongoing

There are no ongoing activities.

Recent events

[View alerts](#)

[Configure](#)

1 hour

3:45 PM

3:42 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:40 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:38 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:38 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Status for clusteroperator/openshift-apiserver changed: Pro...**

3:36 PM **Updated Deployment.apps/apiserver-n openshift-apiserver...**

3:36 PM **Updated ConfigMap/config.openshift-apiserver...**

[View all events](#)

Welcome to the new OpenShift experience!

Introducing a fresh modern look to the console! With this update, we made changes to the user interface to enhance usability and streamline your workflow. This includes improved navigation and visual refinement to help manage your applications and infrastructure more easily.

What do you want to do next?

[Launch tour](#) [Skip tour](#)



Thank you



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